FINITE TIME BLOWUP FOR THE EULER EQUATION

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ABSTRACT. We exhibit finite-time blowup for the three-dimensional incompressible, unforced Euler equations from smooth, compactly supported, divergence-free initial data.

CONTENTS

1. INTRODUCTION

We study the unforced incompressible Euler equations on $\mathbb{R}^3$:

$$\partial_t u + (u \cdot \nabla) u + \nabla p = 0, \qquad \nabla \cdot u = 0, \qquad u(\cdot, 0) = u_0. \tag{1.1}$$

Here $u(t,x) \in \mathbb{R}^3$ is the velocity and $p(t,x) \in \mathbb{R}$ is the pressure. Spatial norms below are over $\mathbb{R}^3$, unless a different domain is specified. We prove that a smooth initial velocity can develop a singularity in finite time. We take the initial datum in

$$C^\infty_{c,\sigma}(\mathbb{R}^3) = \{u_0 \in C^\infty_c(\mathbb{R}^3;\mathbb{R}^3) : \nabla \cdot u_0 = 0\} \tag{1.2}$$

and write $T_*(u_0)$ for the maximal lifespan of its smooth Euler solution. Local existence gives $T_*(u_0) > 0$ [24].

Theorem 1.1. *There exists $u_0 \in C^\infty_{c,\sigma}(\mathbb{R}^3)$ such that $0 < T_*(u_0) < \infty$. Its smooth Euler solution satisfies*

$$\limsup_{t \uparrow T_*} \|\nabla u(t)\|_{L^\infty} = \infty, \qquad \int_0^{T_*} \|\operatorname{curl} u(t)\|_{L^\infty}\, dt = \infty.$$

1.1. **Historical context.** The Beale–Kato–Majda criterion [1] states that a smooth three-dimensional Euler solution continues past time T if $\int_0^T \|\operatorname{curl} u(t)\|_{L^\infty}\, dt < \infty$. Finite-time breakdown therefore requires this integral to diverge.

Elgindi [15] proved finite-time blowup on $\mathbb{R}^3$ for axisymmetric velocities without swirl in $C^{1,\alpha}$, for sufficiently small $\alpha > 0$. Elgindi, Ghoul, and Masmoudi [16] proved stability of this blowup and obtained finite-energy examples with the same regularity. Córdoba, Martínez-Zoroa, and Zheng [13] proved blowup from finite-energy $C^{1,\alpha}$ initial velocities that are smooth away from the origin. Their construction places vorticity in infinitely many regions approaching the origin, separated by regions where the vorticity vanishes. Elgindi and Pasqualotto [19] proved blowup from finite-energy $C^{1,\alpha}$ initial velocities that are smooth away from a circle. Recent preprints of Chen and Shkoller [5, 29] prove blowup on $\mathbb{R}^3$ for finite-energy axisymmetric velocities without swirl in $C^{1,\alpha}$, for every $0 < \alpha < 1/3$.

For each fixed $0 < \varepsilon < 1/2$, Córdoba and Martínez-Zoroa [10, Theorem 1.1] constructed finite-energy solutions to the forced Euler equations on $\mathbb{R}^3$, with velocity in $C^{3,1/2}$ before blowup and an external force uniformly bounded in $C^{1,1/2-\varepsilon} \cap L^2$. For these solutions, $\int_0^t \|\nabla u(s)\|_{L^\infty}\, ds$ diverges as $t \uparrow T < \infty$. For the incompressible porous media equation, Córdoba and Martínez-Zoroa [11] constructed singularities from smooth initial data by successive amplification of oscillatory layers, using approximations of increasing order to keep every spatial derivative of the source uniformly bounded.

For smooth initial data, Chen and Hou [6, 7] proved blowup in an axially periodic cylinder with an impermeable wall, combining analysis with rigorous numerical estimates. Theorem 1.1 treats smooth, compactly supported initial velocities on $\mathbb{R}^3$.

Growth of derivatives also underlies strong ill-posedness at critical regularity. Bourgain and Li [2] used large deformation of the particle flow to prove norm inflation for two-dimensional Euler in the critical velocity space H^2. Elgindi and Jeong [17] gave a construction in which localized vorticity generates a hyperbolic flow that stretches the vorticity gradient. Elgindi and Masmoudi [18] proved strong ill-posedness in the velocity spaces $C^k \cap L^2$ for integers $k \geq 1$, by estimating singular integral terms along the particle flow.

A related approach starts from oscillatory approximate solutions, estimates their difference from exact solutions, and combines spatially separated examples. Córdoba and Martínez-Zoroa [9] used this approach to prove ill-posedness and loss of regularity for the inviscid surface quasi-geostrophic equation. For two-dimensional Euler, Córdoba, Martínez-Zoroa, and Ożański [12] constructed global solutions with initial velocity in H^r, $1 < r < 2$, that lose a positive amount of Sobolev regularity at every positive time. Their approximate solutions use a radial background flow to shear an angularly oscillating perturbation and amplify its vorticity derivatives. Jeong, Martínez-Zoroa, and Ożański [23] constructed three-dimensional Euler solutions with a continuously decreasing Sobolev regularity threshold, starting from vorticity in H^s, $0 < s < 3/2$. Their mechanism combines hyperbolic deformation with vortex stretching.

For weak solutions, which satisfy Euler's equations in the sense of distributions, uniqueness and conservation of kinetic energy can fail. De Lellis and Székelyhidi [26] constructed nonunique bounded weak solutions. They later constructed continuous solutions on the three-dimensional torus with decreasing kinetic energy [27]. On the same periodic domain, Isett [22] proved that, for every $0 < \beta < 1/3$, there are solutions with spatial C^β regularity whose kinetic energy is not constant. Buckmaster, De Lellis, Székelyhidi, and Vicol [4] constructed solutions in this regularity range whose kinetic energy equals any prescribed smooth positive function of time.

Craik and Criminale [14] constructed exact finite-amplitude waves on affine background flows, exploiting the cancellation of the wave's quadratic self-interaction. Fabijonas and Holm [20] proposed iterated superpositions of such waves. Le Dizès and Leblanc [25] showed that their construction generally satisfies the equations only along a single trajectory, preventing its use as the background for the next iteration.

The linearized Euler equation describes the evolution of a small perturbation of a given flow. For perturbations with a rapidly varying phase, Lifschitz and Hameiri and Friedlander and Vishik [28, 21] studied the leading equations for the phase gradient and the velocity amplitude perpendicular to it. At the nonlinear level, Cheverry [8] constructed oscillatory approximate solutions of incompressible Euler and analyzed the additional phase corrections required by their nonlinear evolution. We write the leading linear system in (2.4) and use it to select oscillations whose velocity gradients grow along particle trajectories. Proposition 3.1 corrects these oscillations to exact smooth Euler solutions; Proposition 4.1 shows that their new gradients can amplify subsequent oscillations.

2. The amplification mechanism and proof order

We now outline the iteration used to prove Theorem 1.1. The construction in Sections 3 to 5 will produce exact smooth odd Euler solutions U^j, with pressures p_j, on nested time intervals. At stage j, we add a localized oscillation and corrections to U^{j-1} to obtain U^j. We call U^{j-1} the *parent flow* and U^j the *child flow* of this stage. The child becomes the parent at stage $j+1$. Their initial data have support in a fixed ball. We choose times $t_j \uparrow T_\infty < \infty$, with U^j defined through t_j, such that

$$|\nabla U^j(t_j, 0)| \longrightarrow \infty, \qquad \sum_j \|U^j(0) - U^{j-1}(0)\|_{H^m} < \infty \quad \text{for every fixed } m. \tag{2.1}$$

Throughout the construction we also maintain $\nabla^2 p_j(t,x) \le K_+ I$, with a constant K_+ for all stages. We choose the initial perturbations by solving displacement boundary value problems. This upper bound is used in proving those problems uniquely solvable; see (3.11).

2.1. **Flow deformation and wave amplification.**

Fix a stage and write $u = U^{j-1}$ for the parent velocity and $p = p_{j-1}$ for its pressure. We follow the oscillations in particle coordinates, where their phase remains fixed. A particle starting at a has trajectory $X(t,a)$ given by

$$\partial_t X(t,a) = u(t, X(t,a)), \qquad X(0,a) = a.$$

Along this trajectory, set

$$F = \nabla_a X, \qquad M(t,a) = \nabla u(t, X(t,a)), \qquad H(t,a) = \nabla^2 p(t, X(t,a)). \tag{2.2}$$

Incompressibility gives $\det F = 1$. Differentiating the trajectory equation and $\partial_{tt} X = -\nabla p(t,X)$ gives

$$F_t = MF, \qquad F_{tt} = -HF. \tag{2.3}$$

Thus the pressure Hessian controls the acceleration of the deformation. The corresponding vorticity equation is

$$(\partial_t + u\cdot\nabla)\omega = (\omega\cdot\nabla)u, \qquad \omega = \operatorname{curl} u.$$

This equation describes vorticity stretching along a particle trajectory. For a localized oscillation, we also track its phase and velocity amplitude. We call such an oscillation a packet.

Write $a = \ell y$, where $\ell > 0$ is the localization length, and choose a unit vector m_0 and a large frequency k. The phase is $k m_0 \cdot y$. Its physical gradient is $(k/\ell)m$, where $m(t,y) = F(t,\ell y)^{-T} m_0$. The principal velocity coefficient $v(t,y)$ satisfies

$$m_t = -M^T m, \qquad v_t = -Mv + 2m\frac{m\cdot Mv}{|m|^2}, \qquad m\cdot v = 0, \tag{2.4}$$

with M evaluated at $(t, \ell y)$. The last term in the velocity equation preserves $m\cdot v = 0$, so v remains orthogonal to the phase normal m. For an amplitude $\alpha > 0$, a nonnegative compactly supported cutoff $\chi_1(y)$, and a smooth zero-mean periodic profile f_δ with sufficiently small $\delta > 0$, the leading velocity increment is

$$w_{\text{lead}}(t, X(t, \ell y)) = \frac{\ell \alpha}{k} \chi_1(y) v(t,y) f_\delta(k m_0 \cdot y). \tag{2.5}$$

Differentiating the phase produces the leading gradient $\alpha \chi_1 v \otimes m\, f_\delta'$. Increasing k reduces the velocity by a factor k^{-1}, while leaving this leading gradient unchanged.

We choose all parent flows to be odd, $u(t,-x) = -u(t,x)$, so $X(t,0) = 0$. Along this fixed central trajectory, the parent gradient contains a rank-one shear hqp^T, with scalar amplitude h and orthogonal unit vectors p, q. This matrix sends p to hq and vanishes on vectors perpendicular to p. The shear acts together with the rest of the parent gradient to rotate m and amplify v. At the target time t_j, when the new shear reaches its prescribed size, the directions $m/|m|$ and $v/|v|$ define the normal and transverse velocity of the next frame. Choosing α makes the new leading rank-one gradient the parent shear for the next packet; Proposition 4.1 quantifies both the growth and the transfer of directions.

Córdoba and Martínez-Zoroa also use a vorticity layer to amplify a more localized layer in their forced Euler construction [10, §1.2].

Products of zero-mean periodic functions need not have zero mean. The nonlinear corrections therefore include a phase-independent part as well as an oscillatory part. To construct both, we first regard the phase as an independent variable $\theta \in \mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$ and write

$$\langle A \rangle_\theta = \frac{1}{2\pi} \int_0^{2\pi} A(t, y, \theta)\, d\theta. \tag{2.6}$$

The mean is this phase average; the oscillatory part has zero average. The linear mean and transverse problems in (3.19) and (3.27) determine their successive coefficients, including the spatially nonlocal mean pressure. After enough coefficients have been added, the error left in Euler's equation is small. Lemma 3.3 constructs a further correction that cancels this error. We then evaluate the resulting fields at $\theta = k m_0 \cdot y$. This evaluation is the restriction to the physical phase graph.

2.2. **Initial smallness and control of pressure.** At an amplification stage j, the activation time is $t_0 = t_{j-1}$: it separates the boundary value problem used to choose the wave from its forward evolution toward the target t_j. The auxiliary displacement coefficient η determines the velocity coefficient by $v = \eta_t - M\eta$ along a parent trajectory. To obtain the initial increment bounds (3.17) and the Sobolev summability in (2.1), we prescribe this displacement at activation. If $t_0 > 0$, we solve a displacement boundary value problem on $[0, t_0]$, with zero displacement at time zero and a prescribed endpoint at t_0. We call this interval the packet's history. The endpoint is chosen to select the growing transverse solution after activation, while the history estimates control the initial velocity increment. At $t_0 = 0$, the transverse velocity is prescribed directly.

In particle coordinates, the uniform upper bound on the pressure Hessians is

$$H \le K_+ I, \quad \text{that is,} \quad \eta^T H \eta \le K_+ |\eta|^2.$$

Together with the initial gradient bounds and the short time interval, it gives coercivity of the displacement form whose interior term is $\int (|\eta_t|^2 - \eta^T H \eta)\, dt$; see (3.11). Large negative eigenvalues of H increase this form. We consequently control the positive eigenvalue contribution of each new packet, using the leading increment

$$\Delta H_{\text{lead}} = -2\alpha \chi_1 (m \cdot M v) \frac{m \otimes m}{|m|^2} f_\delta'(\theta). \tag{2.7}$$

After a short initial part of the amplification interval, the geometric construction gives $m \cdot Mv > 0$. The profile is chosen with $f_\delta'(0) = \delta^{-1}$ and $f_\delta' \geq -C$. Its large positive derivative produces the desired gradient and a negative semidefinite leading pressure increment; the uniformly bounded negative derivative limits the positive pressure increment. The scale choices make these positive contributions and the errors in (3.14) summable. On the history interval and the short initial part of amplification, the full pressure increments satisfy summable bounds.

The mean displacement boundary condition enforces compact support of the initial mean correction; the initial oscillatory correction is localized by the cutoff. Thus all initial data have a common compact support. Together with (2.1), this gives convergence to $u_0 \in C^\infty_{c,\sigma}(\mathbb{R}^3)$. If its Euler solution were smooth past T_∞, stability would compare it with every U^j up to t_j, contradicting the growing gradients. This gives $T_*(u_0) \leq T_\infty$.

2.3. **The stages of the proof.** Amplification separates the packet's initial size from the shear it later produces. At the central target point, its leading gradient is $\alpha\delta^{-1} v \otimes m$. We choose α so that $\alpha\delta^{-1}|v||m| = h_{\text{child}}$, the desired size of the next shear; see (4.13). The history comparison (4.12) in Proposition 4.1 contains an exponentially decaying factor. We choose the scales so that this factor outweighs the parent coefficient bounds and the derivatives introduced by high frequency, making the initial increments summable in every fixed Sobolev norm.

The target directions $m/|m|$ and $v/|v|$ must also form a frame in which the new shear can amplify the next packet. Proposition 4.1 establishes this transfer and the sign $m \cdot Mv > 0$ after the short initial part of amplification. The sign makes the large positive part of f_δ' contribute a negative semidefinite leading pressure increment. The scale choices make the positive pressure contributions and correction errors summable, preserving the common upper Hessian bound. Proposition 3.1 realizes the growing wave as an exact Euler perturbation: it constructs the nonlinear corrections and bounds their effect on the gradient, pressure, and initial data. We prove this packet estimate first under growth assumptions, then establish those assumptions and the frame transfer for the parent shear. The resulting exact flow supplies the parent for the next stage. Figure 1 shows where these results enter the iteration.

1. **Construct the mean and oscillatory corrections.** In Section 3, a smooth parent flow supplies the coefficients of two linear inverses. Lemma 3.2 controls their derivatives, hence the expansion in inverse frequency and the cancellation of successive coefficients.
2. **Obtain an exact Euler solution.** Section 3.6 removes the remaining error and restricts the solution to the physical phase graph. Proposition 3.1 collects the resulting gradient, pressure, initial-data, and particle-map estimates used in the iteration.
3. **Prove amplification and transfer of the frame.** Proposition 4.1 supplies the growing transverse solution and propagator bound required by the packet construction. It also shows that the frame at the target time satisfies the geometric conditions for another application.
4. **Choose the scales and iterate.** Sections 5.2, 5.3 and 5.5 choose and verify the amplitudes, frequencies, localization lengths, and time intervals. The seed and inductive step in Sections 5.6 and 5.7 produce the solutions U^j with the bounds in (2.1).
5. **Pass to the limiting initial datum.** Section 6 proves smooth convergence of the initial data, applies Euler stability under the assumption of continuation past T_∞, and proves the divergence assertions in Theorem 1.1: unboundedness of $\|\nabla u(t)\|_\infty$ as $t \uparrow T_*$, and $\int_0^{T_*} \|\operatorname{curl} u(t)\|_\infty \, dt = \infty$.

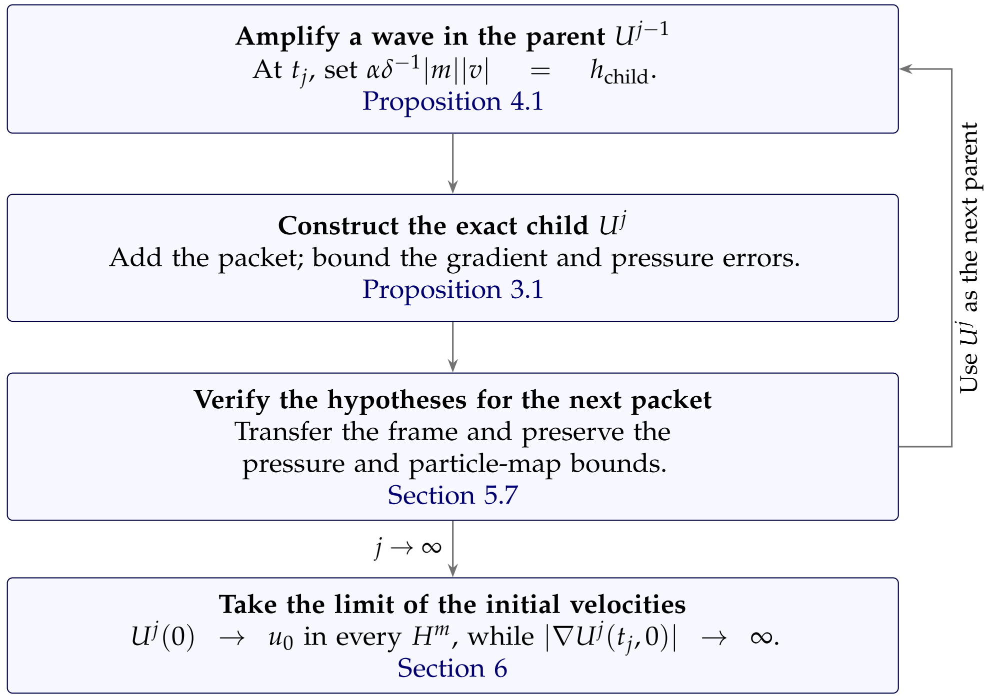


FIGURE 1. Construction of U^j from the smooth parent U^{j-1}. Proposition 4.1 supplies a growing wave with the target shear and directions needed for repetition; Proposition 3.1 adds the wave and its corrections to obtain an exact Euler solution. The return arrow means that, after the checks in Section 5.7, U^j becomes the parent at stage $j+1$. The downward limit uses the initial-data summability and gradient growth in (2.1), with $t_j \uparrow T_\infty$. Euler stability in Section 6 then gives $T_*(u_0) \le T_\infty$.

2.4. **Notation and conventions. Particle labels and components.** We write a for a physical particle label and $y = a/\ell$ for its normalization, where $0 < \ell \le 1$. In §4, the physical label is written a, since $a = B_{pq}(t_0)$ denotes the scalar in (4.4). For a coefficient $b(t, a)$, the normalized coefficient $b_\ell(t, y) = b(t, \ell y)$ satisfies $\partial_y^I b_\ell = \ell^{|I|}(\partial_a^I b)(t, \ell y)$; thus normalization does not increase supremum derivative bounds. A dot on a quantity in particle coordinates denotes differentiation in physical time at a fixed label. For an orthonormal frame (p, q, n), we write $B_{ij} = i \cdot Bj$ and $v_i = i \cdot v$, for $i, j \in \{p, q, n\}$.

Derivative norms. The packet estimates use $s = 6$ and $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$ with normalized measure. An *ordered word* I of length n is an ordered sequence of n coordinate derivatives, and ∂^I is their composition. For functions on $\mathbb{R}^3$ or $\mathbb{R}^3 \times \mathbb{T}$, set

$$|f|_n = \sum_{|I|=n} \|\partial^I f\|_{H^s}, \qquad |a|_{n,\infty} = \sum_{|I|=n} \|\partial^I a\|_{W^{s,\infty}}. \tag{2.8}$$

Here a denotes a coefficient, which need not decay at infinity. On a time interval, a norm with a supremum in time means the sum of the supremum for each word; the same wordwise convention applies to L^2 in time. The sum of wordwise suprema is at most d^n times the supremum of the word sum in dimension d, which only changes a fixed growth constant in a factorial estimate.

Gevrey bounds. An order-two Gevrey bound for f means

$$|f|_n \le CR^n (n!)^2 \qquad (n \ge 0),$$

with C, R independent of n; coefficient bounds use $|a|_{n,\infty}$ instead. Such bounds allow nonzero compactly supported cutoffs. The product rules in §3.1 and the linear estimates in Lemma 3.2 preserve this form after fixed enlargements of R, allowing us to control the successive coefficients of the oscillatory expansion. For $\rho R < 1$, the weighted derivative norm (3.43) satisfies

$$\|f\|_\rho = \sum_{n\ge 0} \frac{\rho^n |f|_n}{(n!)^2} \le \frac{C}{1-\rho R}.$$

We use this norm in Lemma 3.3 to control the correction simultaneously at all spatial derivative orders. Constants in estimates stated only at a fixed derivative order may depend on that order.

Polynomial constants. The parameter $P \ge 2$ collects finitely many coefficient and scale bounds. In the packet estimates, P^c denotes a fixed power that may be enlarged finitely many times. Its exponent may depend on s, the dimension, and the fixed exponent c_0 in the packet data bounds of §3.2, but is independent of the frequency, the number of expansion terms, and the derivative order. In §4, K_h controls factorial Sobolev bounds for $X - \mathrm{id}$, X_t, and X_{tt}, as in (3.15). Powers K_h^c may also be enlarged a fixed finite number of times. Their exponents depend on the fixed low-order bounds, not on the stage or the exponent in the preceding stage's estimate. Constants described as fixed in §5 are independent of the stage index; §5.2 specifies their order of choice.

Notation index. The entries below locate the definitions used in each construction. Symbols attached to a packet or a stage retain the scope of that construction.

Notation	Meaning	Definition
Flow, phase, and primary oscillation		
$C^\infty_{c,\sigma}$, $T_*(u_0)$	Smooth compactly supported divergence-free data on $\mathbb{R}^3$, and the maximal smooth lifespan of the solution from u_0.	(1.2)
X, F, M, H	Particle map, deformation $F = \nabla_a X$, velocity gradient $M = \nabla u$, and pressure Hessian $H = \nabla^2 p$, with the last two evaluated along the parent trajectory.	(2.2)
a, a	Physical particle label; a is used in §4, where a is a scalar coefficient.	§2.4
ℓ, y, Ξ	Localization length, rescaled label $a = \ell y$, and $\Xi(t, y) = \ell^{-1} X(t, \ell y)$; also $F = \nabla_y \Xi$.	(3.5)
θ, $\langle\cdot\rangle_\theta$	Independent angle in $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$, with average $(2\pi)^{-1}\int_0^{2\pi} \cdot\, d\theta$. Physical evaluation uses $\theta = km_0 \cdot y$.	(2.6)
k, κ	Packet frequency in the rescaled label and its inverse $\kappa = k^{-1}$; the physical phase gradient is $(k/\ell)m$.	(3.7)
m_0, m, D_m, v	Fixed unit phase normal, transported normal $m = F^{-T} m_0$, $D_m = \|m\|^2$, and primary transverse velocity coefficient $m \cdot v = 0$.	(2.4), (3.7)
α, f_δ, δ	Amplitude, odd zero-mean periodic profile, and its concentration parameter; $f'_\delta(0) = \delta^{-1}$. The leading velocity is $(\ell\alpha/k)\chi_1 v f_\delta$.	(2.5), (3.4)
d_i, $\widetilde{d}_i$, D_i	$d_i = \sum_j (F^{-1})_{ji}\partial_{y_j}$, $\widetilde{d}_i = d_i + km_i\partial_\theta$, and $D_i = \kappa\partial_{y_i} + (m_0)_i\partial_\theta$. Physical graph derivatives are $\ell^{-1}\widetilde{d}_i$.	(3.7); (3.46)
Norms and packet corrections		
s, I, $\|f\|_n$, $\|a\|_{n,\infty}$	$s = 6$; I is an ordered derivative word. Sum $\\|\partial^I f\\|_{H^s}$, respectively $\\|\partial^I a\\|_{W^{s,\infty}}$, over $\|I\| = n$, on $\mathbb{R}^3$ or $\mathbb{R}^3 \times \mathbb{T}$ as specified.	(2.8)
P, P^c, P_j	Data-size bound $P \geq 2$, fixed powers of P, and the bound chosen at stage j. The exponent c in P^c is independent of frequency, expansion length, and derivative order.	§2.4; (5.10)
R, d, $h(t)$ in shift bounds	Derivative growth constant, nonnegative integer shift, and positive time profile; the derivative bound is $h(t)R^{n+d}((n+d)!)^2$. This h is a general profile.	(3.1)
ρ, $\\|f\\|_\rho$	Radius and weighted derivative norm $\sum_{n\geq 0} \rho^n \|f\|_n/(n!)^2$.	(3.43)
χ_1, χ_o	Fixed even cutoffs: $\operatorname{supp}\chi_1 \subset \{\|y\| < 1/2\}$, $\chi_1(0) = 1$; $\chi_o = 1$ on $\|y\| \leq 1$, supported in $\|y\| \leq 2$.	§3.1
$R_\perp$, K, G, K_R	Orthonormal basis matrix for $m_0^\perp$, $K = F^T F$, $G = K^{-1}$, and $K_R = R_\perp^T K R_\perp$.	(3.10)
K_h, $g(t)$	Factorial Sobolev bound K_h for $X - \mathrm{id}, X_t, X_{tt}$; positive profile $g(t_0) = 1$ weighting the transverse propagator on $\|y\| \leq 1/2$ by $g(t)/g(s')$, $t_0 \leq s' \leq t \leq S$.	(3.15); §3.2 (iii)
K_+, B_e, B_c, L	Upper pressure-Hessian bound $H \leq K_+ I$, lower initial symmetric gradient bounds $-B_e I$ for $\|\ell y\| \geq r$ and $-B_c I$ for $\|\ell y\| < r$, and coefficient L of the initial boundary operator.	(3.11)

Notation	Meaning	Definition
$\mathcal{N}$, $\mathcal{A}$	Newton operator $(-\Delta_y)^{-1}$ and initial-support operator $\mathrm{curl}_y\,\chi\mathcal{N}\chi\,\mathrm{curl}_y$, where $\chi(y) = \chi_o(\ell y)$.	(3.20)
A_p, B_p, Q_p, C_p	Zero-angle-mean transverse and angle-independent velocity coefficients at inverse-frequency order p; vector potential $Q_p = -\partial_\theta^{-1}(m \times A_p)/D_m$ and curl remainder $C_p = d \times Q_p$. The angle primitive has zero mean.	(3.33)
N_k, W^a, q^a	Expansion length $N_k = \lfloor k^\vartheta \rfloor$, $\vartheta = 10^{-6}$, and approximate normalized velocity and pressure increments; physical increments carry factors ℓ and ℓ^2.	(3.34)
Amplification geometry		
p, q, n, B_{ij}	Orthonormal frame of an older flow: normalized ray, normalized transverse velocity, and $n = p \times q$; $B_{ij} = i \cdot Bj$ for $i, j \in \{p, q, n\}$.	(4.2)
B, h, h_*, E	Older central gradient, current shear amplitude, its activation value, and error in $M(t, 0) = B + hqp^T + E$, with $h(t_0) = h_*$.	(4.3)
a, β, ε	Frozen coefficients $a = B_{pq}(t_0)$, $\beta = B_{nq}(t_0)/a$, and $\varepsilon = \sqrt{a/h_*}$. The physical label in §4 is a.	(4.4)
τ, x, x_{prev}, x_{tar}	Rescaled times $\tau = a(t - t_0)/\varepsilon$, $x = \sqrt{\beta}\,\tau$, comparison scale $x_{\mathrm{prev}} \asymp \beta^{-1/2}$, and target value x_{tar}. Here x is not spatial position.	(4.4)
p_2, q_2, n_2, A_{tar}	Child frame $p_2 = m/\lvert m\rvert$, $q_2 = v/\lvert v\rvert$, $n_2 = p_2 \times q_2$, and central target magnitude $A_{\mathrm{tar}} = \lvert m(t_{\mathrm{tar}}, 0)\rvert\lvert v(t_{\mathrm{tar}}, 0)\rvert$.	Proposition 4.1
Iteration and limiting datum		
U^j, p_j, M_j, H_j	Exact stage solution and its pressure, with $M_j = \nabla U^j$, $H_j = \nabla^2 p_j$. At stage $j \geq J$, the parent is U^{j-1} and the older solution is U^{j-2}; $U^{J-2} = U_B$ is the base.	§5, §5.4
x_j, h_j, k_j, ℓ_j, δ_j, α_j	Stage scale parameter, target shear, frequency, localization length, profile concentration, and amplitude $\alpha_j = \delta_j h_j/(\lvert m(t_j, 0)\rvert\lvert v(t_j, 0)\rvert)$ for $j \geq J$.	(5.3), (5.18)
t_j, S_j, W_j	Target time, retained horizon, and available width. Stage $j \geq J$ activates at $t_0 = t_{j-1}$, with $t_{J-1} = 0$, and retains $S_j = t_j + 2W_{j+1}$.	(5.7), (5.8)
T_∞	Finite accumulation time $\lim_j t_j$; the limiting datum satisfies $T_*(u_0) \leq T_\infty$.	(6.4), (6.6)

3. A localized oscillation over a smooth Euler flow

In this section we prove Proposition 3.1, the packet estimate used in the iteration outlined in §2. We work under the hypotheses in §3.2; the next section will establish the required wave growth and propagator bounds for a parent shear. Starting from the given smooth Euler solution u, we construct another exact solution u^{new} by adding the leading oscillation (2.5) and correction terms. The resulting gradient and pressure-Hessian differences are (3.13) and (3.14) in Claim 1 of Proposition 3.1.

Two linear problems supply the correction terms. Equation (3.19) constructs angle-independent velocities; its boundary condition (3.25) imposes compact support at time zero.

Equation (3.29) constructs transverse velocity coefficients with zero angle mean and retains the spatial support of its forcing.

We first seek approximate velocity and pressure increments as finite series in k^{-1}, with θ independent of y; these are (W^a, q^a) in (3.34). Substitution into Euler's equation leaves an error, called the residual. Its normalized form is bounded in (3.45). Under (3.12), Lemma 3.3 constructs correction fields on $\mathbb{R}^3 \times \mathbb{T}$ with zero initial velocity. Restricting to $\theta = km_0 \cdot y$ then gives the physical velocity and pressure in (3.47)–(3.48).

3.1. **Norms, cutoffs, and factorial estimates.** We first fix the derivative bounds and cutoff functions used to control all orders of the oscillatory expansion with a common growth constant. We use the derivative norms $|f|_n$ and $|a|_{n,\infty}$ of §2.4, which sum the H^6 and $W^{6,\infty}$ norms over ordered derivative words. The data bound $P \geq 2$ and fixed powers P^c follow the conventions in the same section. For $R \geq 1$ and an integer $d \geq 0$, we say that a function has *shift d* with profile $h(t) > 0$ if

$$\sum_{|I|=n} \sup_t h(t)^{-1} \|\partial^I f(t)\|_{H^s} \leq R^{n+d}((n+d)!)^2 \qquad (n \geq 0). \tag{3.1}$$

In the linear estimates we also use the version of (3.1) obtained by replacing each $\sup_t h(t)^{-1}\|\partial^I f(t)\|_{H^s}$ by $\|h^{-1}\partial^I f\|_{L^2_t H^s}$. Here R is the constant controlling derivative growth: a larger R gives a weaker bound. Later we sum these derivatives with weights involving a small radius ρ, chosen proportional to R^{-1}. The shift d records the losses from differentiation, products, and inverse operators while R remains fixed across the expansion.

Here are the two elementary rules used to keep a common growth constant for the entire construction. The Sobolev product inequality in dimension at most four and Leibniz's rule show that, if f_i has shift d_i and profile h_i for $i = 1, 2$, with the same R, then $f_1 f_2$ satisfies the bound for shift $d_1 + d_2$ and profile $h_1 h_2$, up to a fixed multiplicative constant, because

$$\frac{\binom{n}{l}((l+d_1)!)^2((n-l+d_2)!)^2}{((n+d_1+d_2)!)^2} = \frac{\binom{n}{l}}{\binom{n+d_1+d_2}{l+d_1}^2} \leq \binom{n}{l}^{-1}, \qquad \sum_{l=0}^{n} \binom{n}{l}^{-1} \leq 3. \tag{3.2}$$

The same proof applies to multiplication $W^{s,\infty} \times H^s \to H^s$. A derivative in y or θ raises the shift by 1 and preserves the profile. Multiplication by a coefficient a satisfying

$$\sum_{|I|=n} \sup_t \|\partial^I a(t)\|_{W^{s,\infty}} \leq P^c R_c^n (n!)^2, \qquad R_c = P^c,$$

costs an additional shift when R is a sufficiently large fixed power of P.

The second rule estimates an unknown sequence of derivative bounds Z_n from a prescribed sequence F_n. Suppose these sequences are nonnegative and satisfy the triangular estimate

$$Z_n \leq P^c \left(F_n + \sum_{l=1}^{n} \binom{n}{l} R_c^l (l!)^2 Z_{n-l} \right). \tag{3.3}$$

For an integer $d \geq 1$, suppose $F_n \leq R^{n+d-1}((n+d-1)!)^2$ for every $n \geq 0$. Induction then gives $Z_n \leq R^{n+d}((n+d)!)^2$ for every $n \geq 0$. Indeed, division by $R^{n+d}((n+d)!)^2$ bounds the data term by P^c/R and the remaining sum by $P^c \sum_{l \geq 1}(R_c/R)^l$ times the largest previous normalized bound. We choose R so that their sum is at most one. This choice depends on the estimate at the fixed Sobolev index producing (3.3), and never on the expansion length.

The cutoff χ_1 localizes the oscillatory velocity in the rescaled label y. The cutoff χ_o, evaluated later at ℓy, will impose support of the initial mean velocity in the physical label.

We fix even smooth cutoffs χ_1, χ_o on $\mathbb{R}^3_y$ with

$$0 \le \chi_1 \le 1, \quad \operatorname{supp} \chi_1 \subset \{|y| < 1/2\}, \quad \chi_1(0) = 1,$$
$$\chi_o = 1 \text{ on } \{|y| \le 1\}, \quad \operatorname{supp} \chi_o \subset \{|y| \le 2\},$$

and derivative bounds $C^{n+1}(n!)^2$. Such cutoffs follow, for example, from the extension by zero of $\exp(-1/z)$ for $z > 0$. Cauchy's estimate on the complex circle of radius $z/2$ gives $n!(2/z)^n \exp(-c/z) \le C^{n+1}(n!)^2$; integration, products, and fixed rescalings give the asserted cutoffs. The leading gradient and pressure-Hessian increments depend on f_δ', as in (3.13) and (3.14). For sufficiently small $\delta > 0$, we therefore fix a smooth odd periodic function f_δ of mean zero such that

$$f_\delta'(0) = \delta^{-1}, \qquad -C \le f_\delta' \le \delta^{-1}, \qquad \|\partial_\theta^n f_\delta\|_{H^{s+2}(\mathbb{T})} \le \delta^{-c}(C\delta^{-c})^n (n!)^2. \tag{3.4}$$

To see this explicitly, take an even nonnegative bump $g \le 1$ constructed as above, with derivative bounds $C^{n+1}(n!)^2$, supported in a small subinterval of $(-\pi, \pi)$, with $g(0) = 1$, and put $c_g = \langle \delta^{-1} g(\theta/\delta) \rangle_\theta$. This number is independent of δ. The even function

$$h_\delta(\theta) = \frac{\delta^{-1} g(\theta/\delta) - c_g}{1 - c_g \delta}$$

has mean zero, maximum δ^{-1} at zero, and a uniform lower bound. Its odd periodic primitive is f_δ. The derivative estimates follow from the cutoff estimates and the bounded mean-zero integration operator.

3.2. **Coordinates and hypotheses.** We write the velocity increment as $\ell W(t, y, \theta)$ in the particle coordinates $x = X(t, \ell y)$, where X is the parent particle map, ℓ is the localization length, and y is the rescaled particle label. The angle $\theta \in \mathbb{T}$ is initially an independent variable. To obtain the physical increment we restrict to $\theta = k m_0 \cdot y$, where k is the frequency and m_0 is a fixed unit phase normal. The coefficient and propagation assumptions below are the inputs to Proposition 3.1.

Fix the hypothesis exponent c_0. Let u be a smooth odd Euler velocity on $\mathbb{R}^3 \times [0, S]$, where $0 < S \le 1$, with pressure p, and let $X(t, a)$ be its particle map:

$$X_t(t, a) = u(t, X(t, a)), \qquad X(0, a) = a.$$

The map $X(t, \cdot)$ preserves volume and $X(t, 0) = 0$. Fix $0 < \ell \le 1$ and define

$$\Xi(t, y) = \ell^{-1} X(t, \ell y), \quad F = \nabla_y \Xi, \quad M = (\nabla u)(t, X(t, \ell y)), \quad H = (\nabla^2 p)(t, X(t, \ell y)). \tag{3.5}$$

Differentiation of the particle equation and of Euler's equation gives

$$\det F = 1, \qquad F(0) = I, \qquad F_t = MF, \qquad F_{tt} = -HF. \tag{3.6}$$

Set $d_i = \sum_j (F^{-1})_{ji} \partial_{y_j}$. These are the commuting derivatives in the coordinates Ξ; the Jacobian identity gives $d \cdot W = \operatorname{div}_y(F^{-1} W)$.

Fix a constant unit vector m_0 and a frequency $k > 0$, and put

$$m = F^{-T} m_0, \qquad D_m = |m|^2, \qquad \widetilde{d} = d + k m \partial_\theta, \qquad \kappa = k^{-1}. \tag{3.7}$$

We first regard θ as an independent periodic variable. The *graph restriction* of a function $a(t, y, \theta)$ is $a(t, y, k m_0 \cdot y)$. Its derivative with respect to the normalized physical coordinate $\Xi = x/\ell$ is the graph restriction of $\widetilde{d} a$. Its physical derivative is the graph restriction of $\ell^{-1} \widetilde{d} a$. In particular the components of $\widetilde{d}$ commute. Adding the physical velocity ℓW and pressure $\ell^2 q$, evaluated on this graph at $x = X(t, \ell y)$, preserves Euler's equation whenever W, q satisfy, on $\mathbb{R}^3 \times \mathbb{T}$,

$$W_t + MW + W \cdot \widetilde{d}W + \widetilde{d}q = 0, \qquad \widetilde{d} \cdot W = 0. \tag{3.8}$$

Fix $t_0 \in [0, S)$ and a constant 3×2 matrix $R_\perp$ whose columns are an orthonormal basis of $m_0^\perp$. Also fix $\alpha > 0$. The time t_0 separates the history interval $[0, t_0]$, when $t_0 > 0$, from forward evolution on $[t_0, S]$. We first prescribe the primary velocity coefficient v. If $t_0 > 0$, choose a constant $\xi_T \in \mathbb{R}^2$ with $|\xi_T| \leq P^{c_0}$. For each fixed y, choose $\xi = \xi(t, y)$ with $\xi(0, y) = 0$ and $\xi(t_0, y) = \xi_T$ so that the following action, with $\eta = FR_\perp \xi$, is stationary under variations $\xi \mapsto \xi + \varepsilon b$ satisfying $b(0) = b(t_0) = 0$:

$$\int_0^{t_0} \left(|\eta_t|^2 - H\eta \cdot \eta \right) dt.$$

Set $v = FR_\perp \xi_t$ on this interval. If $t_0 = 0$, prescribe $v(0)$ to be a constant unit vector in $m_0^\perp$. In either case, continue v from t_0 by

$$v_t = -Mv + 2m \frac{m \cdot Mv}{D_m}. \tag{3.9}$$

Under (3.11), this fixed-endpoint variational problem has a unique solution; the argument is given with (3.27). Thus the endpoint data determine v on $[0, t_0]$.

For vectors $z \in \mathbb{R}^3$ and $b \in \mathbb{R}^2$, the squared lengths $|Fz|^2$ and $|FR_\perp b|^2$ are quadratic forms with matrices K and K_R, respectively. Introduce these matrices and the inverse G:

$$K = F^T F, \qquad G = K^{-1} = F^{-1} F^{-T}, \qquad K_R = R_\perp^T K R_\perp. \tag{3.10}$$

The following assumptions concern these coefficients, the time interval, and the forward propagation of the primary equation.

(i) **Coefficient and scale bounds.** For each coefficient $A \in \{F, F^{-1}, F_t, F_{tt}, M, H\}$, assume

$$\sum_{|I|=n} \sup_{0 \leq t \leq S} \|\partial_y^I A(t)\|_{W^{s,\infty}(\mathbb{R}^3)} \leq P^{c_0} (P^{c_0})^n (n!)^2 \qquad (n \geq 0).$$

The matrices K, G, K_R have eigenvalues bounded below by P^{-c_0}, and the scalar D_m satisfies $D_m \geq P^{-c_0}$. The inverses $K^{-1}, G^{-1}, K_R^{-1}, D_m^{-1}$ satisfy coefficient bounds of the same form. The inverses of S and, when positive, t_0 are at most P^{c_0}. Also $\ell^{-1}, \delta^{-1} \leq P^{c_0}$.

(ii) **Quadratic-form bounds and coercivity.** For nonnegative K_+, B_e, B_c, the quadratic-form bounds

$$H \leq K_+ I, \qquad \operatorname{sym} M(0) \geq -B_e I \quad \text{if } |\ell y| \geq r, \qquad \operatorname{sym} M(0) \geq -B_c I \quad \text{if } |\ell y| < r$$

hold, where $0 < r < 1/4$. Choose a scalar $L \geq 0$ for the displacement boundary term in (3.21). This term compensates for the possible negative initial gradient inside $|\ell y| < r$. With the dimensional constants C_1, C_2 in (3.23), assume

$$\frac{K_+ S^2}{2} + B_e S + C_2 B_c r^3 S \leq \frac{1}{2}, \qquad L \geq C_1 B_c, \qquad 0 \leq L \leq P^{c_0}. \tag{3.11}$$

Alternatively, if $\operatorname{sym} M(0) \geq -B_e I$ everywhere, the interior bound may be omitted and $B_c = 0$ used in (3.11).

(iii) **Transverse propagation and amplitude.** For each y with $|y| \leq 1/2$, express a transverse solution of (3.9) as $A(t, y) = F(t, y) R_\perp a(t, y)$, where $a(t, y) \in \mathbb{R}^2$ is its vector of transverse coefficients. For $t_0 \leq s' \leq t \leq S$, assume that the linear map $a(s', y) \mapsto a(t, y)$

has operator norm at most $P^{c_0} g(t)/g(s')$, where $g : [t_0, S] \to (0, \infty)$ and $g(t_0) = 1$. The amplitude satisfies $\alpha, \sup_{[t_0,S]} \alpha g \le P^{c_0}$.

All the data are smooth.

For the packet estimate, retain these smooth data and assume §3.2 (i)–(iii).

Proposition 3.1. *There is a finite exponent $Q = Q(c_0)$, uniform over the smooth packet data satisfying §3.2 (i)–(iii), such that for every $k \ge k_0(c_0)$ satisfying*

$$P^Q \le k^{\vartheta/100}, \qquad \vartheta = 10^{-6}, \tag{3.12}$$

there is a smooth odd exact Euler solution u^{new} on $[0, S]$, with pressure p^{new}.

1. **Gradient and pressure increments.** *At $x = X(t, \ell y)$, with $\theta = km_0 \cdot y$, its increments satisfy*

$$\nabla u^{\text{new}} - \nabla u = \alpha \chi_1 v \otimes m\, f'_\delta(\theta) + O(k^{-1/4}), \tag{3.13}$$

$$\nabla^2 p^{\text{new}} - \nabla^2 p = -2\alpha\chi_1 (m \cdot Mv) \frac{m \otimes m}{D_m} f'_\delta(\theta) + O(k^{-1/4}). \tag{3.14}$$

The errors are uniform globally in space and on $[0, S]$.

2. **Particle-map bounds.** *Suppose in addition that, for every integer $n \ge 0$, the parent particle map obeys*

$$\sum_{|I|=n} \sup_{0\le t\le S} \|\partial_a^I (X - \mathrm{id}, X_t, X_{tt})\|_{H^s} \le K_h^{n+1} (n!)^2, \qquad 1 \le K_h \le P^{c_0}. \tag{3.15}$$

Then, for every integer $n \ge 0$, the particle map X^{new} of u^{new} obeys

$$\sum_{|I|=n} \sup_{0\le t\le S} \|\partial_a^I (X^{\text{new}} - \mathrm{id}, X_t^{\text{new}}, X_{tt}^{\text{new}})\|_{H^s} \le (k^{C_*})^{n+1} (n!)^2, \qquad C_* = 10(s+2). \tag{3.16}$$

3. **Initial increment.** *The initial increment splits as $u_{\text{high},0} + u_{\text{mean},0}$, with supports in $\{|a| \le \ell/2\}$ and $\{|a| \le 2\}$, respectively. For each fixed integer $m \ge 0$,*

$$\|u_{\text{high},0}\|_{H^m} \le \ell^{-m} \alpha k^m P^{c_m}, \qquad \|u_{\text{mean},0}\|_{H^m} \le \ell^{-m} k^{-2} P^{c_m}. \tag{3.17}$$

If $t_0 = 0$ and $L = 0$, then $u_{\text{mean},0} = 0$. Let ϕ_δ be the mean-zero periodic primitive of f_δ. In this case, the initial oscillatory increment is

$$u_{\text{high},0}(a) = -\frac{\ell^2 \alpha}{k^2} \operatorname{curl}_a \left[\chi_1(a/\ell)(m_0 \times v(0)) \phi_\delta(km_0 \cdot a/\ell) \right]. \tag{3.18}$$

Proof strategy for Proposition 3.1. We first solve the linear equation for the angle-independent correction, imposing compact support of its initial velocity through a displacement boundary condition. We then solve the linear equation for the zero-mean correction subject to $m \cdot A = 0$. The estimates in Lemma 3.2 allow these two corrections to be constructed successively at each power of k^{-1}. After truncation, the remaining error satisfies (3.45). Finally, Lemma 3.3 removes that error with a correction having zero initial velocity, so the initial support is preserved.

3.3. **A linear inverse with compactly supported initial velocity.** To construct the mean part of the packet in Proposition 3.1, let $f \in L^2((0, S); L^2(\mathbb{R}^3; \mathbb{R}^3))$ be angle independent. We seek an angle-independent velocity B and pressure $\bar{q}$ satisfying

$$B_t + MB + d\bar{q} = f, \qquad d \cdot B = 0, \qquad \operatorname{supp} B(0) \subset \{|\ell y| \le 2\}. \tag{3.19}$$

Only $B(0)$ is required to have compact support; the pressure and the mean velocity at positive times may be nonlocal. Introduce an auxiliary displacement $\eta = Fz$, where $\operatorname{div}_y z = 0$ and

$z(S) = 0$. The terminal condition fixes the additive constant in z. Since $F_t = MF$, its associated velocity is

$$B = \eta_t - M\eta = Fz_t, \qquad d \cdot B = \operatorname{div}_y z_t = 0.$$

We will impose $z_t(0) = L\mathcal{A}z(0)$, with $\mathcal{A}$ chosen to have compactly supported output. Introduce the Newton operator $\mathcal{N} = (-\Delta_y)^{-1}$, the cutoff $\chi(y) = \chi_o(\ell y)$, and the nonnegative operator

$$\mathcal{A} = \operatorname{curl}_y \chi \mathcal{N} \chi \operatorname{curl}_y . \tag{3.20}$$

The outer curl makes $\mathcal{A}v$ divergence free, and the outer cutoff gives $\operatorname{supp} \mathcal{A}v \subset \{|\ell y| \leq 2\}$. For smooth v,

$$\langle \mathcal{A}v, v \rangle = \|\mathcal{N}^{1/2} \chi \operatorname{curl}_y v\|_2^2 \geq 0.$$

On the Hilbert space

$$\mathcal{V} = \{z \in H^1((0,S); L^2_\sigma(\mathbb{R}^3)) : z(S) = 0\}, \qquad \|z\|_{\mathcal{V}} = \|z_t\|_{L^2_t L^2_y},$$

where L^2_σ denotes distributionally divergence-free fields, define the bilinear form in the first line below. We seek $z \in \mathcal{V}$ satisfying the second line for every $b \in \mathcal{V}$:

$$\begin{aligned} a(z,b) &= \int_0^S \int_{\mathbb{R}^3} \big((Fz)_t \cdot (Fb)_t - HFz \cdot Fb \big)\, dy\, dt + \langle M(0)z(0), b(0) \rangle + L\langle \mathcal{A}z(0), b(0) \rangle, \\ a(z,b) &= -\int_0^S \langle f, Fb \rangle\, dt \qquad (b \in \mathcal{V}). \end{aligned} \tag{3.21}$$

We first prove that $\mathcal{A}$ is bounded on L^2, then prove $a(z,z) \geq P^{-c}\|z\|_{\mathcal{V}}^2$. For a scalar multiplier ψ with $\|\psi\|_\infty + \|\nabla\psi\|_3 < \infty$,

$$\|\psi \operatorname{curl} z\|_{\dot H^{-1}} \leq C\big(\|\psi\|_\infty + \|\nabla\psi\|_3\big)\|z\|_2. \tag{3.22}$$

Indeed, pairing with φ and integrating by parts gives $\langle z, \psi \operatorname{curl} \varphi + \nabla\psi \times \varphi \rangle$. Hölder's inequality and $\|\varphi\|_6 \leq C\|\nabla\varphi\|_2$ prove (3.22). For two scalar multipliers ψ_1, ψ_2 satisfying these bounds, use

$$\langle v, \operatorname{curl} \psi_1 \mathcal{N} \psi_2 \operatorname{curl} z \rangle = \langle \mathcal{N}^{1/2}\psi_1 \operatorname{curl} v, \mathcal{N}^{1/2}\psi_2 \operatorname{curl} z \rangle.$$

Applying (3.22) to both factors proves that $\operatorname{curl} \psi_1 \mathcal{N} \psi_2 \operatorname{curl}$ is bounded on L^2. For our cutoff,

$$\|\partial^I \chi\|_\infty + \|\nabla \partial^I \chi\|_3 \leq C^{n+1}(n!)^2 \qquad (|I| = n).$$

Here the factor ℓ^{n+1} in the second term is multiplied by the L^3 scaling factor ℓ^{-1}. By Leibniz's rule, $[\partial^I, \mathcal{A}]z$ is a sum of terms

$$\operatorname{curl}(\partial^J \chi)\mathcal{N}(\partial^K \chi) \operatorname{curl} \partial^{I'} z, \qquad |J| + |K| + |I'| = |I|, \quad |J| + |K| \geq 1.$$

The pairing estimate bounds each operator on L^2. Applying up to s further derivatives gives

$$\| \operatorname{curl}(\partial^J \chi)\mathcal{N}(\partial^K \chi) \operatorname{curl} \|_{H^s \to H^s} \leq C^{|J|+|K|+1}\big((|J| + |K|)!\big)^2.$$

These bounds control the boundary terms when we differentiate (3.21).

The boundary contribution from $M(0)$ can be negative on $\{|\ell y| < r\}$. To absorb it into $L\langle \mathcal{A}z(0), z(0) \rangle$, we bound the L^2 mass of $z(0)$ in this region by the boundary form and a small multiple of $\|z(0)\|_2^2$. Put $Z = \|\mathcal{N}^{1/2}\chi \operatorname{curl} z(0)\|_2$ and $w = \mathcal{N} \operatorname{curl}(\chi \operatorname{curl} z(0))$. The Fourier multiplier of $\mathcal{N}^{1/2}\operatorname{curl}$ has norm at most one, so $\|w\|_2 \leq Z$. Since $\operatorname{div}_y z(0) = 0$ and $\chi = 1$ on $|y| \leq \ell^{-1}$,

$$-\Delta(z(0) - w) = 0 \qquad \text{on } |y| < \ell^{-1}.$$

The harmonic interior estimate on the concentric ball $|y| < (2\ell)^{-1}$ gives

$$\sup_{|y|<(2\ell)^{-1}} |z(0) - w|^2 \le C\ell^3 \|z(0) - w\|^2_{L^2(|y|<\ell^{-1})}.$$

On $|y| < r/\ell$, use $|z(0)|^2 \le 2|w|^2 + 2|z(0) - w|^2$ and $\|w\|_2 \le Z$. Integrating the supremum bound over this ball, whose volume is $Cr^3\ell^{-3}$, and using $r < 1/4$ gives dimensional constants C_1, C_2 such that

$$\|z(0)\|^2_{L^2(|\ell y|<r)} \le C_1 Z^2 + C_2 r^3 \|z(0)\|^2_2. \tag{3.23}$$

Let $E = \int_0^S \|(Fz)_t\|_2^2\, dt$. Since $z(S) = 0$,

$$F(t)z(t) = -\int_t^S (Fz)_\tau(\tau)\, d\tau.$$

Cauchy–Schwarz and $F(0) = I$ give

$$\int_0^S \|Fz\|_2^2\, dt \le \frac{S^2}{2} E, \qquad \|z(0)\|_2^2 \le SE,$$

The quadratic-form assumptions give

$$\langle M(0)z(0), z(0)\rangle \ge -B_e\|z(0)\|_2^2 - B_c\|z(0)\|^2_{L^2(|\ell y|<r)}.$$

Using $H \le K_+ I$ and (3.23), we obtain

$$a(z,z) \ge \left(1 - \frac{K_+ S^2}{2} - B_e S - C_2 B_c r^3 S\right) E + (L - C_1 B_c) Z^2 \ge \frac{1}{2} E.$$

The final inequality is (3.11). Moreover $z_t = F^{-1}(Fz)_t - F^{-1}F_t z$ and terminal integration imply $\|z_t\|^2_{L^2_t L^2} \le P^c E$. Thus the bounded bilinear form is coercive on $\mathcal{V}$, whether or not $M(0)$ is symmetric. The Lax–Milgram theorem gives a unique solution of (3.21).

We now show that the variational solution produces a velocity satisfying (3.19), including its initial support condition. Let $\mathbb{P}_\sigma$ be the L^2 orthogonal projection onto the divergence-free fields L^2_σ. Testing with compact support in time first gives

$$\left(\mathbb{P}_\sigma F^T F z_t\right)_t = \mathbb{P}_\sigma\left[F^T f + ((F^T F)_t - 2F^T F_t) z_t\right].$$

The terms containing z cancel because $F_{tt} = -HF$. On L^2_σ, the operator $T(t) = \mathbb{P}_\sigma F^T F$ is positive with inverse norm P^c. The identity $T(t)^{-1} - T(s)^{-1} = T(t)^{-1}(T(s) - T(t))T(s)^{-1}$ shows that its inverse is continuously differentiable in time. The displayed differential equation gives $Tz_t \in H^1((0,S); L^2_\sigma)$. Multiplication by the continuously differentiable inverse $T(t)^{-1}$ therefore gives $z_t \in H^1((0,S); L^2_\sigma)$, and

$$\mathbb{P}_\sigma(F^T F z_{tt}) = \mathbb{P}_\sigma(F^T f - 2F^T F_t z_t). \tag{3.24}$$

Integrating (3.21) by parts now gives

$$\langle -z_t(0) + L\mathcal{A}z(0), b(0)\rangle = 0 \qquad (b \in \mathcal{V}).$$

Here $F(0) = I$ and $F_t(0) = M(0)$ cancel the two terms containing $M(0)z(0)$. The trace $b(0)$ ranges over L^2_σ, and both $z_t(0)$ and $\mathcal{A}z(0)$ belong to this space. Hence

$$z_t(0) = L\mathcal{A}z(0). \tag{3.25}$$

Set

$$B = Fz_t, \qquad d\bar{q} = f - B_t - MB. \tag{3.26}$$

By (3.24), there is a distribution $\bar{q}$ with $\nabla_y \bar{q} = F^T(f - B_t - MB) \in L^2$, hence $d\bar{q} = f - B_t - MB$. Moreover $d \cdot B = \operatorname{div}_y z_t = 0$. Finally $B(0) = L\mathcal{A}z(0)$ has the required compact support. Only the initial mean velocity is localized; the mean field at positive times is allowed to be nonlocal.

3.4. **The transverse inverse and its derivative bounds.** Let $R_\perp$ be the constant 3×2 matrix whose orthonormal columns span $m_0^\perp$, and set $m = F^{-T}m_0$. For a force $f(t, y, \theta)$ with zero angle mean, we seek A, π satisfying

$$A_t + MA + m\partial_\theta \pi = f, \qquad m \cdot A = 0.$$

This inverse constructs corrections generated by the force f. Their displacement endpoints vanish when $t_0 > 0$, and their initial velocity vanishes when $t_0 = 0$. The primary coefficient v in (3.9) instead solves the homogeneous equation with prescribed terminal displacement or initial velocity. When $t_0 > 0$, we first construct A on $[0, t_0]$. At each fixed (y, θ), seek $\xi \in H_0^1((0, t_0); \mathbb{R}^2)$ and set $\eta = FR_\perp \xi$. Require the following identity for every $b \in H_0^1((0, t_0); \mathbb{R}^2)$:

$$\int_0^{t_0} (\eta_t \cdot \zeta_t - H\eta \cdot \zeta)\, dt = -\int_0^{t_0} f \cdot \zeta\, dt, \qquad \zeta = FR_\perp b, \quad b(0) = b(t_0) = 0. \tag{3.27}$$

The equation contains no derivatives in y or θ. Since $\eta(t_0) = 0$,

$$\int_0^{t_0} (|\eta_t|^2 - H\eta \cdot \eta)\, dt \geq \left(1 - \frac{K_+ t_0^2}{2}\right) \int_0^{t_0} |\eta_t|^2\, dt \geq \frac{1}{2} \int_0^{t_0} |\eta_t|^2\, dt.$$

Together with the coefficient bounds, this proves coercivity in $\|\xi_t\|_{L^2(0,t_0)}$. The use of an upper bound on the pressure Hessian to control this quadratic action has a precedent in Brenier's short-time action-minimization criterion for Euler flows [3, Section 2]. Integrating (3.27) by parts and using $F_{tt} = -HF$ gives, with $K_R = R_\perp^T F^T F R_\perp$,

$$K_R \xi_{tt} = R_\perp^T F^T (f - 2F_t R_\perp \xi_t), \qquad A = FR_\perp \xi_t. \tag{3.28}$$

For zero endpoint data, uniqueness gives $\xi(\cdot, y, \theta) = 0$ whenever $f(\cdot, y, \theta) = 0$. The coefficients are independent of θ, so averaging the variational equation gives $\langle \xi \rangle_\theta = 0$ whenever $\langle f \rangle_\theta = 0$. Thus A preserves spatial support and zero angle mean.

Equation (3.28) gives $(FR_\perp)^T(f - A_t - MA) = 0$. The columns of $FR_\perp$ span $m^\perp$, so this residual is parallel to m. Differentiating $m \cdot A = 0$ and using $m_t = -M^T m$ gives $m \cdot A_t = m \cdot MA$. The coefficient of m in the residual is therefore $(m \cdot f - 2m \cdot MA)/D_m$, and

$$A_t + MA + m\partial_\theta \pi = f, \qquad \pi = \partial_\theta^{-1}\left(\frac{m \cdot f - 2m \cdot MA}{D_m}\right). \tag{3.29}$$

Here ∂_θ^{-1} is defined on zero-mean functions by $\widehat{\partial_\theta^{-1} h}(j) = (ij)^{-1}\widehat{h}(j)$ for $j \neq 0$, with zero coefficient at $j = 0$. For $t \geq t_0$, write $A = FR_\perp a$ and solve the following first-order equation. Use $a(t_0) = \xi_t(t_0)$ when $t_0 > 0$, and $a(0) = 0$ when $t_0 = 0$:

$$K_R a_t + 2R_\perp^T F^T F_t R_\perp a = R_\perp^T F^T f. \tag{3.30}$$

The following estimate collects the inverses (3.19) and (3.29), with the shift bounds defined in (3.1). The constraint $m \cdot A = 0$ removes the leading phase divergence, but A need not satisfy $\widetilde{d} \cdot A = 0$. The lemma therefore also bounds the vector potential Q_A and curl correction C_A defined below; $A + k^{-1}C_A$ will be exactly divergence free. The integer shift index d is distinct from the differential operator in $d\bar{q}$ and $d\pi$.

For these estimates, retain the parent flow u, its pressure p and particle map X, and the data $P, S, \ell, k, t_0, m_0, R_\perp, F, M, H$ from §3.2. Assume all data are smooth and satisfy the coefficient, inverse, time, and scale bounds in §3.2 (i), the quadratic-form and coercivity conditions in §3.2 (ii), and the propagator and amplitude bounds with profile g in §3.2 (iii). Use the cutoff χ_o fixed in §3.1 and the shift convention (3.1).

Lemma 3.2. *For the mean inverse (3.21) and transverse inverse (3.29), with smooth data satisfying §3.2 (i)–(iii), there exists a choice $R = P^c$ with the following properties for every integer $d \geq 10$ and every $h_0 > 0$.*

- *(a)* **Mean inverse.** *An angle-independent smooth force f of shift $d - 10$ with constant profile h_0 gives, through (3.21), the fields $B, B_t, d\bar{q}$ of shift d with the same profile, where $B, d\bar{q}$ are defined by (3.26). These fields satisfy (3.19); the initial trace is $B(0) = L\mathcal{A}z(0)$.*
- *(b)* **Transverse inverse.** *Let a smooth force f have zero angle mean and spatial support in $|y| \leq 1/2$. Use zero displacement at both ends of the history interval when $t_0 > 0$, and zero initial velocity when $t_0 = 0$, followed by (3.30). Write A, π for the resulting velocity and pressure coefficients in (3.29). Suppose the force has shift $d - 10$, separately on these intervals, with profile*

$$h(t) = \begin{cases} h_0, & 0 \leq t \leq t_0, \\ h_0 g(t), & t_0 \leq t \leq S. \end{cases}$$

Define the vector-potential coefficient Q_A and its curl correction C_A by

$$Q_A = -\partial_\theta^{-1} \frac{m \times A}{D_m}, \qquad C_A = d \times Q_A.$$

Then $A, A_t, \pi, Q_A, C_A, C_{A,t}, d\pi$ have shift d with profile h. The solution is transverse, $m \cdot A = 0$, and these fields retain zero angle mean. Any fixed closed spatial set containing the support of the force at every time also contains the support of these fields.

The choice of R is independent of the shift and the number of expansion terms.

When $t_0 > 0$, prescribed terminal displacement $\xi(t_0) = \xi_T$ is handled by writing $\xi = \xi_0 + t\xi_T/t_0$, where $\xi_0(0) = \xi_0(t_0) = 0$. Then ξ_0 satisfies (3.28) with force $f - 2F_t R_\perp \xi_T/t_0$, and

$$A = FR_\perp \left(\xi_{0,t} + \frac{\xi_T}{t_0} \right).$$

The displacement and velocity bounds include the derivative bounds of ξ_T/t_0, and the preserved support set must also contain $\operatorname{supp} \xi_T$. If $\langle \xi_T \rangle_\theta = 0$, the pressure construction and zero-angle-mean conclusions also apply. Taking $f = 0$ and the terminal data prescribed in §3.2 recovers the history displacement and velocity of the primary coefficient v.

Proof. We first obtain the estimates at the fixed Sobolev order s, then control additional derivatives by (3.3). Time traces and the forward propagator give the stated time suprema. For the mean form, consider

$$a(z, b) = \int_0^S (\langle g_1, b_t \rangle + \langle g_0, b \rangle)\, dt + \langle g_b, b(0) \rangle, \qquad b \in \mathcal{V},$$

where $g_1, g_0 \in L^2((0,S); L^2(\mathbb{R}^3; \mathbb{R}^3))$ and $g_b \in L^2(\mathbb{R}^3; \mathbb{R}^3)$. The original forcing corresponds to $g_1 = 0$, $g_0 = -F^T f$, and $g_b = 0$. Since $b(S) = 0$,

$$\|b\|_{L_t^2 L^2} \leq S \|b_t\|_{L_t^2 L^2}, \qquad \|b(0)\|_2 \leq S^{1/2} \|b_t\|_{L_t^2 L^2}.$$

Coercivity therefore yields

$$\|z_t\|_{L_t^2 L^2} \leq P^c (\|g_1\|_{L_t^2 L^2} + \|g_0\|_{L_t^2 L^2} + \|g_b\|_2).$$

Spatial difference quotients preserve L^2_σ and the terminal condition. Subtracting the translated form puts every coefficient difference on the right side. At derivative level j, those terms contain coefficient derivatives in $W^{j,\infty}$ and previously controlled derivatives of z_t of order at most $j-1$. Derivatives of z and their initial values are controlled by $\partial_y^I z(t) = -\int_t^S \partial_y^I z_t(\tau)\,d\tau$. For the boundary commutators, apply (3.22) with derivatives of χ as the multipliers. Taking weak limits of difference quotients proves the next derivative bound. Induction through the fixed orders $j = 0, \dots, s$ gives

$$\|z_t\|_{L^2_t H^s} \le P^c\big(\|g_1\|_{L^2_t H^s} + \|g_0\|_{L^2_t H^s} + \|g_b\|_{H^s}\big).$$

For the general weak data above, set

$$\begin{aligned} Z_n &= \sum_{|I|=n} \|\partial_y^I z_t\|_{L^2_t H^s}, \\ F_n &= \sum_{|I|=n} \big(\|\partial_y^I g_1\|_{L^2_t H^s} + \|\partial_y^I g_0\|_{L^2_t H^s} + \|\partial_y^I g_b\|_{H^s}\big). \end{aligned}$$

Apply the fixed-s estimate to each differentiated equation. Every commutator places a nonempty subset of the additional derivatives on a coefficient, giving (3.3). For the original forcing $g_0 = -F^T f$, Leibniz's rule gives

$$\sum_{|I|=n} \|\partial_y^I (F^T f)\|_{L^2_t H^s} \le P^c \sum_{l=0}^{n} \binom{n}{l} R_c^l (l!)^2 \sum_{|J|=n-l} \|\partial_y^J f\|_{L^2_t H^s}.$$

After normalization by the claimed factorial bound, the terms with $l \ge 1$ contain the geometric factors $(R_c/R)^l$ used in the commutator estimate. They are included in the choice $R = P^c$. The product rule (3.2) also gives a bound with an additional shift. Finite higher regularity is first obtained by difference quotients, so the factorial induction does not assume the existence of the derivatives it estimates.

To estimate $T(t)^{-1}$, write $T(t)v = h$ with $v, h \in L^2_\sigma$. For a spatial multi-index β,

$$T(t)\partial_y^\beta v = \partial_y^\beta h - \mathbb{P}_\sigma \sum_{0<\gamma\le\beta} \binom{\beta}{\gamma} \partial_y^\gamma (F^T F)\, \partial_y^{\beta-\gamma} v.$$

The projection commutes with spatial derivatives. The L^2 coercivity of $T(t)$ starts the induction, and this identity gives its fixed-s and factorial estimates. Consequently (3.24) yields $z_{tt} \in L^2_t H^s$. For $w \in H^1((0,S); H^s)$, use the following trace inequality, with all unmarked norms taken in H^s:

$$\sup_t \|w(t)\| \le C\big(S^{-1/2}\|w\|_{L^2_t} + S^{1/2}\|w_t\|_{L^2_t}\big) \tag{3.31}$$

Apply this inequality to $w = \partial_y^I z_t$ for each derivative word I. The strong equation also gives

$$\|z_{tt}(t)\|_{H^s} \le P^c\big(\|f(t)\|_{H^s} + \|z_t(t)\|_{H^s}\big),$$

and its differentiated equations give the corresponding factorial bounds in the time supremum when f is bounded in time. For every spatial derivative word I, the bounds $\partial_y^I z_t \in H^1((0,S); H^s)$ give continuity in H^s and a well-defined initial trace.

For the transverse history problem, apply the coercivity and difference-quotient arguments to ξ on $[0, t_0]$. Its strong equation is (3.28), with K_R^{-1} replacing $T(t)^{-1}$, and there is no boundary operator to commute. Angle derivatives commute with the equation because its coefficients are independent of θ. Apply (3.31) with S replaced by t_0; the factors involving t_0^{-1} are bounded by P^c.

Write (3.30) as $a_t + \mathsf{C}a = \mathsf{D}f$, where

$$\mathsf{C} = 2K_R^{-1}R_\perp^T F^T F_t R_\perp, \qquad \mathsf{D} = K_R^{-1}R_\perp^T F^T.$$

The assumed propagator bound and $g(t_0) = 1$ give

$$\sup_{t_0 \le t \le S} \frac{\|a(t)\|_2}{g(t)} \le P^c \left(\|a(t_0)\|_2 + \int_{t_0}^{S} \frac{\|f(t)\|_2}{g(t)}\, dt \right).$$

For a spatial multi-index β,

$$(\partial_y^\beta a)_t + \mathsf{C}\partial_y^\beta a = \partial_y^\beta(\mathsf{D}f) - \sum_{0<\gamma\le\beta} \binom{\beta}{\gamma} (\partial_y^\gamma \mathsf{C})\partial_y^{\beta-\gamma} a.$$

Thus each differentiated equation has the original propagator, with lower derivatives of a added to its forcing. Apply the Duhamel estimate successively through order s. After division by g, each time integral is bounded by the corresponding weighted supremum because $S \le 1$. These finitely many steps cost P^c. Additional derivative words give (3.3) with profile g. Angle derivatives commute with the coefficients. All these fields remain supported in $|y| \le 1/2$, where the propagator estimate is assumed. In particular we never use a Gronwall factor exponential in the undifferentiated matrix norm.

The inverse coefficient bounds can also be recovered from the coefficient bounds and $\|J^{-1}\|_\infty \le P^c$. For a coefficient matrix J being inverted and $|\beta| \ge 1$, differentiating $JJ^{-1} = I$ gives

$$\partial^\beta J^{-1} = -J^{-1} \sum_{0<\gamma\le\beta} \binom{\beta}{\gamma} (\partial^\gamma J)\partial^{\beta-\gamma} J^{-1}.$$

Induction through the fixed $W^{s,\infty}$ orders gives a bound P^c. Further derivative words give the convolution in (3.3); choosing a sufficiently large fixed power $R = P^c$ proves the factorial bounds for J^{-1}.

To make the derivative accounting explicit, an input force of shift $d - 10$ gives successively the following available shifts:

quantity	z_t in L_t^2	z_{tt} in L_t^2	z_t in L_t^∞	z_{tt} in L_t^∞	$B, B_t, d\bar{q}$
shift	$d-9$	$d-8$	$d-7$	$d-6$	$\le d-4$.

(3.32)

The first column follows from the weak estimate and (3.3); the second follows from (3.24). The trace inequality gives the third column, and the strong equation in the time supremum gives the fourth. For the last column, use $B = Fz_t$, $B_t = F_t z_t + F z_{tt}$, and $d\bar{q} = f - B_t - MB$. For the transverse history interval, replace z_t, z_{tt} by ξ_t, ξ_{tt} and use (3.28). At t_0, its forward data $a(t_0) = \xi_t(t_0)$ have shift $d - 7$. The forward inverse gives $d - 6$, multiplication to obtain A costs a shift, and (3.29) gives A_t with at most one more. Hence π, Q_A have shift at most $d - 4$, $Q_{A,t}$ at most $d - 3$, $C_A, d\pi$ at most $d - 2$, and $C_{A,t}$ at most $d - 1$. Adding the bounds from the history and forward intervals costs a fixed factor, which the final shift absorbs. For the mean velocity, the initial support follows from $B(0) = L\mathcal{A}z(0)$. For the transverse fields, the equations act separately at each y, and their coefficients are independent of θ. Spatial support and zero angle mean are therefore preserved by the inverse and by the formulas for π, Q_A, C_A. Finally, $A = FR_\perp a$ gives $m \cdot A = m_0 \cdot R_\perp a = 0$, with $a = \xi_t$ on the history interval. $\square$

3.5. **Cancelling the coefficients of the oscillatory expansion.** We construct the finite approximation (3.34) and prove (3.44) and (3.45). The quadratic term can have nonzero angle mean even when the leading oscillation has zero mean. At each power of $\kappa = k^{-1}$, we therefore solve the mean equation before the oscillatory equation, using Lemma 3.2 for the required support and growth bounds. The next subsection removes the remaining momentum residual with a correction whose initial velocity is zero.

Let $N_k = \lfloor k^{\vartheta} \rfloor$. The index p below denotes a power of κ, not a derivative order. For $1 \le p \le N_k$, we seek zero-mean transverse fields A_p and angle-independent fields B_p satisfying $m \cdot A_p = 0$ and $d \cdot B_p = 0$. For each A_p, define a vector potential Q_p and its curl remainder C_p. We also define V_p, which collects the terms multiplying κ^p in the velocity expansion:

$$Q_p = -\partial_\theta^{-1} \frac{m \times A_p}{D_m}, \qquad C_p = d \times Q_p, \qquad V_p = A_p + B_p + C_{p-1}, \quad C_0 = 0. \tag{3.33}$$

The role of C_p is exact incompressibility. Indeed $m \times \partial_\theta Q_p = A_p$, so

$$\kappa^p A_p + \kappa^{p+1} C_p = \kappa^{p+1} \widetilde{d} \times Q_p.$$

The approximation is

$$W^a = \sum_{p=1}^{N_k} \big(\kappa^p (A_p + B_p) + \kappa^{p+1} C_p \big), \qquad q^a = \sum_{p=1}^{N_k} \big(\kappa^p \bar{q}_p + \kappa^{p+1} \pi_p \big), \tag{3.34}$$

and $\widetilde{d} \cdot W^a = 0$ identically. We set $B_1 = 0$, $\bar{q}_1 = 0$, and $A_1 = \alpha \chi_1 v f_\delta$. Define $\pi_1 = -2\partial_\theta^{-1}((m \cdot MA_1)/D_m)$ by (3.29) with zero forcing. When $t_0 > 0$, multiply the primary displacement coordinate ξ by $\alpha \chi_1 f_\delta$. This factor is independent of time, and (3.28) contains no label or angle derivatives. The multiplied displacement therefore solves the history equation and gives the velocity coefficient A_1.

Here is the precise coefficient extraction. Extend the notation by $V_{N_k+1} = C_{N_k}$ and $V_p = 0$ outside $1 \le p \le N_k + 1$. The coefficient of κ^p in the quadratic term with label derivatives is $\sum_{i+j=p} V_i \cdot dV_j$; in the term with a phase derivative it is $\sum_{i+j=p+1} (V_i \cdot m) \partial_\theta V_j$. Thus for $2 \le p \le N_k$ the forcing for $A_p + B_p$ is

$$f_p = -C_{p-1,t} - MC_{p-1} - d\pi_{p-1} - \sum_{i+j=p} V_i \cdot dV_j - \sum_{i+j=p+1} (V_i \cdot m) \partial_\theta V_j. \tag{3.35}$$

There is no phase-derivative term of order zero, since the positive indices i, j cannot satisfy $i + j = 1$. The term of order one vanishes because $V_1 = A_1$ and $m \cdot A_1 = 0$. In the label-derivative sum, both indices are at most $p - 1$, so all coefficients are already known. In the phase-derivative sum, order-p coefficients can occur only when $(i, j) = (1, p)$ or $(p, 1)$. The first term vanishes because $V_1 \cdot m = 0$. In the second, $V_p \cdot m = B_p \cdot m + C_{p-1} \cdot m$, so the only undetermined nonzero term is $-(B_p \cdot m) \partial_\theta A_1$. Separate it by writing

$$f_p = f_p^{\text{known}} - (B_p \cdot m) \partial_\theta A_1, \qquad \bar{f}_p = \langle f_p^{\text{known}} \rangle_\theta. \tag{3.36}$$

Here f_p^{known} depends only on coefficients already constructed, and $\langle h \rangle_\theta = (2\pi)^{-1} \int_0^{2\pi} h \, d\theta$. The undetermined term has zero angle mean. We first solve

$$B_{p,t} + MB_p + d\bar{q}_p = \bar{f}_p, \qquad d \cdot B_p = 0, \tag{3.37}$$

using (3.21), with displacement zero at S and initial trace (3.25). With this B_p fixed, we solve

$$\begin{gathered} A_{p,t} + MA_p + m\partial_\theta \pi_p = f_p^{\text{known}} - \bar{f}_p - (B_p \cdot m)\partial_\theta A_1, \\ m \cdot A_p = 0. \end{gathered} \tag{3.38}$$

using the transverse inverse. When $t_0 > 0$, its displacement coordinate vanishes at both 0 and t_0. When $t_0 = 0$, impose $A_p(0) = 0$. Forward continuation is given by (3.30). Thus each order is constructed from the preceding orders, with the mean solved before the oscillation.

The leading coefficient A_1 is supported in $|y| \le 1/2$. At every later order, terms involving only angle-independent coefficients disappear when the angle mean is subtracted. Every remaining term in the transverse forcing contains an oscillatory coefficient, or a derivative of such a coefficient, already supported in this set. This also holds for the term involving the newly solved B_p, since it contains $\partial_\theta A_1$. The transverse inverse and the formulas for Q_p and C_p preserve this support. Thus all oscillatory coefficients, including their curl remainders, are supported in $|y| \le 1/2$.

When $t_0 > 0$, the history inverse gives the velocity coefficient A_p on $[0, t_0]$, and the forward problem uses $A_p(t_0)$ as its initial value. The history and forward pieces of A_p therefore agree at t_0 and satisfy the same first-order velocity equation. Their first derivatives agree there; induction in time derivatives proves smooth matching. The mean strong equation gives the same time regularity as its forcing. Thus all coefficients are smooth. The oddness of u makes F, M, H even in y. Under $(y, \theta) \mapsto (-y, -\theta)$, both d and ∂_θ reverse parity. Uniqueness of the linear inverses then proves inductively that A_p, B_p, C_p are odd and $Q_p, \pi_p, \bar{q}_p$ are even, the latter pressure being defined up to a constant.

We prove the following coefficient bounds by induction on p, using a common growth constant $R = P^c$ independent of N_k. Set

$$\begin{gathered} a_p = 100p - 80 \quad (p \ge 1), \qquad b_p = 100p - 140 \quad (p \ge 2), \\ \gamma(t) = \begin{cases} \alpha, & t \le t_0, \\ \alpha g(t), & t \ge t_0, \end{cases} \qquad H_0 = \max(1, \alpha, \sup \alpha g). \end{gathered} \tag{3.39}$$

With a fixed $R = P^c$, the fields

$$A_p, A_{p,t}, \pi_p, Q_p, C_p, C_{p,t}, d\pi_p$$

have shift a_p and profile γH_0^{2p-2}; the fields $B_p, B_{p,t}, d\bar{q}_p$ have shift b_p and constant profile H_0^{2p-2}.

The shift costs of the two inverses are given in Lemma 3.2 and (3.32). For $t_0 > 0$, the primary endpoint data $\alpha\chi_1 f_\delta \xi_T$, after division by α, have derivative norms bounded by $P^c(P^c)^n(n!)^2$. The affine extension $t\alpha\chi_1 f_\delta \xi_T / t_0$ and these linear estimates require fewer than $a_1 = 20$ shifts. For $t_0 = 0$, the same conclusion follows from the corresponding derivative bounds for the initial data $\alpha\chi_1 v(0) f_\delta$ after division by α.

For the induction step, fix $2 \le p \le N_k$ and assume the bounds at lower orders. We first estimate the mean force and obtain the bound for B_p; that current-order bound is then available in the transverse force. To apply Lemma 3.2, the mean force must have shift $b_p - 10 = 100p - 150$, and the zero-mean force must have shift $a_p - 10 = 100p - 90$. The table gives shift bounds for the forcing terms and indicates whether their angle mean, their zero-mean part, or both may be nonzero. The bounds include the displayed derivatives and multiplication by the known coefficients.

forcing term at order p	shift bound	part of the forcing
$V_i \cdot dV_j, i+j=p$	$100p-158$	mean or oscillatory
$(B_i \cdot m)\partial_\theta A_j,\ i+j=p+1$	$100p-118$	oscillatory
$(B_i \cdot m)\partial_\theta C_{j-1},\ i+j=p+1$	$\leq 100p-118$	oscillatory
$(C_{i-1} \cdot m)\partial_\theta(A_j + C_{j-1}),\ i+j=p+1$	$\leq 100p-158$	mean or oscillatory
$C_{p-1,t},\ MC_{p-1},\ d\pi_{p-1}$	$\leq 100p-178$	oscillatory.

Every summand of V_i has shift at most a_i. The first line therefore starts with $a_i + a_j = 100p - 160$ and uses two further shifts for differentiation and multiplication by F^{-1} in d. The second starts with $b_i + a_j = 100p - 120$ and uses two further shifts for ∂_θ and multiplication by m. Terms with $A_i \cdot m$ vanish, and pure mean terms have smaller shifts. Each displayed bound lies below its required force shift, including the bound for the term involving the newly solved B_p.

The target profiles for the mean and oscillatory forces at order p are H_0^{2p-2} and $\gamma(t)H_0^{2p-2}$, respectively. For $i+j=p$, the angle mean of $V_i \cdot dV_j$ is bounded with the first profile, and its zero-mean part is bounded with the second. For $i+j=p+1$, the product $(B_i \cdot m)\partial_\theta A_j$ has profile $\gamma(t)H_0^{2p-2}$. Replacing B_i by C_{i-1} introduces another factor $\gamma(t)$ and reduces the exponent of H_0 by two. The resulting profile $\gamma(t)^2 H_0^{2p-4}$ is bounded by either target profile because $\gamma(t) \leq H_0$ and $H_0 \geq 1$. Terms containing further curl remainders have smaller exponents of H_0. The linear terms $C_{p-1,t}$, MC_{p-1}, and $d\pi_{p-1}$ have profile $\gamma(t)H_0^{2p-4}$ and therefore satisfy the oscillatory target. Angle averaging and subtraction of the average are bounded operations.

There are $O(p)$ products in each forcing. Let d_{target} denote $b_p - 10$ for the mean force and $a_p - 10$ for the oscillatory force. The table allows every product to be bounded with shift $d_{\text{target}} - 1$. Increasing this shift to d_{target} multiplies the available bound by $R(n + d_{\text{target}})^2$. Since $d_{\text{target}} \geq 25p$, choosing $R = P^c$ sufficiently large covers the number of products and the fixed powers of P from the product estimates. The choice of R needed for (3.3) is also independent of p. Applying the two inverse estimates now proves the claimed coefficient bounds with a growth constant independent of the expansion order.

The coefficients are now constructed and bounded at a fixed value of R. It remains to show that truncating the expansion leaves a residual small enough to correct by exact evolution. After orders $1, \ldots, N_k$ have canceled, the residual in (3.8) contains orders $N_k + 1, \ldots, 2N_k + 2$, including $C_{N_k,t}$ and $d\pi_{N_k}$. Their shifts are at most $110(p+1)$ at order p. The elementary inequality

$$(n+d)! \leq 2^{n+d} n! d! \tag{3.40}$$

separates derivative order from expansion order. Uniformly in time, it bounds the derivative norm $|\cdot|_n$ of every such coefficient by

$$(P^c)^n (n!)^2 \left[P^c (CN_k)^{300}\right]^{p+1}.$$

Indeed, a shift index $d \leq 110(p+1)$ and $p \leq 2N_k + 2$ give

$$(d!)^2 \leq (CN_k)^{220(p+1)}.$$

There are at most CN_k products at each order. Their sum is covered by

$$(CN_k)\,(CN_k)^{220(p+1)} \leq (CN_k)^{300(p+1)};$$

fixed powers of P and fixed constants raised to $p+1$ are absorbed in the factor $(P^c)^{p+1}$. The same factorial splitting gives this bound for $|V_p|_n$, $|V_{p,t}|_n$, and $|\nabla_{y,\theta} V_p|_n$ for $1 \le p \le N_k + 1$.

We estimate the first two velocity coefficients separately to obtain bounds independent of N_k. The coefficients of κ and κ^2 in W^a are A_1 and $A_2 + B_2 + C_1$. In (3.39), their shift indices are $a_1 = 20$, $a_2 = 120$, and $b_2 = 60$, with C_1 having shift a_1. These indices are independent of N_k. For each such fixed index d, (3.40) gives

$$R^{n+d}((n+d)!)^2 \le P^c(P^c)^n(n!)^2.$$

Their profiles are also bounded by fixed powers of P, so the same bound applies to these first two velocity coefficients. Since $300\vartheta < 0.01$, enlarging Q in (3.12) ensures

$$P^c(CN_k)^{300} \le k^{0.01}. \tag{3.41}$$

Define the lifted approximate velocity and its transport coefficient by

$$z^a = \kappa^{-1}F^{-1}W^a, \qquad b^a = (\kappa z^a, m_0 \cdot z^a). \tag{3.42}$$

Here z^a is the velocity increment in parent label coordinates, rescaled by κ^{-1}. The first three components of b^a transport y, and its last component transports θ on $\mathbb{R}^3 \times \mathbb{T}$. To sum the derivative bounds, choose a sufficiently small radius $\rho = P^{-c}$ and define

$$\|f\|_\rho = \sum_{n\ge 0} \frac{\rho^n |f|_n}{(n!)^2}. \tag{3.43}$$

The growth bounds above are summable when ρR is sufficiently small; thus ρ is the norm radius corresponding to the growth constant R. After decreasing ρ by a fixed factor, the sums over derivative orders n converge. Summing over expansion orders p using (3.41) then gives

$$\|z^a\|_\rho + \|z^a_t\|_\rho + \|\nabla_{y,\theta} z^a\|_\rho \le P^c, \qquad \|b^a\|_\rho \le k^{-1}P^c + O(k^{-1.9}). \tag{3.44}$$

Indeed $m_0 \cdot F^{-1}A_p = m \cdot A_p = 0$, so the transverse coefficients do not contribute to the last component of b^a. The mean terms and curl remainders first occur at order κ in z^a. Their coefficients at that order have bounds independent of N_k. Terms of order $p \ge 3$ in W^a contribute to z^a a sum bounded by $\sum_{p\ge 3} k^{1-p+0.01(p+1)}$. The same calculation gives $\|W^a - \kappa A_1\|_\rho \le k^{-2}P^c$. Define the normalized residual by

$$r^a = \kappa^{-1}F^{-1}\big(W^a_t + MW^a + W^a \cdot \widetilde{d}W^a + \widetilde{d}q^a\big).$$

The residual coefficient bounds and (3.41) give

$$\|r^a\|_\rho \le P^c \sum_{p=N_k+1}^{2N_k+2} k^{1-p+0.01(p+1)}.$$

Since $N_k = \lfloor k^\vartheta \rfloor$, enlarging the fixed threshold $k_0(c_0)$ if necessary gives

$$\|r^a\|_\rho \le \exp(-0.7k^\vartheta \log k) =: \mathcal{R}_k. \tag{3.45}$$

All radius shrinkings and all powers of P used so far are fixed.

3.6. **Correcting the approximate solution.** The normalized residual r^a of the approximation (3.34) satisfies (3.45). Differentiating along the physical graph introduces a factor of k, which complicates estimates for the correction. On $\mathbb{R}^3_y \times \mathbb{T}_\theta$, the transport coefficient b^a is small by (3.44). We solve there for a correction e to the lifted velocity z^a, with zero initial value, and restrict the resulting solution to the physical graph afterward. Throughout this subsection, $s = 6$, $\kappa = k^{-1}$, and

$$D_i = \kappa\partial_{y_i} + (m_0)_i\partial_\theta, \qquad G = F^{-1}F^{-T}, \qquad K = G^{-1} = F^T F. \tag{3.46}$$

The divergence associated with D is $\operatorname{div}_D v := \sum_{i=1}^3 D_i v_i = \kappa \operatorname{div}_y v + m_0 \cdot \partial_\theta v$. Thus a field is divergence-free with respect to D precisely when $\operatorname{div}_D v = 0$. The notation $D \times v$ similarly denotes the curl obtained by replacing each coordinate derivative with D_i. These operators satisfy

$$\widetilde{d} = \kappa^{-1}F^{-T}D, \qquad \widetilde{d}\cdot(\kappa F z) = \operatorname{div}_D z.$$

Thus writing $W = \kappa F z$ expresses incompressibility through the constant-coefficient derivatives D_i. For a vector field v and an integer $r \geq 0$, write

$$|v|_{n,r} = \sum_{I\in\{1,2,3,4\}^n} \|\partial^I v\|_{H^r}.$$

A word I specifies an ordered list of derivatives in (y,θ); for $n = 0$ this is just the H^r norm. Coefficient norms use $W^{r,\infty}$ in place of H^r. All constants in the Gevrey energy estimates below are independent of the cutoff in derivative order and of the auxiliary viscosity.

The preceding estimates give a radius P^{-c} at which $z^a = \kappa^{-1}F^{-1}W^a$, $\partial_t z^a$, and the first derivatives of z^a have weighted norm at most P^c. At that radius,

$$b^a = (\kappa z^a, m_0\cdot z^a), \qquad \sum_{n\geq 0}\frac{\rho^n}{(n!)^2}|b^a|_{n,s} \leq B_0 \leq k^{-1/2}.$$

We decrease the radius by a fixed factor when necessary. The normalized residual r^a has weighted H^s norm at most

$$\mathcal{R}_k = \exp(-0.7k^\vartheta \log k).$$

The coefficient bounds, including the fixed number of extra derivatives used below, have the form $P^c R_0^n (n!)^2$, where $R_0 = P^c$.

Take the smooth packet data and hypotheses in §3.2, and impose the smallness condition (3.12), with $Q = Q(c_0)$ fixed sufficiently large and $k \geq k_0(c_0)$ sufficiently large. Let (W^a, q^a) be the truncated expansion (3.34) constructed in the preceding subsection, and let z^a and r^a be its lifted velocity and normalized residual, satisfying (3.44) and (3.45). Set $\Delta_k = \exp(-k^{\vartheta/2})$.

Lemma 3.3. *For the truncated velocity and pressure (W^a, q^a) in (3.34), there exist a smooth velocity correction e and a smooth vector pressure correction p_e on $[0,S]\times\mathbb{R}^3_y\times\mathbb{T}_\theta$ satisfying*

$$e(0) = 0, \qquad \operatorname{div}_D e = 0, \qquad D\times p_e = 0,$$

and, at a fixed radius $\rho_* = P^{-c}$,

$$\sup_{0\leq t\leq S}\left(\|e(t)\|_{\rho_*} + \|e_t(t)\|_{\rho_*} + \|p_e(t)\|_{\rho_*}\right) \leq P^c\Delta_k.$$

There is a smooth scalar $q_e(t,y)$ with

$$\nabla_y q_e(t,y) = \kappa p_e(t,y,km_0\cdot y).$$

The velocity and pressure defined by

$$u^{\mathrm{new}}(t, X(t,\ell y)) = u(t, X(t,\ell y)) \\ + \ell\,[W^a + \kappa F e](t, y, km_0\cdot y). \tag{3.47}$$

and

$$p^{\mathrm{new}}(t, X(t,\ell y)) = p(t, X(t,\ell y)) \\ + \ell^2\big[q^a(t, y, km_0\cdot y) + q_e(t,y)\big]. \tag{3.48}$$

form an exact Euler solution on $[0,S]$. The velocity u^{new} is smooth and odd, and its initial increment is

$$u^{\mathrm{new}}(0,\ell y) - u(0,\ell y) = \ell W^a(0, y, km_0\cdot y).$$

Proof. We first construct the pressure inverse and estimate the correction on $\mathbb{R}^3\times\mathbb{T}$. We then remove the auxiliary viscosity and restrict to the physical graph to obtain the asserted Euler solution.

The pressure equation. Let $\mathcal{G}$ be the L^2 closure of the fields $D\phi$, with ϕ smooth, periodic in θ, and compactly supported in y. Its orthogonal complement consists exactly of fields satisfying $\operatorname{div}_D v = \sum_i D_i v_i = 0$ in distributions. The orthogonal projection $P_{\mathcal{G}}$ has Fourier symbol

$$\frac{(\kappa\xi + jm_0)\otimes(\kappa\xi + jm_0)}{|\kappa\xi + jm_0|^2}, \qquad (\xi, j)\in\mathbb{R}^3\times\mathbb{Z},$$

with an arbitrary value on its measure-zero singular set. Thus it commutes with derivatives and has norm at most one on every H^r. For a given vector field f, we seek $\mathfrak{p}\in\mathcal{G}$ such that $f - G\mathfrak{p}\in\mathcal{G}^\perp$, equivalently

$$\operatorname{div}_D(G\mathfrak{p} - f) = 0.$$

Since G is symmetric and $G \ge P^{-c} I$, the operator $P_{\mathcal{G}}G|_{\mathcal{G}}$ is invertible on $\mathcal{G}$ by coercivity. We denote its pressure solution operator by

$$\mathcal{T} f = (P_{\mathcal{G}}G|_{\mathcal{G}})^{-1}P_{\mathcal{G}} f.$$

Here $\mathfrak{p} = \mathcal{T} f\in\mathcal{G}$ denotes the vector output of the pressure inverse; the physical pressure is scalar. Translation preserves $\mathcal{G}$. Applying spatial difference quotients to $P_{\mathcal{G}}(G\mathfrak{p}) = P_{\mathcal{G}} f$ therefore proves the inverse bounds through the fixed Sobolev order s. Differentiating a word of n additional derivatives then gives, for $r = s-1, s$,

$$|\mathfrak{p}|_{n,r} \le P^c\left(|f|_{n,r} + \sum_{l=1}^{n}\binom{n}{l}R_0^l(l!)^2|\mathfrak{p}|_{n-l,r}\right). \tag{3.49}$$

Indeed, every commutator places at least one additional derivative on G; the remaining derivatives fall on $\mathfrak{p}$. The same argument at any fixed larger index gives a finite inverse bound, without a claim that its constant is uniform in that index.

Substitute $W = \kappa F z$ into (3.8) and multiply by $\kappa^{-1}F^{-1}$. Using $\widetilde{d} = \kappa^{-1}F^{-T}D$ gives

$$z_t + 2F^{-1}F_t z + z\cdot Dz + \kappa F^{-1}(z\cdot\nabla_y F)z + G\mathfrak{p} = 0, \qquad \operatorname{div}_D z = 0. \tag{3.50}$$

Here $\mathfrak{p} = \kappa^{-2}Dq$ is a vector pressure variable. In particular, $W_t + MW = \kappa F z_t + 2\kappa F_t z$, and $W\cdot\widetilde{d}W = \kappa F(z\cdot Dz) + \kappa^2(z\cdot\nabla_y F)z$, which verify the factors in this equation. The approximate vector pressure $\mathfrak{p}^a = \kappa^{-2}Dq^a$ belongs to $\mathcal{G}$. Its oscillatory part is a D-derivative of a Sobolev function. Its angle-independent part is an L^2 gradient in y, hence belongs to the $j=0$ component of $\mathcal{G}$.

Write $z = z^a + e$ and $\mathfrak{p} = \mathfrak{p}^a + p_e$, with $e(0) = 0$. Subtracting the approximate equation, whose left side is r^a, from (3.50) gives the correction equation. For $0 < \nu \le 1$, we regularize it by adding $\nu\Delta_{y,\theta} e$:

$$e_t + b \cdot \nabla_{y,\theta} e + Z(e) + G p_e = -r^a + \nu\Delta_{y,\theta} e, \qquad \operatorname{div}_D e = 0, \tag{3.51}$$

where

$$\begin{aligned} b &= (\kappa(z^a + e), m_0 \cdot (z^a + e)), \\ Z(e) &= e \cdot Dz^a + 2F^{-1}F_t e \\ &\quad + \kappa F^{-1}\left[(z^a + e) \cdot \nabla_y F\right](z^a + e) - \kappa F^{-1}(z^a \cdot \nabla_y F) z^a. \end{aligned}$$

Thus Z contains no derivative of e, and $\operatorname{div} b = 0$. Applying $P_{\mathcal{G}}$ to (3.51) determines p_e. We isolate the derivative of e carried by the transport term by writing $p_e = p_0 + p_1$, where

$$p_0 = -\mathcal{T}(Z(e) + r^a), \qquad p_1 = -\mathcal{T}(b \cdot \nabla_{y,\theta} e).$$

The viscosity contributes no pressure because $\operatorname{div}_D(\Delta e) = \Delta(\operatorname{div}_D e) = 0$; the operators D_i have constant coefficients and commute with Δ.

An estimate with a finite derivative cutoff. We first derive an a priori estimate for smooth solutions of (3.51) on their interval of existence, uniformly in the derivative cutoff and the auxiliary viscosity. A slowly decreasing norm radius will absorb the transport and pressure commutators while remaining positive on $[0, S]$. For each word I of additional derivatives, define $E_I \ge 0$ by the following formula, where $\lambda \in \mathbb{N}_0^4$ indexes the base derivatives and the integral is over $\mathbb{R}^3 \times \mathbb{T}$:

$$E_I^2 = \sum_{|\lambda| \le s} \int \langle K \partial^\lambda \partial^I e, \partial^\lambda \partial^I e \rangle, \qquad E_n = \sum_{|I| = n} E_I.$$

The bounds on K give $P^{-c}|e|_{n,s} \le E_n \le P^c |e|_{n,s}$. For $m \ge 0$, a decreasing radius $\rho(t) > 0$, and $w_n = \rho^n/(n!)^2$, set

$$X_m = \sum_{n=0}^{m} w_n E_n, \qquad Y_m = \sum_{n=0}^{m} n w_n E_n.$$

Choose $\rho \le (P^c R_0)^{-1}$. After multiplication by w_n and summation in (3.49), its commutator terms are bounded by $P^c \sum_{l \ge 1} (\rho R_0)^l$ times the corresponding weighted pressure norm. Absorbing this quantity into the left side and applying the multiplication estimates gives

$$\sum_{n=0}^{m} w_n |p_0|_{n,s} \le P^c (X_m + X_m^2 + \mathcal{R}_k), \tag{3.52}$$

$$\sum_{n=0}^{m} w_n |p_1|_{n,s-1} \le P^c (B_0 + X_m) X_m. \tag{3.53}$$

In the second line the derivative on e is contained in its base H^s norm. At base index s we instead use the shifted estimate

$$\sum_{j=0}^{m-1} (j+1) w_{j+1} |p_1|_{j,s} \le P^c (B_0 + X_m) Y_m. \tag{3.54}$$

To check its derivative count explicitly, the term with l derivatives on b and $j - l$ on ∇e has weight ratio

$$\frac{(j+1) w_{j+1} \binom{j}{l}}{w_l (j - l + 1) w_{j-l+1}} = \binom{j+1}{l}^{-1} \le 1.$$

For the commutator terms in (3.49), the same ratio compares the weight at pressure order j with the weight at order $j-l$. Their sum is bounded by $P^c \sum_{l\geq 1}(\rho R_0)^l$ times the left side of (3.54), and is absorbed there. Consequently this estimate needs no velocity or pressure derivative beyond the cutoff m.

For $n = |I|$ and $|\lambda| \leq s$, apply $\partial^\lambda \partial^I$ to (3.51) and pair in $L^2(\mathbb{R}^3 \times \mathbb{T})$ with $K\partial^\lambda\partial^I e$. The highest pressure term vanishes:

$$\langle K\partial^\lambda\partial^I e, G\partial^\lambda\partial^I p_e\rangle = \langle \partial^\lambda\partial^I e, \partial^\lambda\partial^I p_e\rangle = 0.$$

Here $KG = I$, the differentiated velocity satisfies $\operatorname{div}_D(\partial^\lambda\partial^I e) = 0$, and the differentiated pressure belongs to $\mathcal{G}$. If all additional derivatives fall on p_e, a commutator with a base derivative on G uses only $|p_e|_{n,s-1}$. Terms with $l \geq 1$ additional derivatives on G use $|p_e|_{j,s}$, $j = n - l$. For their p_1 part, the relevant ratio is

$$\frac{w_{j+l}\binom{j+l}{l}}{w_l(j+1)w_{j+1}} = \rho^{-1}\frac{j+1}{\binom{j+l}{l}} \leq \rho^{-1}.$$

The weighted sum of the derivatives of G with $l \geq 1$ is bounded by $P^c \sum_{l\geq 1}(\rho R_0)^l \leq P^c \rho R_0$. Combining this bound with (3.54) gives the contribution $P^c R_0(B_0 + X_m)Y_m$. The p_0 terms and the terms with all additional derivatives on p_e are controlled by (3.52) and (3.53).

For transport, put $h = \partial^\lambda\partial^I e$. Since $\operatorname{div} b = 0$, integration by parts gives

$$\langle Kh, b\cdot\nabla h\rangle = -\tfrac{1}{2}\int \langle (b\cdot\nabla K)h, h\rangle.$$

For terms with all additional derivatives on e, the base commutators satisfy

$$\|[\partial^\lambda, b\cdot\nabla]\partial^I e\|_2 \leq C\|\nabla b\|_{H^{s-1}}\|\nabla\partial^I e\|_{H^{s-1}},$$

by the product estimate in H^{s-1}. A term with $l \geq 1$ additional derivatives on b uses $|b|_{l,s}$ and $|e|_{j+1,s}$, where $j = n - l$. The preceding weight ratio therefore bounds their sum by $P^c\rho^{-1}(B_0 + X_m)Y_m$. After summation in the weighted energies, differentiating K in time contributes at most $P^c X_m$. For the heat term, integration by parts with the same h gives

$$\nu\langle Kh, \Delta h\rangle = -\nu\sum_a \langle K\partial_a h, \partial_a h\rangle - \nu\sum_a\langle(\partial_a K)h, \partial_a h\rangle \leq -\tfrac{1}{2}\nu P^{-c}\|\nabla h\|_2^2 + \nu P^c\|h\|_2^2.$$

Since $Z(e)$ contains no derivative of e, the multiplication estimates give

$$\sum_{n=0}^m w_n|Z(e)|_{n,s} \leq P^c(X_m + X_m^2).$$

Moreover, differentiating the weights contributes $\sum_{n=0}^m w_n' E_n = (\rho'/\rho)Y_m$. Summing the energy estimates with these contributions yields

$$X_m' \leq P^c(X_m + X_m^2 + \mathcal{R}_k) + \left[\frac{\rho'}{\rho} + P^c(\rho^{-1} + R_0)(B_0 + X_m)\right]Y_m. \tag{3.55}$$

At zeros of an individual energy we first use $(E_I^2 + \epsilon^2)^{1/2}$, then send ϵ to zero at fixed m. All constants in this inequality are independent of m and ν.

Choose

$$\rho(0) = (P^c R_0)^{-1}, \qquad \rho' = -CP^c(B_0 + \Delta_k).$$

The fixed exponent in this choice is enlarged to cover all preceding constants. The scale condition $P^Q \leq k^{\vartheta/100}$, for a sufficiently large fixed Q, ensures $\rho(t) \geq \rho(0)/2$ on $[0, S]$.

Under the bootstrap assumption $X_m \le \Delta_k$, the bracket in (3.55) is nonpositive. Since $X_m(0) = 0$, Gronwall's inequality then gives

$$X_m(t) \le P^c S \exp(P^c S)\mathcal{R}_k < \tfrac{1}{2}\Delta_k.$$

Indeed,

$$\log(P^c S \exp(P^c S)\mathcal{R}_k) = \log(P^c S) + P^c S - 0.7k^{\vartheta} \log k.$$

The scale condition makes the first two terms negligible relative to $k^{\vartheta} \log k$. For sufficiently large k, this expression is less than $-k^{\vartheta/2} - \log 2 = \log(\Delta_k/2)$. Continuity closes the bootstrap uniformly in m and ν.

Existence and removal of viscosity. We construct the correction (e, p_e) asserted in Lemma 3.3 by solving (3.51) for $\nu > 0$ and then passing to $\nu \downarrow 0$. To eliminate the pressure and construct a parabolic equation on the divergence-free subspace, set $\mathcal{P}(t) = I - G\mathcal{T}$. The identity $P_{\mathcal{G}} G\mathcal{T} = P_{\mathcal{G}}$ shows that $\mathcal{P}(t)$ takes values in $\{v : \operatorname{div}_D v = 0\}$; it fixes every field in this space. Moreover, $\mathcal{P}(t)G\mathfrak{p} = 0$ for $\mathfrak{p} \in \mathcal{G}$. Apply it to the non-heat terms in (3.51). For each fixed $q = s + m$, the projected nonlinearity is locally Lipschitz $H^q \to H^{q-1}$. The heat estimate

$$\|e^{\nu t\Delta} f\|_{H^q} \le C\big(1 + (\nu t)^{-1/2}\big)\|f\|_{H^{q-1}}$$

gives a local mild solution by contraction and continuation while its H^q norm remains bounded. Write e^{ν} for the viscous solution, and set

$$b^{\nu} = (\kappa(z^a + e^{\nu}), m_0 \cdot (z^a + e^{\nu})), \qquad f^{\nu} = -\mathcal{P}(t)\big(b^{\nu} \cdot \nabla e^{\nu} + Z(e^{\nu}) + r^a\big).$$

Then

$$\partial_t e^{\nu} - \nu\Delta e^{\nu} = f^{\nu}, \qquad e^{\nu}(0) = 0.$$

No time derivative of $\mathcal{P}(t)$ enters this equation. Its range is fixed, and $[D_i, \Delta] = 0$, so the heat flow preserves $\operatorname{div}_D e^{\nu} = 0$.

On a compact existence interval $I = [t_1, t_2]$, heat maximal regularity gives

$$\|e^{\nu}\|_{L^2(I;H^{q+1})} + \|\partial_t e^{\nu}\|_{L^2(I;H^{q-1})} \le C_{\nu,q,I}\left(\|e^{\nu}(t_1)\|_{H^q} + \|f^{\nu}\|_{L^2(I;H^{q-1})}\right).$$

The constant may depend on ν; only finiteness is used here. For the inhomogeneous term, the Fourier kernel satisfies

$$\int_0^{\infty} \nu|\xi, j|^2 e^{-\nu|\xi,j|^2 t}\, dt \le 1.$$

Young's inequality in time gives the two-derivative gain in L^2; low frequencies are bounded on I. Since $e^{\nu} \in L^{\infty}(I; H^q) \cap L^2(I; H^{q+1})$, product estimates place the unprojected source in $L^2(I; H^q)$. The pressure inverse gives the same regularity for p_e^{ν}:

$$b^{\nu} \cdot \nabla e^{\nu} + Z(e^{\nu}) + r^a, \quad p_e^{\nu} \in L^2(I; H^q).$$

For derivative words J with $|J| \le m$ and $|\lambda| \le s$, the preceding bounds give

$$\partial^{\lambda}\partial^J e^{\nu} \in L^2(I; H^1), \qquad \partial^{\lambda}\partial^J \partial_t e^{\nu} \in L^2(I; H^{-1}).$$

Multiplication by K preserves H^1, so the differentiated equation can be tested against $K\partial^{\lambda}\partial^J e^{\nu}$. This justifies the energy estimate through the H^{-1}–H^1 pairing, or equivalently by smoothing. The uniform energy bound extends e^{ν} to S, and uniqueness at $q = s$ identifies the solutions constructed with different derivative cutoffs.

For every fixed integer $j \ge 0$, the energy bounds and pressure inverse give

$$\sup_{0<\nu\le 1}\left(\|e^{\nu}\|_{L_t^{\infty} H^{j+2}} + \|\partial_t e^{\nu}\|_{L_t^{\infty} H^j} + \|p_e^{\nu}\|_{L_t^{\infty} H^j}\right) \le C_j.$$

Here the two extra derivatives bound the heat term. Local Sobolev compactness and time equicontinuity give a sequence $\nu_l \downarrow 0$ such that, for every finite j,

$$\begin{aligned} e^{\nu_l} &\longrightarrow e \quad \text{in } C([0,S]; H^j_{\mathrm{loc}}), \\ p_e^{\nu_l} &\rightharpoonup^* p_e \quad \text{in } L^\infty([0,S]; H^j), \\ \nu_l \Delta e^{\nu_l} &\longrightarrow 0 \quad \text{in } L^\infty([0,S]; H^j). \end{aligned}$$

The pressure limit remains in the closed gradient space $\mathcal{G}$. Passing to the limit in (3.51) gives

$$e_t + b \cdot \nabla e + Z(e) + Gp_e = -r^a, \qquad \operatorname{div}_D e = 0, \qquad e(0) = 0,$$

where $b = (\kappa(z^a + e), m_0 \cdot (z^a + e))$. For each fixed t and finite derivative cutoff m, local convergence and the uniform global H^{s+m} bound imply weak convergence in H^{s+m}. Lower semicontinuity gives

$$X_m(e(t)) \le \liminf_{l\to\infty} X_m(e^{\nu_l}(t)) \le \Delta_k.$$

Monotone convergence as $m \to \infty$ proves the full weighted bound. The limit equation and pressure formula give

$$e \in W_t^{1,\infty} H^j \quad (j \ge 0), \qquad p_e = -\mathcal{T}\big(Z(e) + r^a + b \cdot \nabla e\big).$$

Because $e \in W_t^{1,\infty} H^{j+1}$, the source in the pressure formula is continuous in H^j. Continuity of $\mathcal{T}(t)$ at this fixed Sobolev index gives $p_e \in C_t H^j$. Repeated differentiation of the inviscid correction equation and the pressure formula gives smoothness in time.

It remains to bound p_e and e_t in the weighted norm. Their formulas contain $\nabla_{y,\theta} e$, so we shrink the radius to control this additional derivative. Let $\rho_{\min} = \rho(0)/2 \le \inf_{[0,S]} \rho(t)$ and $\rho_* = \rho_{\min}/2$. The cost of an additional derivative at this smaller radius is bounded by

$$\sup_{n\ge 0} \frac{\rho_*^n ((n+1)!)^2}{\rho_{\min}^{n+1} (n!)^2} = \rho_{\min}^{-1} \sup_{n \ge 0} (n+1)^2 2^{-n} \le C \rho_{\min}^{-1} = P^c.$$

The pressure inverse and the limit equation therefore yield

$$\sup_{0 \le t \le S} \big(\|e(t)\|_{\rho_*} + \|p_e(t)\|_{\rho_*} + \|e_t(t)\|_{\rho_*} \big) \le P^c \Delta_k.$$

The viscous equation is invariant under joint reflection $(y, \theta) \mapsto (-y, -\theta)$, with the prescribed odd parity of z^a, r^a and even parity of F, G. Uniqueness makes e^ν odd, and this parity passes to the limit.

Restriction to the physical graph. Put $v^{\mathrm{diag}}(y) = v(y, km_0 \cdot y)$. The chain rule gives $\nabla_y v^{\mathrm{diag}} = \kappa^{-1} (Dv)^{\mathrm{diag}}$ componentwise. In particular, $D \times p_e = 0$ implies $\nabla_y \times p_e^{\mathrm{diag}} = 0$. Since $\mathbb{R}^3$ is simply connected, there is a smooth potential q_e satisfying

$$\nabla_y q_e = \kappa p_e^{\mathrm{diag}}, \qquad dq_e = \kappa F^{-T} p_e^{\mathrm{diag}}.$$

Indeed,

$$\kappa^{-1} F^{-1} dq_e = G p_e^{\mathrm{diag}},$$

which is the pressure term in the graph restriction of the correction equation. Adding the restricted approximate and correction equations therefore gives (3.8) on the physical graph for $W = W^a + \kappa F e$, with scalar pressure $(q^a)^{\mathrm{diag}} + q_e$. Moreover,

$$\widetilde{d} \cdot W = \widetilde{d} \cdot W^a + \operatorname{div}_D e = 0.$$

Thus the velocity and pressure in (3.47) and (3.48) solve Euler's equation in physical coordinates. For every scalar or vector component v,

$$\int_{\mathbb{R}^3} |v(y, km_0 \cdot y)|^2 \, dy \le C \int_{\mathbb{R}^3} \|v(y,\cdot)\|^2_{H^1(\mathbb{T})} \, dy.$$

Applying this estimate and the chain rule to $W^a + \kappa Fe$ and its derivatives gives finite spatial Sobolev norms for the graph-restricted velocity increment. The same argument applies to the gradient of the pressure increment, using the estimated gradients of the mean pressure potentials. The volume-preserving change of coordinates and the bounded derivatives of F, F^{-1} transfer these increment bounds to physical space. Higher spatial bounds and the equation give time continuity and smoothness at every finite order. The correction vanishes initially, so the prescribed initial perturbation is retained. This proves Lemma 3.3 and supplies the exact solution asserted in Proposition 3.1. The remaining subsections prove its derivative, particle-map, and initial-data conclusions. □

3.7. **The leading gradient and pressure Hessian.**

Proof of Claim 1 of Proposition 3.1. To prove (3.13) and (3.14), we must control the errors after differentiating velocity once and pressure twice. Each derivative on the physical graph can cost a factor of k. For the exact normalized velocity $W = W^a + \kappa Fe$, differentiating the leading term gives

$$\widetilde{d}_j(\kappa A_{1,i}) = \alpha\chi_1 v_i m_j f'_\delta + \kappa d_j(\alpha\chi_1 v_i) f_\delta.$$

The second term is at most $k^{-1}P^c$. The coefficient estimates following (3.44) give $\|W^a - \kappa A_1\|_\rho \le k^{-2}P^c$ at a radius P^{-c}. A graph derivative, at a smaller radius, therefore contributes at most $k^{-1}P^c$. The correction κFe contributes at most $P^c\Delta_k$, where $\Delta_k = \exp(-k^{\vartheta/2})$ is the bound obtained in Lemma 3.3. These bounds prove (3.13) by (3.12).

For the pressure, (3.29) gives

$$\pi_1 = -2\alpha\chi_1 \frac{m \cdot Mv}{D_m} \partial_\theta^{-1} f_\delta.$$

The term in $\widetilde{d}_i\widetilde{d}_j(\kappa^2\pi_1)$ with both derivatives on the angle is therefore

$$\kappa^2 k^2 m_i m_j \partial_\theta^2 \pi_1 = -2\alpha\chi_1 (m \cdot Mv) \frac{m_i m_j}{D_m} f'_\delta.$$

Each remaining term has at most one phase derivative $km_i\partial_\theta$, hence is bounded by $k^{-1}P^c$. The higher oscillatory pressures begin with $\kappa^3\pi_2$ and their second graph derivatives sum to the same bound. The mean pressures begin with $\kappa^2\bar{q}_2$; using the estimated gradient $d\bar{q}_p$ and a further derivative d_i bounds their Hessians directly, without estimating the potentials. The normalized error-pressure gradient is $dq_e = \kappa F^{-T} p_e$ on the graph. A further graph derivative costs at most kP^c, giving $P^c\Delta_k$. This proves (3.14). Restoring the physical factors ℓ for velocity and ℓ^2 for pressure cancels the factors from one and two physical derivatives, respectively. □

3.8. **Factorial bounds for the new particle map.**

Proof of Claim 2 of Proposition 3.1. We next prove the particle-map estimate (3.16). Here we assume the parent particle-map bound (3.15). Estimating the new velocity's physical Lipschitz norm and then exponentiating would lose the required frequency bound. We instead use the small transport field b on $\mathbb{R}^3 \times \mathbb{T}$ defined below. For $z = z^a + e$, put

$$b = (\kappa z, m_0 \cdot z).$$

Its derivative word sums in L^∞ and L^2 have bounds $BR_b^n(n!)^2$, where $B \le 2k^{-1/2}$ and $R_b = P^c$. The corresponding bounds for b_t have size and growth constant bounded by fixed powers of P. By increasing Q, we ensure $BR_b \ll 1$. Let $\Phi(t,y,\theta)$ be the flow of b on $\mathbb{R}^3 \times \mathbb{T}$, so $\partial_t\Phi = b(t,\Phi)$ and $\Phi(0,y,\theta) = (y,\theta)$. This flow preserves four-dimensional volume because $\operatorname{div}_{y,\theta} b = 0$. Use the lift $\widehat{\Phi}$ to $\mathbb{R}^3 \times \mathbb{R}$ with $\widehat{\Phi}(0) = \mathrm{id}$. Its displacement is periodic in the angular variable and defines $\Phi - \mathrm{id}$ on $\mathbb{R}^3 \times \mathbb{T}$.

We give the composition estimate used for Φ. In the ordered derivative formula for a composition, group a word of length n into q nonempty blocks with sizes $j_1, \dots, j_q$ summing to n. When we sum over the ordered sizes, the combinatorial coefficient is $n!/(q!\prod_i j_i!)$. On inserting the factorial bounds for the outer qth derivative and the inner j_ith derivatives, and dividing by $(n!)^2$, the remaining factor is

$$\frac{q!\prod_{i=1}^q j_i!}{n!} \le 1. \tag{3.56}$$

It is one when every $j_i = 1$. Increasing an index j_i multiplies the ratio by $(j_i+1)/(n+1) \le 1$, proving the inequality for all block sizes. Sums over component directions are controlled by the word sums themselves.

Fix a derivative cutoff $J \ge 1$ and define, for $\zeta \ge 0$,

$$U(t,\zeta) = d\zeta + \sum_{n=1}^{J} \frac{\|\partial^n(\Phi(t) - \mathrm{id})\|_\infty}{(n!)^2}\zeta^n.$$

Here $\|\partial^n h\|_\infty$ denotes the sum of the L^∞ norms over derivative words of length n. The notation $\|\partial^n h\|_2$ uses the corresponding sum of L^2 norms. Choose the dimensional constant d to bound the first-derivative word sum of the identity. Thus U bounds the derivative series of Φ through order J. The integral equation for Φ and (3.56), compared through degree J, give

$$U(t,\zeta) \le d\zeta + \int_0^t B\frac{R_bU(t',\zeta)}{1 - R_bU(t',\zeta)}\,dt' \qquad \text{if } R_bU(t',\zeta) < 1 \text{ for } 0 \le t' \le t. \tag{3.57}$$

At $\zeta = (4dR_b)^{-1}$, the initial right side is $1/(4R_b)$. If $R_bU \le 1/2$, the integral increases R_bU by at most BR_bS. The smallness $BR_b \ll 1$ closes the bootstrap, giving $R_bU \le 1/2$. Returning to the integral equation gives $\|\partial^n(\Phi - \mathrm{id})\|_\infty \le CBS\zeta^{-n}(n!)^2$ for $1 \le n \le J$; the order-zero bound follows by direct integration of b. The estimates are uniform in the finite truncation, so passing to all orders is justified.

For the L^2 estimate, place $(\partial^q b) \circ \Phi$ in L^2 and the derivatives of Φ in L^∞. Volume preservation gives $\|(\partial^q b) \circ \Phi\|_2 = \|\partial^q b\|_2$. Consequently, at the value of ζ chosen above,

$$\sum_{n=1}^{J} \frac{\|\partial^n(\Phi(t) - \mathrm{id})\|_2}{(n!)^2}\zeta^n \le \int_0^t B\frac{R_bU(t',\zeta)}{1 - R_bU(t',\zeta)}\,dt' \le BS.$$

This gives $\|\partial^n(\Phi - \mathrm{id})\|_2 \le BS\zeta^{-n}(n!)^2$ for $1 \le n \le J$, uniformly in J. At order zero, the integral equation and volume preservation give $\|\Phi - \mathrm{id}\|_2 \le BS$ directly. Finally,

$$\Phi_t = b \circ \Phi, \qquad \Phi_{tt} = (b_t + (b \cdot \nabla_{y,\theta})b) \circ \Phi.$$

The preceding composition and product estimates at a smaller radius P^{-c} give size and growth constant bounded by fixed powers of P for these two fields in both L^2 and L^∞.

Write $\widehat{\Phi}(t, y_0, \theta_0) = (y(t), \theta(t))$, using the angular lift defined above. Along this trajectory,

$$\frac{d}{dt}\big(\theta(t) - km_0 \cdot y(t)\big) = m_0 \cdot z - k\kappa m_0 \cdot z = 0.$$

Thus $\theta_0 = km_0 \cdot y_0$ implies $\theta(t) = km_0 \cdot y(t)$, so the flow preserves the physical phase graph. Let $\pi_y : \mathbb{R}^3 \times \mathbb{T} \to \mathbb{R}^3$ be projection onto the first three coordinates, and define the rescaled map

$$Y(t, a) = \ell\, \pi_y \Phi\left(t, \frac{a}{\ell}, \frac{km_0 \cdot a}{\ell}\right). \tag{3.58}$$

Here the last argument is interpreted in $\mathbb{T}$, and $Y(0, a) = a$. In the normalized label coordinate y, the velocity underlying this flow is $\kappa z(t, y, km_0 \cdot y)$. Its divergence is

$$\operatorname{div}_y \big(\kappa z(t, y, km_0 \cdot y)\big) = \kappa \operatorname{div}_y z + m_0 \cdot \partial_\theta z = \operatorname{div}_D z = 0.$$

Thus Y preserves three-dimensional volume. Each derivative of a graph restriction costs at most Ck; its L^2 bound uses the one-dimensional trace inequality in θ and at most one further derivative on the product space, without an additional graph factor. Rescaling then gives $Y - \mathrm{id}, Y_t, Y_{tt}$ polynomial size and growth constant $k\ell^{-1}P^c$ in both L^2 and L^∞.

The new physical particle map is

$$X^{\mathrm{new}}(t, a) = X(t, Y(t, a)). \tag{3.59}$$

Set $y = Y(t, a)/\ell$ and $\theta = km_0 \cdot y$. The definition of Y and graph invariance give $Y_t(t, a) = \ell\kappa z(t, y, \theta)$, while $(\nabla_a X)(t, Y(t, a)) = F(t, y)$. Hence

$$\frac{d}{dt} X(t, Y(t, a)) = u(t, X(t, Y(t, a))) + \ell\kappa F(t, y) z(t, y, \theta) = u^{\mathrm{new}}(t, X(t, Y(t, a))).$$

The initial value is $X(0, Y(0, a)) = a$, which verifies (3.59).

For the final composition, write the inner map as $g = \mathrm{id} + h$. Suppose

$$\|\partial^n h\|_\infty \le A_i R_i^n (n!)^2, \qquad \|\partial^n f\|_\infty \le A_o R_o^n (n!)^2 \quad (n \ge 0).$$

The derivative series of g is bounded by $(d + 2A_i R_i)\zeta$ when $\zeta \le (2R_i)^{-1}$. Choose ζ also to satisfy $R_o(d + 2A_i R_i)\zeta \le 1/2$. Then (3.56) gives

$$\|\partial^n (f \circ g)\|_\infty \le C A_o \zeta^{-n} (n!)^2.$$

If g is a volume-preserving diffeomorphism and the assumed bounds for f hold in L^2, the conclusion holds in L^2 as well: place the derivative of f in L^2 and retain L^∞ bounds for the derivatives of g.

Apply this estimate with $g = Y$. For the displacement, use

$$X^{\mathrm{new}} - \mathrm{id} = (Y - \mathrm{id}) + (X - \mathrm{id}) \circ Y.$$

For its time derivatives, use

$$\begin{aligned} X^{\mathrm{new}}_t &= X_t \circ Y + (\nabla X \circ Y) Y_t, \\ X^{\mathrm{new}}_{tt} &= X_{tt} \circ Y + 2(\nabla X_t \circ Y) Y_t + (\nabla^2 X \circ Y)[Y_t, Y_t] + (\nabla X \circ Y) Y_{tt}. \end{aligned}$$

In the factors $\nabla X \circ Y$, write $\nabla X = I + \nabla(X - \mathrm{id})$. The remaining outer functions are spatial derivatives of $X - \mathrm{id}$, X_t, or X_{tt} of order at most two, so the parent hypothesis bounds their sizes and growth constants by fixed powers of P. For $g = Y$, the preceding estimates give $A_i \le P^c$ and $R_i \le kP^c$. The radius in the composition estimate can therefore be chosen with $\zeta^{-1} \le kP^c$. Applying the product estimate to the displayed time derivatives gives size P^c and growth constant kP^c for $X^{\mathrm{new}} - \mathrm{id}$, X^{new}_t, and X^{new}_{tt}. Passing from L^2 to H^s

costs s shifts, hence at most $C_s^{n+1} P^c (kP^c)^{n+s} (n!)^2$. Condition (3.12), after a further fixed enlargement of Q, bounds this by $(k^{C_*})^{n+1}(n!)^2$ with $C_* = 10(s+2)$. This proves (3.16) without a frequency-dependent exponential. □

3.9. **Initial support and the completion of the packet estimate.**

Proof of Claim 3 of Proposition 3.1. It remains to prove the initial-data bounds (3.17) and the support assertions in Proposition 3.1. The correction e in Lemma 3.3 has $e(0) = 0$. Set $y_a = a/\ell$ and $\theta_a = km_0 \cdot a/\ell$, with θ_a interpreted in $\mathbb{T}$, and define

$$\begin{aligned} u_{\mathrm{high},0}(a) &= \ell \sum_{p=1}^{N_k} \big(\kappa^p A_p + \kappa^{p+1} C_p\big)(0, y_a, \theta_a), \\ u_{\mathrm{mean},0}(a) &= \ell \sum_{p=2}^{N_k} \kappa^p B_p(0, y_a). \end{aligned} \tag{3.60}$$

Since $X(0,a) = a$ and $B_1 = 0$, formula (3.47) gives $u^{\mathrm{new}}(0,a) - u(0,a) = u_{\mathrm{high},0}(a) + u_{\mathrm{mean},0}(a)$.

At time zero every oscillatory coefficient retains the factor α, including coefficients solved using a positive history interval. For fixed m, apply (3.40) only at $n \le m+s+1$. All dependence on m enters a prefactor P^{c_m}; the remaining expansion-order factor is $[P^c(CN_k)^{300}]^{p+1}$, with c independent of m. Thus (3.41) sums these bounds with the same choice of Q. For $\mathcal{T}_k f(y) = f(y, km_0 \cdot y)$, the chain rule and the embedding $H^1(\mathbb{T}) \hookrightarrow L^\infty(\mathbb{T})$ give

$$\|\mathcal{T}_k f\|_{H^m(\mathbb{R}^3)} \le C_m k^m \|f\|_{H^{m+1}(\mathbb{R}^3 \times \mathbb{T})}.$$

The extra derivative is taken before restriction to the graph. Physical rescaling satisfies

$$\|\ell g(\cdot/\ell)\|_{H^m} \le \ell^{5/2-m} \|g\|_{H^m} \le \ell^{-m} \|g\|_{H^m}.$$

Applying these estimates to the oscillatory sum gives the first bound in (3.17).

The mean initial velocity begins at $\kappa^2 B_2(0)$ and has no angle derivatives. Orders $p \ge 3$ sum as $\sum_{p\ge 3} k^{-p+0.01(p+1)}$; the separate fixed-order estimate bounds $\kappa^2 B_2(0)$ by $k^{-2}P^{c_m}$. This proves the second initial bound. The oscillatory supports are $|a| \le \ell/2$, and (3.25) gives the mean supports $|a| \le 2$. The exact correction has zero initial value. If $t_0 = 0$ and $L = 0$, the prescribed forward initial conditions give $A_p(0) = 0$ for every forced $p \ge 2$, hence also $C_p(0) = 0$; the boundary condition gives $B_p(0) = 0$. The initial increment is therefore $\ell(\kappa A_1 + \kappa^2 C_1)(0, y_a, \theta_a)$: the primary oscillation and its curl correction. Since $F(0) = I$ and $v(0)$ is constant when $t_0 = 0$, $Q_1(0, y, \theta) = -\alpha \chi_1(y)(m_0 \times v(0))\phi_\delta(\theta)$. The identity $\ell\kappa^2(\widetilde{d} \times Q_1)(0, y_a, \theta_a) = \ell^2\kappa^2 \operatorname{curl}_a[Q_1(0, y_a, \theta_a)]$ at time zero gives (3.18).

Only finitely many polynomial exponents have entered the proof: the fixed-s inverses, the common growth constant, the separation of derivative and expansion orders, the weighted error estimate, and the graph and flow estimates. The only factorial growth with the expansion length was absorbed by $(CN_k)^{300(p+1)}$, with $300\vartheta < 0.01$. Thus a single finite $Q(c_0)$ ensures every use of (3.12). The constants c_m in the initial bounds may depend on the fixed Sobolev order m; their summation over expansion orders uses the same Q. This completes the proof of Proposition 3.1. □

4. Amplification and transfer of the wave geometry

The packet construction in Proposition 3.1 is now proved, conditional on bounds for the primary wave and its transverse propagator. In this section we prove those bounds for a parent shear of the form (4.3), and prove that the resulting gradient can amplify the next packet. The scale choices and the verification that the packet and parent hypotheses hold at every stage remain for §5.

Claim 1 of Proposition 3.1 gives a leading gradient proportional to $v \otimes m$ in (3.13) and a pressure-Hessian increment with coefficient $m \cdot Mv$ in (3.14). Here M is the parent velocity gradient from (3.5), $m = F^{-T} m_0$ is the transported phase normal, and v solves (3.9). We must control both the size and the directions of m, v so that the new gradient can serve as the parent shear for the next packet.

The result proved here, Proposition 4.1, supplies the pressure sign, the new frame $p_2 = m/|m|$, $q_2 = v/|v|$, and bounds comparing the gradient amplitude factor $|m||v|$ at earlier times with its target value. It also bounds the linear map taking arbitrary transverse data from time s to time t, as required by §3.2 (iii). The proof follows m, v in the moving frame of the parent shear (4.3).

4.1. **The parent field and the statement to be proved.** We first specify the parent shear and state the estimates needed for the next packet. We retain the particle map and the matrices F, M, H from (3.5), with the label and component conventions in §2.4. A *ray* is a nonzero vector m satisfying $\dot{m} = -M^T m$; it is the normal to a transported phase surface. A transverse velocity is a vector v with $m \cdot v = 0$ satisfying

$$\dot{v} = -Mv + 2m \frac{m \cdot Mv}{|m|^2}. \tag{4.1}$$

The last term enforces transversality: direct differentiation gives $\frac{d}{dt}(m \cdot v) = 0$.

Let the parent be a smooth odd Euler flow on $[0, S]$, $0 < S \leq 1$, with the particle map and matrices F, M, H in (3.5), localization length $0 < \ell \leq 1$, and activation time $t_0 \in [0, S)$. Impose the following hypotheses.

(i) **Parent shear and frame.** Let $B(t)$ be the gradient at the origin of an older Euler solution. At stage j of the iteration, B is the gradient of U^{j-2}, while U^{j-1} is the parent. Take orthonormal vectors $p(t), q(t)$ obtained by normalizing, respectively, a ray and a transverse velocity for B, and put $n = p \times q$. The normalization equations are

$$\dot{p} = -B^T p + B_{pp} p, \qquad \dot{q} = -Bq + 2B_{pq} p + B_{qq} q. \tag{4.2}$$

Suppose that on an activation interval $[t_0, S]$ the current parent satisfies

$$M(t, 0) = B(t) + h(t) q(t) p(t)^T + E(t), \qquad \|E(t)\| \leq E_*, \qquad h(t_0) = h_*, \qquad \frac{\dot{h}}{h} = -B_{pp} - B_{qq}. \tag{4.3}$$

The scalar h is the shear strength, and E is the remaining gradient error. The rank-one matrix hqp^T sends the old normal p into the old transverse direction q. We assume $\|B\| \leq CG_*$ and $\|\dot{B}\| \leq CG_*^2$, where $G_* \geq 1$.

(ii) **Particle-map bounds.** Let $K_h \geq 2$ bound $X - \mathrm{id}, X_t, X_{tt}$ in physical particle labels as in (3.15). Sobolev embedding, finitely many additional derivatives, and $F^{-1} = \mathrm{cof}(F)^T$ give bounds K_h^c for $F, F^{-1}, F_t, F_{tt}, M, H$ and their first physical-label derivatives. At derivative order $j \geq 0$, their Gevrey multiplier bounds have the form $K_h^c (K_h^c)^j (j!)^2$, with the exponent convention in §2.4.

(iii) **Scales and error bounds.** We freeze the frame coefficients B_{pq} and B_{nq} at t_0 and rescale time using their values. Choose a positive comparison parameter x_{prev} and set

$$\begin{gathered} a = B_{pq}(t_0) \in [1/2, 2], \qquad \beta = \frac{B_{nq}(t_0)}{a}, \qquad \frac{1}{2} \leq \beta x_{\text{prev}}^2 \leq 2, \\ \varepsilon = \sqrt{a/h_*}, \qquad \tau = \frac{a(t-t_0)}{\varepsilon}, \qquad x = \sqrt{\beta}\,\tau. \end{gathered} \tag{4.4}$$

Thus x_{prev} is comparable to $\beta^{-1/2}$, while τ and x measure elapsed physical time in units ε/a and $\varepsilon/(a\sqrt{\beta})$, respectively. Here $0 < \beta \leq \beta_0$ for a sufficiently small absolute constant β_0, and the target is $x = x_{\text{tar}} \geq 2$. Its physical time is $t_{\text{tar}} = t_0 + \varepsilon x_{\text{tar}}/(a\sqrt{\beta})$. We assume $0 \leq \tau \leq \Theta$, where $\Theta \geq 2 + \beta^{-1/2} + x_{\text{tar}}/\sqrt{\beta}$, and require

$$t_{\text{tar}} \leq S \leq t_{\text{tar}} + (\varepsilon/a)\Theta^{-60}.$$

Thus the interval reaches the target and extends at most Θ^{-60} units of τ beyond it. Set

$$e_* = C\left(\varepsilon\Theta G_*^2 + E_* + \frac{\ell K_h^c}{\varepsilon}\right), \qquad e_*\Theta^{60} \leq x_{\text{prev}}^{-10}. \tag{4.5}$$

The three summands account for variation of the older gradient and frame, the central error E, and the difference between central and neighboring coefficients after rescaling by ε. The smallness condition controls their accumulated effect on the ray and velocity equations. The fixed constant C in this definition is chosen large enough for the estimates below. Equivalently, we may use a sufficiently small fixed constant on the right of the smallness condition.

(iv) **History and activation conditions ($t_0 > 0$).** Assume $H(t, y) \leq K_+ I$ on $[0, t_0]$ for all labels y, together with the packet coercivity condition (3.11). Also assume, on $[0, t_0]$,

$$\|M\|_\infty \leq C_M h_*, \qquad \|H\|_\infty \leq C_H h_* h_{\text{old}}, \qquad 1 \leq h_{\text{old}} \leq h_*, \qquad t_0^{-1} \leq h_*. \tag{4.6}$$

Here h_{old} controls the allowed size of the history pressure Hessian relative to the current shear scale h_*. At activation write $\overline{B} = B(t_0) + E(t_0)$ and require

$$\overline{B}_{pp} < 0, \qquad \|\overline{B}\| \leq \zeta h_*. \tag{4.7}$$

The fixed number $\zeta > 0$ is chosen sufficiently small in terms of C_M, C_H. Finally $t_0^{-1} \leq K_h^c$ when $t_0 > 0$. The conditions on $\overline{B}$ will allow us to choose a stationary history displacement whose terminal velocity has $v_p(t_0, 0) \leq 0 < v_q(t_0, 0)$.

Choose $m_0 = F(t_0, 0)^T n(t_0)/|F(t_0, 0)^T n(t_0)|$ and $m = F^{-T} m_0$. Then $m(t_0, 0)$ is a positive multiple of $n(t_0)$, so the central transverse velocity at activation lies in $\operatorname{span}\{p(t_0), q(t_0)\}$. Fix a constant 3×2 matrix $R_\perp$ whose columns form an orthonormal basis of $m_0^\perp$.

Proposition 4.1. *For every parent flow satisfying the four hypotheses in §4.1, there is a primary transverse velocity v with the following properties. When $t_0 > 0$, the construction chooses a constant endpoint $\xi_T \in \mathbb{R}^2$. On $[0, t_0]$, the velocity is $v = FR_\perp \xi_t$, where $FR_\perp \xi$ is the stationary displacement with $\xi(0) = 0$ and $\xi(t_0) = \xi_T$ from §3.2.*

1. **Activation data and pressure sign.** *Its central activation data satisfy*

$$v_q(t_0, 0) = 1, \qquad -C \leq v_p(t_0, 0) \leq 0. \tag{4.8}$$

For $t_0 = 0$ we take $v(t_0) = q$. On every label $|y| \leq 1/2$,

$$m \cdot Mv > 0 \qquad (\tau \geq 1). \tag{4.9}$$

2. **Frame transfer and compression.** *At* $(t, y) = (t_{\text{tar}}, 0)$ *define* $p_2 = m/|m|$, $q_2 = v/|v|$, $n_2 = p_2 \times q_2$ *and* $a_2 = p_2 \cdot Mq_2$, $\beta_2 = (n_2 \cdot Mq_2)/a_2$. *Then*

$$\begin{aligned} \frac{a_2}{a} &= 1 + O\left(x_{\text{tar}}^{-4} + \frac{\beta}{x_{\text{tar}}^2} + \frac{\sqrt{\beta}}{x_{\text{tar}}^3} + e_* \Theta^{40}\right), \\ \beta_2 x_{\text{tar}}^2 &= 1 + O\left(\sqrt{\beta} + x_{\text{tar}}^{-4} + e_* \Theta^{40}\right), \end{aligned} \tag{4.10}$$

The parent gradient also satisfies

$$p_2 \cdot Mp_2 \le -\frac{c\sqrt{h_*}}{x_{\text{prev}} x_{\text{tar}}} + C(G_* + E_*). \tag{4.11}$$

3. **Relative sizes.** *The size estimates below hold for an absolute constant* $b > 0$. *Write* $A_{\text{tar}} = |m(t_{\text{tar}}, 0)||v(t_{\text{tar}}, 0)|$. *Uniformly on the packet support,* $|m||v|/A_{\text{tar}} \le C$ *for* $\tau \ge 1$, *whereas on the history interval, and after activation for* $0 \le \tau < 1$,

$$\frac{|m||v|}{A_{\text{tar}}} \le K_h^c \Theta^c \varepsilon^{-c} \exp(-b x_{\text{prev}}). \tag{4.12}$$

4. **Propagator and normalization.** *There is a positive scalar function* g, *with* $g(t_0) = 1$, *such that the following bound holds uniformly at every fixed label* $|y| \le 1/2$. *Write any transverse solution of* (4.1) *as* $A = FR_\perp a_\perp$. *The solution map* $a_\perp(s) \mapsto a_\perp(t)$ *has operator norm at most* $P^c g(t)/g(s)$ *for* $t_0 \le s \le t \le S$. *For any prescribed child shear* h_{child}, *the normalization*

$$\alpha = \frac{\delta h_{\text{child}}}{A_{\text{tar}}} \tag{4.13}$$

satisfies $\alpha \le P^c e^{-b x_{\text{prev}}}$ *and* $\sup_{[t_0, S]} \alpha g \le P^c$. *Here* P *contains* $K_h, \varepsilon^{-1}, \Theta, h_{\text{child}}, \delta^{-1}$ *and, when applicable,* t_0^{-1}, *or polynomial upper bounds for them.*

The bounds on the transverse propagator and on αg verify §3.2 (iii), which is required to apply Proposition 3.1. At the central label $y = 0$, the phase $km_0 \cdot y$ vanishes and $\chi_1(0) = 1$. Since $f_\delta'(0) = \delta^{-1}$, at $t = t_{\text{tar}}$ the leading gradient increment is

$$\alpha \delta^{-1} v \otimes m = \frac{h_{\text{child}}}{|m||v|} v \otimes m = h_{\text{child}} q_2 p_2^T. \tag{4.14}$$

Thus (3.13), with its stated error, produces a shear in the new frame. On the interval in (4.9), the matrix coefficient $-2\alpha\chi_1(m \cdot Mv)(m \otimes m)/|m|^2$ multiplying f_δ' in (3.14) is negative semidefinite.

4.2. **Choosing the velocity at the activation time.**

Proof of Proposition 4.1. We first prove the activation-data assertion in Claim 1 of Proposition 4.1. For $t_0 > 0$, we choose the activation velocity through the endpoint of its stationary history displacement. We seek an endpoint whose terminal velocity has p-component at most zero and q-component strictly positive, then normalize the latter to 1. The forward analysis uses (4.8) at the central label and the neighboring-label estimates proved below. Let $s_0 > 0$ be defined by $m(t_0, 0) = s_0 n$. The bounds for $F^{\pm 1}$ give $s_0 + s_0^{-1} \le K_h^c$. Suppose $t_0 > 0$. Until we return to neighboring labels, all coefficients are evaluated at the central label. For an endpoint $Y \in n(t_0)^\perp$, consider paths satisfying $m(t) \cdot \eta(t) = 0$, $\eta(0) = 0$, and $\eta(t_0) = Y$, with action

$$\mathcal{E}(\eta) = \int_0^{t_0} (|\eta_t|^2 - H\eta \cdot \eta)\, dt.$$

The constrained variational equation (3.27) constructs the stationary path with these endpoints. Since $\int |\eta|^2 \le (t_0^2/2)\int |\eta_t|^2$, the coercivity condition gives $\mathcal{E}(\eta) \ge \frac{1}{2}\int |\eta_t|^2$ for paths starting at zero. The stationary path minimizes the action among paths with its endpoints.

Write η^Y for the stationary path when its endpoint is displayed explicitly. Define the linear endpoint map $\Lambda : n(t_0)^\perp \to n(t_0)^\perp$ by $\Lambda Y = \operatorname{proj}_{n(t_0)^\perp} \eta_t^Y(t_0)$. Integration by parts gives $\mathcal{E}(\eta^Y) = Y \cdot \Lambda Y$: the stationary action is a quadratic form in the endpoint Y. For stationary paths with endpoints Y, Z, integration by parts also gives

$$Y \cdot \Lambda Z = \int_0^{t_0} (\eta_t^Y \cdot \eta_t^Z - H\eta^Y \cdot \eta^Z)\, dt = Z \cdot \Lambda Y.$$

Thus Λ is symmetric, and the preceding coercive estimate gives $\Lambda \ge 0$. We next bound its entries in units of the shear strength h_*. Set

$$L_0 = t_0^{-1} + \|M\|_\infty + \|H\|_\infty^{1/2} + 1.$$

Use a comparison path η_{cmp} that is zero until $t_0 - L_0^{-1}$ and then equals $L_0(t - t_0 + L_0^{-1}) \operatorname{proj}_{m(t)^\perp} Y$. It remains transverse and has the required endpoints. The derivative of this projection operator is bounded by $C\|M\|_\infty$, so $|\mathcal{E}(\eta_{\mathrm{cmp}})|$ is at most

$$C\left(L_0 + \frac{\|M\|_\infty^2}{L_0} + \frac{\|H\|_\infty}{L_0}\right)|Y|^2 \le C_0 h_* |Y|^2.$$

Minimality gives $Y \cdot \Lambda Y \le \mathcal{E}(\eta_{\mathrm{cmp}})$, and consequently $0 \le \Lambda \le C_0 h_* I$.

Write the trial displacement as $\eta = FR_\perp \xi$. Its primary velocity is $w = \eta_t - M\eta = FR_\perp \xi_t$. We use w before its final normalization. Its terminal value lies in $n(t_0)^\perp$. Write $b = \Lambda_{pq}$, $d = \Lambda_{qq}$ and $u = \Lambda_{pp} - \overline{B}_{pp} > 0$. The endpoint map in the (p, q) coordinates is

$$\begin{pmatrix} w_p \\ w_q \end{pmatrix} = \begin{pmatrix} u & b - \overline{B}_{pq} \\ b - h_* - \overline{B}_{qp} & d - \overline{B}_{qq} \end{pmatrix} \begin{pmatrix} Y_p \\ Y_q \end{pmatrix}.$$

If $b \le h_*/2$, we take $Y = -p$, obtaining

$$w_p = -u \le 0, \qquad w_q = h_* + \overline{B}_{qp} - b \ge h_*/3.$$

If $b > h_*/2$, the positive semidefiniteness of Λ gives $\Lambda_{pp} \ge b^2/\Lambda_{qq} \ge h_*/(4C_0)$. We take $Y = q - (b - \overline{B}_{pq})p/u$. Then $w_p = 0$ and

$$\begin{aligned} w_q &= d - \overline{B}_{qq} - \frac{(b - h_* - \overline{B}_{qp})(b - \overline{B}_{pq})}{u} \\ &= \left(d - \frac{b^2}{u}\right) - \overline{B}_{qq} + \frac{b^2 - (b - h_* - \overline{B}_{qp})(b - \overline{B}_{pq})}{u}. \end{aligned}$$

The term $d - b^2/u$ is nonnegative, since $u \ge \Lambda_{pp}$. The numerator in the last term is

$$bh_* + b(\overline{B}_{pq} + \overline{B}_{qp}) - h_*\overline{B}_{pq} - \overline{B}_{qp}\overline{B}_{pq} \ge h_*^2/2 - C(C_0 + 1)\zeta h_*^2.$$

Together with $u \le (C_0 + \zeta)h_*$, this proves $w_q \ge ch_*$ after fixing ζ. In either case divide η, Y, w by the positive number $w_q(t_0)$, retaining these symbols for the rescaled quantities. The normalized velocity $v = w$ satisfies (4.8), and the normalized endpoint satisfies $|Y| \le C/h_*$. At the central label $\eta = FR_\perp \xi$, so choose $\xi_T = R_\perp^T F(t_0, 0)^{-1} Y$. This gives $\eta(t_0, 0) = Y$, since $m_0 \cdot F(t_0, 0)^{-1} Y = m(t_0, 0) \cdot Y = 0$.

At each neighboring physical label a, prescribe this same coordinate endpoint ξ_T. Its physical displacement endpoint is $F(t_0, \mathsf{a})R_\perp \xi_T$. We need bounds polynomial in K_h for the history velocity and for its variation between labels; the latter control the neighboring activation data. Set $\widetilde{\xi} = \xi - t\xi_T/t_0$, which has zero endpoints. The coercive estimate applied to its equation gives $\|\xi_t\|_{L^2_t} \le K_h^c$. Subtracting (3.27) at labels $\mathsf{a}_1, \mathsf{a}_2$ gives $\|\xi_t(\cdot, \mathsf{a}_1) - \xi_t(\cdot, \mathsf{a}_2)\|_{L^2_t} \le K_h^c|\mathsf{a}_1 - \mathsf{a}_2|$, since every coefficient difference has this same bound. Equation (3.28) supplies the analogous L^2_t bounds for ξ_{tt} and its difference. The time Sobolev inequality converts these into supremum bounds for ξ_t; multiplication by $FR_\perp$ then gives

$$\sup_{[0,t_0]} |v| \le K_h^c, \qquad \sup_{[0,t_0]} |v(t, \mathsf{a}_1) - v(t, \mathsf{a}_2)| \le K_h^c|\mathsf{a}_1 - \mathsf{a}_2|.$$

The localized primary coefficient in the packet construction is $A_1 = \alpha\chi_1 v f_\delta$. Multiplying the history displacement by the same time-independent factor $\alpha\chi_1 f_\delta$ gives a localized displacement whose primary velocity is A_1. This multiplication preserves the stationary equation (3.27), which contains no spatial or angular derivatives. The activation data are now fixed. To complete Claim 1 of Proposition 4.1, we must establish the sign condition (4.9) during forward evolution.

4.3. **The rotating frame and the leading equations.** Starting from the activation data (4.8), we express the ray and velocity equations (2.4) in moving, scaled coordinates. This subsection identifies the ideal system (4.20) and bounds its coefficient error. We prove growth for that system in §4.4 and compare it with the exact evolution in §4.5. In the frame (p, q, n) of (4.2), the large shear hqp^T in (4.3) occupies a single matrix entry. Because this frame moves with the older flow, the change of coordinates also contributes a rotation term to the time derivatives. The scaling $h_*\varepsilon^2 = a$ from (4.4) normalizes the leading shear coefficient in these equations.

Let R_f have columns (p, q, n) and set $S_f = R_f^T \dot{R}_f$. Differentiating (4.2) gives

$$(S_f)_{12} = B_{pq}, \qquad (S_f)_{13} = B_{pn}, \qquad (S_f)_{32} = -B_{nq}, \qquad S_f^T = -S_f.$$

We use this central frame at every packet label $|y| \le 1/2$. The corresponding physical label is $\mathsf{a} = \ell y$, so the physical-label derivative bound shows that its parent gradient differs from $B + hqp^T$ by at most $E_* + \ell K_h^c$. The component columns and the gradient matrix in this frame are

$$\widehat{m} = R_f^T m, \qquad \widehat{v} = R_f^T v, \qquad \widehat{M} = R_f^T M R_f.$$

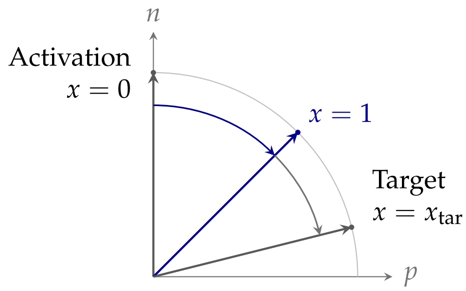
Since $\dot{R}_f^T = -S_f R_f^T$, differentiation gives

$$\dot{\widehat{m}} = -(\widehat{M} - S_f)^T \widehat{m}, \qquad \dot{\widehat{v}} = -(\widehat{M} + S_f)\widehat{v} + 2\widehat{m}\frac{\widehat{m} \cdot \widehat{M}\widehat{v}}{|\widehat{m}|^2}. \tag{4.15}$$

We suppress the hats in the component calculations below. Scalar products and norms are unchanged because the frame is orthonormal. Introduce scaled components by

$$m/s_0 = (P, \varepsilon Q, N), \qquad v = (\varepsilon U, V, \varepsilon W). \tag{4.16}$$

In physical vectors these formulas read $m = s_0(Pp + \varepsilon Qq + Nn)$ and $v = \varepsilon Up + Vq + \varepsilon Wn$. The letters P, Q, N, U, V, W in the scaled equations denote component functions; in particular this P is distinct from the packet data bound in §3. Transversality is the identity $PU + QV + NW = 0$. At $\tau = 0$ and the central label, $(P, Q, N) = (0, 0, 1)$ and $(U, V) = (-\lambda, 1)$, where $0 \le \lambda \le C/\varepsilon$. At neighboring labels, the initial values of these five components differ from their central values by at most $\ell K_h^c/\varepsilon$.



$\dfrac{x^2 p + n}{\sqrt{1+x^4}}$

Normalized projection onto the moving (p, n) plane.

$0 \le x \le 1$: growth of the transverse velocity (Equation (4.22)).

$x = x_{\rm tar} \ge 2$: form $p_2 = m/|m|$ from the full transported phase normal (Equation (4.10)).

FIGURE 2. The phase normal in the ideal system (4.17), projected onto the moving (p, n) plane and normalized to unit length. It turns from n toward p as the scaled time x increases. The marked values are $x = 0$ at activation, $x = 1$ at the end of the interval used to prove (4.22), and the target at which the new frame is calculated in (4.10). The target ray is drawn at $x_{\rm tar} = 2$; the estimates allow any $x_{\rm tar} \ge 2$. The full vector is $x^2 p - 2\varepsilon\sqrt{\beta}\, xq + n$, and the transverse velocity remains perpendicular to this full vector. The rays have unit length; the velocity amplitude is not shown.

A prime now denotes $d/d\tau$. The leading ray system and its solution are

$$P' = -Q, \quad Q' = -2\beta N, \quad N' = 0; \qquad P_0 = \beta\tau^2 = x^2, \quad Q_0 = -2\beta\tau, \quad N_0 = 1. \tag{4.17}$$

Figure 2 shows its normalized projection onto the (p, n) plane. We record why the errors are small despite the large shear. Its scaled coefficient obeys $\varepsilon^2 h/a = 1 + O(\varepsilon\Theta G_*)$. The entries $(M - S_f)_{12}$ and $(M - S_f)_{32}$ are respectively $O(E_* + \ell K_h^c)$ and $2B_{nq} + O(E_* + \ell K_h^c)$. The moving entries B_{pq}, B_{nq} change by $O(\varepsilon\Theta G_*^2)$, including frame rotation. All remaining scaled coefficients contain either an error or a factor $\varepsilon O(G_*)$. Thus the exact ray system differs entrywise from (4.17) by at most Ce_*.

The triangular ideal propagator has norm $C\Theta^2$. Write $Z = (P, Q, N)^T$ and $Z_0 = (P_0, Q_0, 1)^T$. The error equation has forcing bounded by $Ce_*(|Z_0| + |Z - Z_0|)$ and initial error at most Ce_*. Since $|Z_0| \le C\Theta^2$, Duhamel's formula bounds $\sup|Z - Z_0|$ by $Ce_*\Theta^5 + Ce_*\Theta^3 \sup|Z - Z_0|$. The condition $e_*\Theta^3 \ll 1$ absorbs the last term and gives

$$|P - P_0| + |Q - Q_0| + |N - 1| \le Ce_*\Theta^5, \qquad N \ge 1/2. \tag{4.18}$$

The lower bound for N now allows us to solve the transversality identity:

$$W = -\frac{PU + QV}{N}.$$

It remains to estimate the two independent velocity components U, V. Set $D_a = |m|^2/s_0^2 = P^2 + \varepsilon^2 Q^2 + N^2$ and $J = (m \cdot Mv)/(s_0 a)$. Thus D_a is the denominator in the scaled velocity correction, and the pressure sign is the sign of J. For arbitrary (U, V), with W determined by transversality, we have

$$\begin{aligned} |W| &\le C\Theta^2(|U|+|V|), \\ |D_a - (1+P_0^2)| &\le Ce_*\Theta^7, \\ |J - [(P_0+\beta)V + Q_0U]| &\le Ce_*\Theta^5(|U|+|V|). \end{aligned} \tag{4.19}$$

For the last estimate the three leading row contributions are

$$\frac{PB_{pq}V}{a}, \qquad \frac{h\varepsilon^2}{a}QU, \qquad \frac{NB_{nq}V}{a}.$$

Every other term contains an error or $\varepsilon O(G_*)$, multiplied by a factor bounded by $C\Theta^4(|U|+|V|)$. In (4.15), the term $-(M+S_f)v$ contributes $-2V + O(e_*\Theta^2)(|U|+|V|)$ to U', and the term parallel to m contributes $2PJ/D_a$. The main term in V' is $-U$; all remaining terms, including $2\varepsilon^2 QJ/D_a$, are bounded by $Ce_*\Theta^4(|U|+|V|)$. Using $D_a \ge 1/4$ in these rational expressions shows that the exact coefficient matrix for (U,V) differs by at most $Ce_*\Theta^{12}$ from that of the ideal system

$$U' = -2V + \frac{2P_0[(P_0+\beta)V + Q_0U]}{1+P_0^2}, \qquad V' = -U. \tag{4.20}$$

For this ideal system, setting $D = 1+P_0^2$ and using $D' = -2P_0Q_0$ gives

$$(DV')' = 2(1-\beta P_0)V. \tag{4.21}$$

4.4. **Growth of the ideal velocity.** For the ideal system (4.20), we first estimate amplification up to $x=1$ and then control the direction ratio U/V up to the target $x = x_{\mathrm{tar}}$. Let V_λ solve (4.21) with $V_\lambda(0)=1$ and $V_\lambda'(0) = \lambda \ge 0$. Its companion velocity component is $U_\lambda = -V_\lambda'$. On $0 \le x \le 1$, the right side is positive while $V_\lambda > 0$, so integration preserves positivity and monotonicity. Since $x = \sqrt{\beta}\,\tau$, this region lasts for a scaled time $\beta^{-1/2}$. The bounds $D \le 2$ and $2(1-\beta P_0) \ge 1$ turn this long interval of growth into quantitative amplification:

$$V_\lambda(\tau) \ge 1 + \frac{\lambda}{2}\tau + \frac{1}{2}\int_0^\tau (\tau - s)V_\lambda(s)\,ds \ge \cosh(\tau/\sqrt{2}).$$

The second inequality follows by iteration of the positive integral kernel. Consequently

$$V_\lambda(x=1) \ge \exp(b_0/\sqrt{\beta}). \tag{4.22}$$

On this interval $l = V_\lambda'/V_\lambda$ satisfies $l' \le 2 - l^2$ and $l \ge 0$. The function $L(\tau) = \sqrt{2}\coth(\sqrt{2}\tau)$ solves $L' = 2 - L^2$ and tends to $+\infty$ as $\tau \downarrow 0$. Scalar comparison therefore gives $l(\tau) \le \sqrt{2}\coth(\sqrt{2}\tau)$, uniformly in λ, for $\tau > 0$.

Beyond $x=1$ we need both a lower bound for the amplitude and an estimate for its direction ratio U/V. A reciprocal variable sends this range to a bounded interval and keeps the right-hand coefficient of the transformed scalar equation positive. Writing $V(x) = V_\lambda(x/\sqrt{\beta})$, (4.21) becomes

$$\big((1+x^4)V_x\big)_x = (2/\beta - 2x^2)V.$$

Set $\rho = 1/x$ and $f(\rho) = V(1/\rho)/\rho$. In particular $V(x) = \rho f(\rho)$ and $V_x = -\rho^2 f - \rho^3 f_\rho$. Substitution yields

$$\big((1+\rho^4)f_\rho\big)_\rho = (2/\beta - 2\rho^2)f.$$

Since $f(1) > 0$ and $f_\rho(1) = -V(1) - V_x(1) < 0$, the identity

$$(1+\rho^4)f_\rho(\rho) = 2f_\rho(1) - \int_\rho^1 (2/\beta - 2s^2)f(s)\,ds$$

preserves $f_\rho < 0$ as we integrate backwards, and hence preserves $f > 0$. Thus $xV(x) = f(1/x)$ is increasing. The direction ratio satisfies $U_\lambda/V_\lambda = -\sqrt{\beta}\,V_x/V$. To express it through f, define

$$z = -\sqrt{\beta}\,f_\rho/f, \qquad \mu(\rho) = \sqrt{\frac{2-2\beta\rho^2}{1+\rho^4}}, \qquad \sigma = 1-\rho.$$

Direct logarithmic differentiation gives

$$\sqrt{\beta}\,z_\sigma = \mu^2 - z^2 + \sqrt{\beta}\frac{4\rho^3}{1+\rho^4}z. \tag{4.23}$$

The first two terms on the right cancel at $z = \mu$. We compare z with this positive root to obtain the direction estimate. At $\rho = 1$, $z = \sqrt{\beta} + l(x=1)$ is uniformly bounded. Equation (4.23) preserves $z \geq 0$ and gives a uniform upper bound. For $w = z - \mu$ it becomes

$$\sqrt{\beta}\,w_\sigma = -(z+\mu)w + \sqrt{\beta}\left(\frac{4\rho^3}{1+\rho^4}z - \mu_\sigma\right).$$

The bracket is bounded and $z + \mu \geq c > 0$. Multiplication by the integrating factor therefore gives $|w| \leq Ce^{-c(1-\rho)/\sqrt{\beta}} + C\sqrt{\beta}$. In particular, uniformly in $\lambda \geq 0$,

$$z^2 = \frac{2}{1+\rho^4} + O(\sqrt{\beta}) \quad (0 \leq \rho \leq 1/2), \qquad r_\lambda := \frac{U_\lambda}{V_\lambda} = \begin{cases} -l, & x \leq 1, \\ \sqrt{\beta}/x - z/x^2, & x \geq 1. \end{cases} \tag{4.24}$$

For $\tau \geq 1$ these formulas bound $|r_\lambda|$ by an absolute constant. The ideal solution now has the required growth and direction estimates. These are the inputs to Section 4.5, where we prove the sign assertion in Claim 1 of Proposition 4.1 and obtain the propagator estimate needed for Claim 4 of Proposition 4.1.

4.5. **Stability relative to the growing solution.** We transfer the ideal estimates from §4.4 to the exact scaled components (4.16). This proves the pressure sign in Claim 1 of Proposition 4.1 and bounds the propagator for (U, V). The conversion to packet coordinates and the normalization required by Claim 4 are completed in §4.6. In this subsection V_0 is the ideal solution with zero initial slope. In the propagator calculation, s, t denote scaled times. A small coefficient error acts throughout an interval on which the ideal solution grows exponentially. Measuring the propagator in units of $V_0(t)/V_0(s)$ accounts for this growth and leaves an error controlled by powers of Θ. We need this comparison for arbitrary transverse data to supply the assumption in §3.2 (iii). For any solution Y of (4.21), that equation implies

$$\left(DV_0^2(Y/V_0)'\right)' = 0. \tag{4.25}$$

Applying this identity to V_λ, whose initial value is 1 and initial slope is λ, gives

$$V_\lambda = V_0(1+\lambda I), \qquad I(\tau) = \int_0^\tau \frac{du}{D(u)V_0(u)^2}, \qquad I(1) \geq c > 0. \tag{4.26}$$

The last bound follows from uniformly bounded ideal coefficients on $0 \leq \tau \leq 1$. Monotonicity before $x = 1$, and monotonicity of xV_0 afterwards, show that $V_0(s)/V_0(u) \leq C\Theta$ for

$s \le u$. Its logarithmic derivative l_0 is uniformly bounded on the whole interval, including $\tau < 1$ since its initial slope is zero.

For arbitrary initial data $Y(s), Y'(s)$, integration of (4.25) from s to t gives

$$Y(t) = \frac{V_0(t)}{V_0(s)} \left[Y(s) + D(s)(Y'(s) - l_0(s)Y(s)) \int_s^t \frac{(V_0(s)/V_0(u))^2}{D(u)}\, du \right]. \tag{4.27}$$

The integral is at most $C\Theta^3$ and $D(s) \le C\Theta^4$. After extracting the same factor $V_0(t)/V_0(s)$, differentiation adds a bounded multiple of the bracket and the term

$$\frac{D(s)}{D(t)}(Y'(s) - l_0(s)Y(s)) \left(\frac{V_0(s)}{V_0(t)}\right)^2,$$

whose size is at most $C\Theta^6(|Y(s)| + |Y'(s)|)$. Since $(U, V) = (-Y', Y)$ for the ideal system, its two-component propagator Φ_0 satisfies

$$\|\Phi_0(t,s)\| \le C\Theta^8 V_0(t)/V_0(s).$$

Let Φ be the exact propagator, and let ΔA be the difference between the exact and ideal coefficient matrices, so that $\|\Delta A\| \le Ce_*\Theta^{12}$. Define $\widetilde{\Phi}(t,s) = V_0(s)\Phi(t,s)/V_0(t)$, and define $\widetilde{\Phi}_0$ similarly. Duhamel's formula becomes

$$\widetilde{\Phi}(t,s) = \widetilde{\Phi}_0(t,s) + \int_s^t \widetilde{\Phi}_0(t,u)\Delta A(u)\widetilde{\Phi}(u,s)\, du.$$

In the supremum norm over scaled times $s \le t$, its integral operator has norm at most $Ce_*\Theta^{8+12+1} = Ce_*\Theta^{21} \ll 1$. Summing the iterated Duhamel terms gives $\|\Phi(t,s)\| \le 2C\Theta^8 V_0(t)/V_0(s)$. For the primary data, including their neighbor error, it also gives

$$|U - U_\lambda| + |V - V_\lambda| \le Ce_*\Theta^{29}(1+\lambda)V_0(t). \tag{4.28}$$

Indeed, $(U - U_\lambda, V - V_\lambda)^T$ solves the exact system with forcing $\Delta A(U_\lambda, V_\lambda)^T$ and initial error of size Ce_*. The ideal propagator bound gives $|(U_\lambda, V_\lambda)| \le C\Theta^8(1+\lambda)V_0$. Thus the initial error contributes $Ce_*\Theta^8 V_0$; Duhamel's formula bounds the forcing contribution by

$$Ce_*\Theta^{12+8+8+1}(1+\lambda)V_0.$$

For $\tau \ge 1$, (4.26) gives $V_\lambda \ge c(1+\lambda)V_0$. Hence

$$V > 0, \qquad V/V_\lambda = 1 + O(e_*\Theta^{29}), \qquad r := U/V = r_\lambda + O(e_*\Theta^{29}).$$

With $w_3 = W/V = -(Pr+Q)/N$, it follows that $w_3 = -P_0 r_\lambda - Q_0 + O(e_*\Theta^{31})$, and therefore

$$J/V = P_0 + \beta + Q_0 r_\lambda + O(e_*\Theta^{33}). \tag{4.29}$$

The leading expression is at least β for $x \le 1$ because $Q_0 r_\lambda \ge 0$. For $x \ge 1$ it equals $x^2 - \beta + 2\sqrt{\beta}z/x > 0$. Condition (4.5), together with $\beta \asymp x_{\mathrm{prev}}^{-2}$, makes the error smaller than half these lower bounds. Since $m \cdot Mv = s_0 aJ$, this proves (4.9). Together with the activation construction, this completes Claim 1 of Proposition 4.1. The relative error estimates also give the component bounds used in the target comparisons below.

4.6. **The next frame and the size comparisons.** Claim 1 of Proposition 4.1 is proved, and a bound for the scaled velocity propagator is available. We now prove the frame and size conclusions in Claims 2 and 3, then convert that propagator bound and apply the normalization to finish Claim 4. The calculations use $r = U/V$ and $w_3 = W/V$ from §4.5, together with the bound for J/V in (4.29).

Frame transfer and compression. At the target, normalizing m and v removes their amplitudes from a_2 and β_2. This reduces the transfer calculation to the component estimates above. With $E_v = 1 + \varepsilon^2(r^2 + w_3^2)$ and $T = M(v/V)/a$, the definitions of p_2, q_2 give

$$\frac{a_2}{a} = \frac{J/V}{\sqrt{D_a}\sqrt{E_v}}, \qquad \beta_2 = \frac{[(P, \varepsilon Q, N) \times (\varepsilon r, 1, \varepsilon w_3)] \cdot T}{(J/V)\sqrt{E_v}}. \tag{4.30}$$

The preceding estimates imply $T_p = 1 + O(e_*\Theta^2)$, $T_n = \beta + O(e_*\Theta^2)$, and $\varepsilon T_q = r_\lambda + O(e_*\Theta^{31})$. The cross product is $(-N + \varepsilon^2 Q w_3, \varepsilon(Nr - Pw_3), P - \varepsilon^2 Qr)$. Its leading dot product is consequently

$$\begin{aligned} -1 + \beta P_0 + (1 + P_0^2) r_\lambda^2 + P_0 Q_0 r_\lambda &= -1 + (1 + \rho^4) z^2 + \beta\rho^2 - 2\sqrt{\beta} z \rho^3 \\ &= 1 + O(\sqrt{\beta}), \qquad \rho = 1/x \le 1/2. \end{aligned}$$

The first equality follows by inserting $P_0 = x^2$, $Q_0 = -2\sqrt{\beta}x$, and $r_\lambda = \sqrt{\beta}/x - z/x^2$; the βx^2 and $\sqrt{\beta}xz$ terms cancel. The accumulated error in this numerator is at most $Ce_*\Theta^{36}$, using $|P| \le C\Theta^2$, $|Q| \le C\Theta$, $|w_3| \le C\Theta^2$ and $\varepsilon \le Ce_*$. Also

$$J/V = x^2 - \beta + 2\sqrt{\beta} z/x + O(e_*\Theta^{33}), \quad D_a = 1 + x^4 + O(e_*\Theta^7), \quad E_v = 1 + O(\varepsilon^2\Theta^4).$$

Substitution in (4.30) proves (4.10). The compression is calculated directly from the shear:

$$p_2 \cdot M p_2 = \frac{h\varepsilon QP}{D_a} + O(G_* + E_*).$$

At the target, (4.18) gives $QP \le -c\sqrt{\beta}x_{\mathrm{tar}}^3$ and $D_a \le Cx_{\mathrm{tar}}^4$, while $h \asymp h_*$ and $h_*\varepsilon = \sqrt{ah_*}$. Since $\sqrt{\beta} \asymp x_{\mathrm{prev}}^{-1}$, this proves (4.11).

Relative sizes. We now prove Claim 3 of Proposition 4.1. For $\tau \ge 1$, uniformly on neighboring labels,

$$|m||v| \asymp s_0\sqrt{1 + x^4}\, V_\lambda.$$

This ideal size is bounded by a fixed multiple of its target value: V_λ increases up to $x = 1$, and afterwards $x^2 V_\lambda = xf(1/x)$ increases. The logarithmic derivative of the latter with respect to τ is $\sqrt{\beta}/x + z/x^2$, which is bounded; the allowed extra interval after the target changes size by a fixed factor. On the history interval $|m||v| \le K_h^c$, and for $\tau < 1$ it is at most $K_h^c\Theta^c\varepsilon^{-c}$. In contrast,

$$V_\lambda(x = x_{\mathrm{tar}}) \ge V_\lambda(x = 1)/x_{\mathrm{tar}} \ge x_{\mathrm{tar}}^{-1} e^{b_0/\sqrt{\beta}}.$$

These inequalities prove (4.12).

Propagator and normalization. To finish Claim 4 of Proposition 4.1, we convert the propagator bound to the packet coordinates and apply the target normalization. Returning to physical times s, t, choose $g(t) = V_\lambda(a(t - t_0)/\varepsilon)$. Write $\tau_s = a(s - t_0)/\varepsilon$ and $\tau_t = a(t - t_0)/\varepsilon$. Since I is increasing, (4.26) implies

$$\frac{V_0(\tau_t)}{V_0(\tau_s)} \le \frac{V_\lambda(\tau_t)}{V_\lambda(\tau_s)} = \frac{g(t)}{g(s)}.$$

The packet coefficient vector corresponding to (U, V) is

$$a_\perp = R_\perp^T F^{-1}\left(\varepsilon U p + V q - \varepsilon\frac{PU + QV}{N} n\right).$$

The inverse map first forms $A = FR_\perp a_\perp$ and then takes $(U, V) = (\varepsilon^{-1} p \cdot A, q \cdot A)$. The bounds on P, Q, N and $F^{\pm 1}$ therefore show that changing coordinates at the two endpoints costs at most $K_h^c \varepsilon^{-1}\Theta^c$. Here P again denotes the packet data bound in Claim 4 of Proposition 4.1. The propagator estimate therefore has the required form $P^c g(t)/g(s)$, even before $\tau = 1$. The target lower bound and (4.13) give $\alpha \le P^c e^{-bx_{\mathrm{prev}}}$. Write $\tau_{\mathrm{tar}} = x_{\mathrm{tar}}/\sqrt{\beta}$. The preceding monotonicity and size estimates give

$$\sup_{[t_0,S]} g \le C x_{\mathrm{tar}} V_\lambda(\tau_{\mathrm{tar}}), \qquad A_{\mathrm{tar}} \ge c s_0 \sqrt{1 + x_{\mathrm{tar}}^4}\, V_\lambda(\tau_{\mathrm{tar}}).$$

In (4.13), these common amplitude factors cancel. Using $s_0^{-1} \le K_h^c$ gives $\sup \alpha g \le P^c$. This completes the proof of Proposition 4.1. □

5. Choosing the scales and iterating the construction

We have proved two conditional construction results. Proposition 3.1 turns a prescribed growing wave into an exact Euler perturbation, and Proposition 4.1 supplies such a wave and the next frame under the geometric hypotheses in §4.1. It remains to choose the scales so that these hypotheses hold at every stage. In this section we make those choices and construct the exact solutions U^j in (2.1), starting from a smooth base flow.

Repeating the amplification requires each new shear to dominate its predecessor before the parent's existence interval ends. Meanwhile, the changes to the initial datum must remain summable in every fixed Sobolev norm. The exponential separation between initial and target sizes in (4.12) allows the scales to meet these demands while retaining the pressure, initial gradient, and particle-map bounds needed for another step. In §6, we will then pass from these distinct solutions to a limiting initial datum and prove that its solution breaks down.

For each approximating exact solution U^i with pressure p_i, write $M_i = \nabla U^i$ and $H_i = \nabla^2 p_i$. Norms of these fields are taken over physical space and the time interval under discussion.

5.1. A smooth starting flow and local existence.

We construct a base solution U_B whose central gradient gives $a = 1$ and $\beta = x_0^{-2}$ in the geometric parameters (4.4). Localizing the vector potential of a linear velocity field gives a compactly supported, divergence-free datum with the required gradient near the origin.

Fix an oriented orthonormal basis (p, q, n), with $n = p \times q$. A large number x_0 will be fixed last. Define the trace-free linear map

$$L_B q = p + x_0^{-2} n, \qquad L_B p = L_B n = 0.$$

Choose a fixed even, compactly supported Gevrey cutoff χ_B, with $\chi_B = 1$ near the origin, as in §3.1, and set

$$u_{B,0}(a) = \operatorname{curl}\left(-\frac{1}{3}\chi_B(a)\, a \times L_B a\right). \tag{5.1}$$

Since $\operatorname{curl}(a \times L_B a) = -3L_B a$, this is a divergence-free, odd field equal to $L_B a$ near zero. Its support and all its factorial Sobolev bounds are uniform for $x_0 \ge 1$.

We record the local theory used both here and at the end of the proof. For smooth divergence-free data, the projected Euler equation has a unique smooth solution on an interval whose length is bounded below in terms of the initial H^3 norm. Its energy estimates are

$$\frac{d}{dt}\|u(t)\|_{H^m} \le C_m \|\nabla u(t)\|_{L^\infty} \|u(t)\|_{H^m}, \qquad m \ge 3, \quad m \in \mathbb{N}. \tag{5.2}$$

For completeness, apply a Fourier projection to $|\xi| \le N$ to the projected equation. This gives a locally Lipschitz ordinary differential equation on the band-limited subspace of H^m, with initial datum $u_N(0) = \Pi_N u_0$. Orthogonality of the projection preserves the transport cancellation. For a commutator product with derivative orders j and $m-j+1$, interpolate between $\nabla u \in L^\infty$ and $\nabla^m u \in L^2$, using

$$\|\nabla^j u\|_{L^{2(m-1)/(j-1)}} \le C_m \|\nabla u\|_{L^\infty}^{1-(j-1)/(m-1)} \|\nabla^m u\|_{L^2}^{(j-1)/(m-1)}.$$

The endpoint $j = 1$ has exponent ∞. The two interpolation powers add to one. The same calculation at each lower differentiated order, together with conservation of the undifferentiated energy, proves (5.2). The embedding $H^3(\mathbb{R}^3) \hookrightarrow C_b^1(\mathbb{R}^3)$ gives a common interval for all truncations; (5.2) then bounds every higher norm there. Their residual in the untruncated equation satisfies

$$\|r_N\|_{L^2} \le CN^{-(m-1)} \|u_N \cdot \nabla u_N\|_{H^{m-1}}.$$

The L^2 difference energy therefore gives convergence in $C_t L_x^2$. Interpolation with the uniform higher bounds gives convergence in every fixed Sobolev order for smooth data. Passing to the projected equation constructs the solution and its continuous time derivatives. The same difference estimate proves uniqueness. The pressure gradient is the ordinary L^2 gradient projection of $-u \cdot \nabla u$.

Let U_B be the solution with initial datum (5.1), and write X_B for its particle map. The velocity U_B has uniform factorial bounds on a fixed short interval. To see this, let $\rho = \rho(t) > 0$ be a time-dependent radius, with $\rho(0)$ below a uniform Gevrey radius of $u_{B,0}$, and put

$$X_\rho = \sum_{r \ge 0} \frac{\rho^r}{(r!)^2} \sum_{|I|=r} \|\partial^I U_B\|_{H^s}, \qquad Y_\rho = \sum_{r \ge 0} \frac{r\rho^r}{(r!)^2} \sum_{|I|=r} \|\partial^I U_B\|_{H^s}, \qquad s = 6.$$

The factorial product estimates and the preceding transport cancellation give, at each finite truncation of these sums,

$$X_\rho' \le CX_\rho^2 + \left(\frac{\rho'}{\rho} + \frac{CX_\rho}{\rho}\right) Y_\rho.$$

The initial value $X_\rho(0)$ is uniformly finite. Choose $\rho' < 0$ with sufficiently large fixed absolute value. On a sufficiently short uniform interval, the coefficient of Y_ρ is nonpositive, X_ρ stays bounded, and ρ stays positive. At a smaller radius, the projected equation bounds $U_{B,t}$ and the material acceleration. Applying the flow composition argument based on (3.56) gives the same kind of factorial bounds for $X_B - \mathrm{id}$, $X_{B,t}$, and $X_{B,tt}$, with a fixed constant K_h. In particular the velocity gradient M_B and pressure Hessian H_B are uniformly bounded. Fix K_B with

$$\lambda_{\max}(H_B(t,x)) \le K_B$$

on this interval. The later choice of time widths will place every construction inside this uniform base interval.

5.2. **The scales and their order of choice.** We now specify the sizes of the packets to be constructed, then choose constants that make these sizes compatible with Propositions 3.1 and 4.1. We index the base flow as $U^{J-2} = U_B$. The first packet produces U^{J-1} with its prescribed shear already present at time zero. For each $j \ge J$, an amplification stage produces U^j from U^{j-1}; we call these the normal stages. The integer J determines where this indexing starts and will be chosen below.

At a normal stage $j \ge J$, h_j is the prescribed central shear size at the target, k_j is the frequency in the scaled label y, and ℓ_j is the physical label scale of the packet support. The parameter δ_j sets the peak derivative $f'_{\delta_j}(0) = \delta_j^{-1}$ of the angular profile. The number x_j is the dimensionless target used in Proposition 4.1; the next stage uses it as x_{prev}. The following formulas define these sequences in terms of four constants J, D_0, D_k, x_0, whose values we choose in the order described below. The first line starts the geometric recursion at $x_{J-1} = x_0$. The next two lines prescribe the first packet, and the last two prescribe the normal stages:

$$\begin{gathered} x_{J-1} = x_0, \qquad x_j = j^2 x_{j-1} \quad (j \geq J), \\ h_{J-1} = x_0^{D_0}, \qquad k_{J-1} = x_0^{D_k}, \\ r = \ell_{J-1} = x_0^{-D_0}, \qquad \delta_{J-1} = x_0^{-(D_0+10)}, \\ \log h_j = \frac{x_{j-1}}{j^5}, \qquad \log k_j = \frac{x_{j-1}}{j^2}, \\ \log \delta_j^{-1} = \frac{x_{j-1}}{j^3}, \qquad \log \ell_j^{-1} = \frac{x_{j-1}}{j^{7/2}} \quad (j \geq J). \end{gathered} \tag{5.3}$$

Here r is the radius of the fixed ball in which we allow large negative eigenvalues of the initial symmetric velocity gradient. It equals the first packet's support scale; all later oscillatory initial supports will lie in this ball. Real frequencies k_j are admissible because the periodic angle profile is restricted to a graph on $\mathbb{R}^3$; spatial periodicity is not imposed.

Bounds needed to apply the geometric proposition again. We will maintain

$$\|M_i\|_{L^\infty} \leq C_M h_i, \qquad \|H_i\|_{L^\infty} \leq C_H h_i h_{i-1} \quad (i \geq J-1), \qquad h_{J-2} = 1. \tag{5.4}$$

These are the gradient and pressure-Hessian bounds required in (4.6) for the next amplification stage. To fix their constants, first choose C_M larger than the base-flow bounds and the absolute constant C in $|m||v|/A_{\text{tar}} \leq C$ from Claim 3 of Proposition 4.1. After the normalization (4.13), this ratio bounds the new leading gradient by a constant times h_i. Choose C_H sufficiently larger than C_M times that constant and the base Hessian bound: the pressure increment in (3.14) has one additional factor of the parent gradient, of size $C_M h_{i-1}$. The scale separation verified below allows these bounds also to include the preceding solution and the packet remainders.

With C_M, C_H fixed, fix C_0 in $0 \leq \Lambda \leq C_0 h_* I$. Here Λ maps the terminal displacement of a stationary path to the transverse part of its terminal derivative, as defined in §4.2. Then choose ζ as in that proof, so that $\|\overline{B}\| \leq \zeta h_*$, together with $\overline{B}_{pp} < 0$, permits the activation velocity $v_p \leq 0 < v_q$. Thus ζ specifies how small the older gradient and inherited error must be relative to the parent shear. The geometric smallness thresholds may depend on C_M, C_H; once those thresholds hold, the constant C in the size ratio above is absolute. This permits the stated order of choices.

Bounds needed for the packet inverses. The packet coercivity condition (3.11) uses an upper bound for the pressure Hessian and lower bounds for the initial symmetric gradient, separately inside and outside $|x| < r$. We will maintain

$$\lambda_{\max}(H_i(t,x)) \leq K_B + 1, \qquad \frac{M_i(0,x) + M_i(0,x)^T}{2} \geq \begin{cases} -B_c I, & |x| < r, \\ -B_e I, & |x| \geq r. \end{cases} \tag{5.5}$$

The base Hessian satisfies $\lambda_{\max}(H_B) \leq K_B$, so the added 1 allows a total positive pressure-Hessian increase of less than 1 from all packets. Choose B_e one larger than a uniform bound for $\|\nabla u_{B,0}\|_{L^\infty}$; this similarly leaves room for the summed initial changes outside the ball. Inside it, the first packet has shear size h_{J-1}, so set

$$B_c = C_M h_{J-1} + 2, \qquad L = C_1 B_c + 1. \tag{5.6}$$

Here C_1 is the constant in (3.11), and L is the coefficient of the displacement boundary term in the mean inverse. This choice gives $L > C_1 B_c$. The remaining coercivity inequality will follow from the short time interval and the small volume r^3 of the region with the larger bound B_c; it is verified in §5.5. Since $h_{J-1} = x_0^{D_0}$, these formulas define B_c and L as functions of the still unchosen scale x_0. Their polynomial growth in x_0 will be included in the packet size estimates.

The exponent required by the packet proposition. The input exponent c_0 in §3.2 controls the derivative bounds for $F, F^{-1}, F_t, F_{tt}, M, H$, the lower bounds for K, G, K_R, D_m, and the derivative bounds for their inverses, as well as the time, scale, propagator, and amplitude bounds. When these quantities are obtained from the parent particle map and Proposition 4.1, their bounds involve finitely many fixed powers of the packet size parameter P. Choose c_0 larger than all those exponents. The packet proposition then supplies a fixed finite $Q = Q(c_0)$ for (3.12). Also fix $\vartheta = 10^{-6}$ from that condition and $C_* = 10(s+2)$ from (3.16). The latter means that a packet built at frequency k_{j-1} provides the parent particle-map bound $K_h = k_{j-1}^{C_*}$ at the next stage. Neither c_0 nor Q will change when we choose J or x_0.

Choosing the four constants in the scale formulas. We can now choose D_0, D_k, J, x_0 in (5.3). Set

$$D_0 = 1000.$$

This gives the first shear $h_{J-1} = x_0^{1000}$ and radius $r = x_0^{-1000}$. The base time interval in (5.7) has length $6J^2 x_0^{2-D_0/2}$, which therefore tends to zero. The first geometric error also decays: its factor $\varepsilon\Theta^{61}$ is at most $C(J)x_0^{122-D_0/2}$, as checked in §5.5. Choose D_k sufficiently large compared with $(D_0+20)Q/\vartheta$ and 1000. Here and below, P_i denotes the size parameter P in the application of the packet proposition that constructs U^i. For the first packet, the size parameter satisfies $P_{J-1} \le C(J)x_0^{D_0+20}$, as verified in §5.5. Once x_0 is sufficiently large, this choice makes $k_{J-1} = x_0^{D_k}$ large enough for $P_{J-1}^Q \le k_{J-1}^{\vartheta/100}$ and makes its $k_{J-1}^{-1/4}$ error sufficiently small for the first amplification stage.

Next choose J large enough for the normal-stage formulas to satisfy three requirements. The calculations in §5.5 verify them uniformly for $j \ge J$ once x_0 is sufficiently large:

1. The new frequency must dominate the packet input bounds: $P_j^Q \le k_j^{\vartheta/100}$. For this, take J sufficiently large compared with Q/ϑ and C_*Q/ϑ. The second comparison accounts for the inherited particle-map bound $k_{j-1}^{C_*}$.
2. The parent coefficients must vary little across the new packet support. Their contribution to (4.5) is $\ell_j K_h^c/\varepsilon$. For every exponent c occurring there, take J sufficiently large compared with $(cC_*)^2$. The exponent $j^{-7/2}$ in $\log \ell_j^{-1}$ then makes this spatial scale small enough to offset the factor K_h^c inherited from the preceding frequency.
3. The exponential reduction before amplification must outweigh the factor $K_h^c\Theta^c\varepsilon^{-c}$ in (4.12). Take J sufficiently large compared with its fixed exponents divided by b, the constant in $e^{-bx_{j-1}}$. This makes the pressure and initial-gradient changes before amplification summable, even after multiplying by the shear sizes $h_j h_{j-1}$, as shown in (5.15).

Increasing J by a fixed factor to meet these requirements is allowed. Finally choose x_0 sufficiently large, depending on all the constants now fixed. This makes the time interval lie inside the base solution's existence interval, gives the required small total pressure and initial changes, and ensures all the scale comparisons in §5.5. Evaluate B_c, L from (5.6) at this chosen x_0. Their values do not require any new choice of c_0, Q, D_k, or J: their dependence on x_0 was already included in the polynomial size bounds used above.

5.3. **The time intervals.** The scale sequences have now been fixed. The physical target times also depend on the parent gradients, so they are determined as we construct the solutions. The first packet over the base flow prepares the parent for stage J. For $j \geq J$, the time t_{j-1} is the activation time, t_j is the new packet's target time, and S_j is the horizon retained for the next stage. The first amplification stage $j = J$ has activation time zero; all stages $j > J$ have a positive history interval. Define the available time widths by

$$W_j = \frac{3x_j x_{j-1}}{\sqrt{h_{j-1}}}, \qquad S_{\text{base}} = 2W_J = 6J^2 x_0^{2-D_0/2}. \tag{5.7}$$

Set $U^{J-2} = U_B$, $t_{J-1} = 0$, and $S_{J-1} = S_{\text{base}}$. We first construct U^{J-1} on this interval. Subsequently U^j will be constructed on $[0, S_j]$, where its predecessor exists at least until $S_{j-1} = t_{j-1} + 2W_j$.

At stage $j \geq J$, compute a, β after U^{j-1} and its shear frame have been constructed. Use the older gradient $B(t) = M_{j-2}(t, 0)$ in (4.4) at $t_0 = t_{j-1}$, with $h_* = h_{j-1}$, $x_{\text{prev}} = x_{j-1}$, and $x_{\text{tar}} = x_j$. The target time and retained horizon are then defined recursively by

$$t_j = t_{j-1} + \frac{x_j}{\sqrt{\beta a h_{j-1}}}, \qquad S_j = t_j + 2W_{j+1}. \tag{5.8}$$

As long as $a, \beta x_{j-1}^2 \in [1/2, 2]$, this gives

$$\frac{W_j}{6} \leq t_j - t_{j-1} \leq \frac{2W_j}{3}. \tag{5.9}$$

5.4. **The estimates maintained at each stage.** At stage $j \geq J$, the packet is built over U^{j-1}, whose primary shear is measured relative to the older solution U^{j-2}. Both solutions therefore enter the induction hypothesis, which consists of the following bounds.

(i) **Parent shear and frame.** The parent U^{j-1} and older solution U^{j-2} are smooth, odd, exact Euler solutions on the common interval $[0, S_{j-1}]$. At the origin on $[t_{j-1}, S_{j-1}]$, the parent has the form (4.3), with older field $B = M_{j-2}$, error $|E| \leq k_{j-1}^{-1/4}$, and $h(t_{j-1}) = h_{j-1}$. The normalized frame evolves according to (4.2).

(ii) **Activation conditions.** At activation, $a, \beta x_{j-1}^2 \in [1/2, 2]$. For $j > J$, the activation inequalities (4.7) also hold. The first stage $j = J$ instead uses the zero-time prescription in Proposition 4.1.

(iii) **Gradient and pressure bounds.** The constructed solutions satisfy the low bounds (5.4) and the one-sided Hessian and initial gradient bounds (5.5) on their respective intervals. The base solution $U^{J-2} = U_B$ has the uniform bounds established above. For $j > J$, the parent's bounds on $[0, t_{j-1}]$, together with $t_{j-1}^{-1} \leq h_{j-1}$, give (4.6).

(iv) **Particle-map bounds.** For $i \in \{j-2, j-1\}$ with $i \geq J-1$, the particle map of U^i satisfies (3.16) with $k = k_i$, restricted to $[0, S_{j-1}]$. For $i = J-2$, use the uniform base-map bound from Section 5.1. In the current packet hypotheses we take $K_h = k_{j-1}^{C_*}$, and at positive-history stages require $t_{j-1}^{-1} \leq K_h^c$.

The first-packet construction below establishes this state for $j = J$. The inductive step constructs U^j on $[0, S_j]$ and establishes the same state for $j+1$. We first check the scale inequalities used in both steps.

5.5. **Verifying the scale requirements.** As the parent shear grows, its coefficient bounds also grow. The packet construction remains applicable provided the new frequency dominates them in the sense of (3.12). Expressing those bounds in terms of the chosen scales makes this condition explicit.

Packet smallness. At a normal stage, let P_j be a sufficiently large fixed constant plus fixed multiples of

$$k_{j-1}^{C_*}, \quad \ell_j^{-1}, \quad \delta_j^{-1}, \quad h_j, \quad h_{j-1}, \quad h_{J-1}, \quad r^{-1}, \quad S_{\text{base}}^{-1}, \quad x_j, \quad x_{j-1}. \tag{5.10}$$

The first entry is the physical parent bound K_h. We may take $\Theta = C(1 + x_j x_{j-1})$, and $\varepsilon^{-1} \le C\sqrt{h_{j-1}} \le Ch_{j-1}$. For $j > J$, the recursion for x_j gives the exact identities

$$\log k_{j-1} = \frac{x_{j-1}}{(j-1)^4}, \qquad \log h_{j-1} = \frac{x_{j-1}}{(j-1)^7}. \tag{5.11}$$

Consequently, for a fixed C' also covering the first-stage power D_k,

$$\frac{\log P_j}{\log k_j} \le C\left\{\frac{1}{j} + \frac{C_*}{j^2} + \frac{j^2\left(\log x_{j-1} + C'\log x_0 + C'\log J\right)}{x_{j-1}}\right\}. \tag{5.12}$$

Choose J and then x_0 so that the right side is less than $\vartheta/(100Q)$. Thus $P_j^Q \le k_j^{\vartheta/100}$ at every stage. The same ratio tends to zero as $j \to \infty$. Uniformity follows directly from $x_j = j^2 x_{j-1}$: for each fixed A, the quantities $j^A \log x_{j-1}/x_{j-1}$ decrease at least geometrically once J and x_0 are sufficiently large. At the first packet stage the parent bound is fixed, and we may take $P_{J-1} \le C(J)x_0^{D_0+20}$ without including k_{J-1}; the choice of D_k proves the required smallness there. Thus the packet condition (3.12) holds at the seed and every normal stage.

Compatible time intervals. The widths satisfy

$$\frac{W_{j+1}}{W_j} = j^2(j+1)^2\sqrt{\frac{h_{j-1}}{h_j}}.$$

For $j > J$, the logarithm of the square-root factor is

$$-\frac{x_{j-1}}{2j^5} + \frac{x_{j-1}}{2(j-1)^7}.$$

For $j = J$ the second term is instead $(D_0/2)\log x_0$. In both cases the negative term dominates all logarithmic factors. In particular, the horizons nest, $S_j \le S_{j-1}$, and the extra scaled time after target satisfies

$$2\sqrt{ah_{j-1}}\, W_{j+1} \le \Theta^{-60}. \tag{5.13}$$

Also $t_J \ge S_{\text{base}}/12$. At positive-history stages, $t_0 = t_{j-1} \ge t_J$, so $t_0^{-1} \le 12S_{\text{base}}^{-1} \le h_{j-1} \le K_h^c$. For the first packet, $S = S_{\text{base}}$; for each normal stage, $S = S_j \ge t_J$, so $S^{-1} \le 12S_{\text{base}}^{-1}$. For normal stages, these bounds are covered by the S_{base}^{-1} entry in (5.10).

Geometric stability and shear separation. The error in (4.5) has three contributions: variation of the older field, the inherited packet error, and variation across the new packet support. Take

$$E_* = k_{j-1}^{-1/4}, \qquad G_* = 1 + h_{j-2}.$$

For $B(t) = M_{j-2}(t,0)$, the older Euler solution satisfies $\dot{B} = -B^2 - H_{j-2}(t,0)$. The bounds (5.4) therefore give $\|B\| \le CG_*$ and $\|\dot{B}\| \le CG_*^2$; when $j = J$, use the uniform base bounds instead. At $j = J$, we have $G_* = 2$, $\varepsilon \le Cx_0^{-500}$, and $\Theta \le CJ^2 x_0^2$. Therefore

$$\varepsilon\Theta^{61}G_*^2 \ll x_0^{-10}.$$

The preceding packet error has arbitrarily strong polynomial decay by the choice of D_k, and the variation across packet labels, represented by $\ell_j K_h^c/\varepsilon$ in (4.5), includes $\exp(-x_0/J^{7/2})$, which decays faster than every fixed inverse power of x_0. At $j = J+1$, the older size is $G_* = 1 + x_0^{D_0}$, while $\log\varepsilon = -x_0/(2J^5) + O(1)$. At subsequent stages,

$$\begin{aligned}\log(\varepsilon G_*^2) &\le -\frac{x_{j-1}}{2(j-1)^7} + \frac{2x_{j-1}}{(j-1)^2(j-2)^7} + C\\ &\le -\frac{2x_{j-1}}{5(j-1)^7} + C.\end{aligned}$$

Thus εG_*^2 decays faster than every fixed inverse power of Θ and x_{j-1}. The contribution from variation across packet labels has logarithm at most

$$-\frac{x_{j-1}}{j^{7/2}} + \frac{cC_* x_{j-1}}{(j-1)^4} + \frac{Cx_{j-1}}{(j-1)^7} + C.$$

Our choice of J makes this at most $-x_{j-1}/(2j^{7/2})$. Finally, the preceding frequency error has logarithm $-x_{j-1}/(4(j-1)^4)$. These comparisons prove (4.5), with any fixed enlargement of its constants. The additional requirements $\beta \le \beta_0$ and $x_{\mathrm{tar}} \ge 2$ hold because $\beta \le 2x_{j-1}^{-2}$ and $x_j \ge J^2 x_0$. The same scale comparisons give the following three inequalities. They allow the new bounds to absorb the preceding gradient and Hessian bounds, preserve the strict quadratic-form inequality (4.7) at the next activation, and keep the oscillatory initial support inside $|x| < r$, respectively:

$$h_j \gg h_{j-1}^2, \qquad \frac{\sqrt{h_{j-1}}}{x_{j-1}x_j} \gg G_* + 1, \qquad \ell_j < r, \tag{5.14}$$

including the first and second normal stages.

Summable pressure and history costs. To preserve the upper Hessian and initial gradient bounds (5.5), we sum the increments in Claim 1 of Proposition 3.1. Throughout the history interval, and after activation while $0 \le \tau < 1$, the size ratio (4.12), the normalization (4.13), and the parent bound (5.4) bound both leading increments by

$$b_j = Ch_j h_{j-1} K_h^c \Theta^c \varepsilon^{-c} e^{-bx_{j-1}}. \tag{5.15}$$

For $j > J$, the logarithm of the prefactor $Ch_j h_{j-1} K_h^c \Theta^c \varepsilon^{-c}$ is bounded by $C(c, C_*)x_{j-1}/(j-1)^4 + C\log(x_j h_{J-1})$. Increasing J and then x_0 gives $b_j \le e^{-bx_{j-1}/2}$; the first normal stage $j = J$ obeys the same conclusion with a possibly smaller fixed exponent.

On $\tau \ge 1$, the sign (4.9) and the lower bound $f'_\delta \ge -C$ in (3.4) control the positive part of the pressure-Hessian increment (3.14). The normalization (4.13) and the size comparison in Claim 3 of Proposition 4.1 bound this cost by $C\delta_j h_j h_{j-1}$. The scales (5.3) give

$$\delta_j h_j h_{j-1} \le \exp\left(-\frac{x_{j-1}}{2j^3}\right). \tag{5.16}$$

The remaining errors are the $k_j^{-1/4}$ remainders in (3.13) and (3.14). The sums of b_j, $\delta_j h_j h_{j-1}$, and $k_j^{-1/4}$, together with the sums of x_{j-1}^{-c} for every fixed $c > 0$ used below, can be made as

small as prescribed by choosing x_0 last. Indeed each positive exponent x_{j-1}/j^A in these decay estimates grows at least geometrically after J. The seed cost is also small: $\delta_{J-1}h_{J-1} = x_0^{-10}$.
Coercivity. Finally, $S_{\text{base}} \to 0$ and $h_{J-1}r^3 S_{\text{base}} \to 0$. Hence, with a strict margin,

$$\frac{(K_B+1)S^2}{2} + B_e S + C_2 B_c r^3 S \le \frac{1}{2} \qquad (0 < S \le S_{\text{base}}). \tag{5.17}$$

This verifies the time and initial-data part of (3.11). None of these comparisons requires an inequality of the form $e^{P_j^c} \ll k_j$.

5.6. **The first packet.** We construct U^{J-1} over $U^{J-2} = U_B$. This seed packet supplies the parent shear and frame required by Section 5.4 for the first amplification stage $j = J$. Apply Proposition 3.1 over U_B on $[0, S_{\text{base}}]$ with

$$\ell = \ell_{J-1}, \quad k = k_{J-1}, \quad \delta = \delta_{J-1}, \quad m_0 = p, \quad v(0) = q, \quad \alpha = \delta_{J-1}h_{J-1}, \quad L = 0.$$

For this application the base initial bound holds everywhere, so we use the coercivity condition with $B_c = 0$. The homogeneous propagator is bounded on this short interval, and we may take $g = 1$. For sufficiently large x_0, the packet support $|a| \le \ell_{J-1}/2$ lies where $\chi_B = 1$, so $M_B(0,a) = L_B$ there. Since $m(0) = p$ and $v(0) = q$, $m \cdot M_B v = p \cdot L_B q = 1$ initially. Along a base particle, $\dot{M}_B = -M_B^2 - H_B$; the base bounds and the ray and velocity equations therefore give

$$m \cdot M_B v \ge \frac{1}{2} \quad \text{on } [0, S_{\text{base}}].$$

The packet hypotheses hold, and we obtain a smooth odd solution U^{J-1}. The output expansions yield

$$\|M_{J-1}\|_\infty \le C_M h_{J-1}, \qquad \|H_{J-1}\|_\infty \le C_H h_{J-1}.$$

Because the leading Hessian increment is a negative multiple of $m \otimes m$ where $f_\delta' \ge 0$, and $f_\delta' \ge -C$ everywhere, its upper-eigenvalue cost is at most

$$C\delta_{J-1}h_{J-1} + k_{J-1}^{-1/4}.$$

At time zero, the mean coefficients $B_p(0)$ and the forced oscillatory coefficients $A_p(0), C_p(0)$ with $p \ge 2$ vanish. By Claim 3 of Proposition 3.1, the remaining primary oscillation and its curl correction are supported in $|x| \le \ell_{J-1}/2 = r/2$. Thus the initial exterior bound outside $|x| < r$ is unchanged; inside that ball, $B_c = C_M h_{J-1} + 2$ leaves a strict margin.

At the fixed origin, the packet expansion gives

$$M_{J-1} = M_B + h(t)q(t)p(t)^T + E(t), \qquad |E(t)| \le k_{J-1}^{-1/4}, \qquad h(t) = \frac{\alpha|m(t)||v(t)|}{\delta_{J-1}}.$$

The normalized ray and velocity form the orthonormal pair $p(t), q(t)$, and $h(0) = h_{J-1}$. Their equations give $h'/h = -(M_B)_{pp} - (M_B)_{qq}$. For the next construction, stage $j = J$, U^{J-1} is the parent and M_B is the older gradient. At its activation time $t_{J-1} = 0$, the prescribed base matrix gives exactly

$$a = 1, \qquad \beta = x_0^{-2}.$$

There is no positive-history endpoint condition at stage $j = J$, since its activation time is $t_{J-1} = 0$. The physical flow bound (3.16) supplies $K_h = k_{J-1}^{C_*}$ for the next packet. Together with the base-flow bounds, these estimates establish the state in Section 5.4 for $j = J$.

5.7. **The inductive step.** Assume the bounds in Section 5.4 at stage $j \geq J$. We construct U^j on $[0, S_j]$ with central target gradient

$$M_j(t_j, 0) = M_{j-1}(t_j, 0) + h_j q_2 p_2^T + O(k_j^{-1/4}),$$

where $p_2 = m(t_j, 0)/|m(t_j, 0)|$ and $q_2 = v(t_j, 0)/|v(t_j, 0)|$ are the new frame in Claim 2 of Proposition 4.1. The normalization (4.13) and Claim 1 of Proposition 3.1 give this gradient. We then verify the bounds of Section 5.4 for $j+1$: the summed costs from §5.5 preserve (5.5), the estimates (4.10) and (4.11) give the next activation parameters, and (3.16) controls the new particle map.

Constructing U^j from U^{j-1}. We apply the packet construction to U^{j-1}, whose leading shear is measured relative to U^{j-2}. Write $B(t) = M_{j-2}(t, 0)$. The origin is fixed by oddness. The gradient equation there gives

$$\dot{B} = -B^2 - H_{j-2}, \qquad \|B\| \leq CG_*, \qquad \|\dot{B}\| \leq CG_*^2, \qquad G_* = 1 + h_{j-2}.$$

For $j = J$, these bounds follow from the uniform base estimates. Set $t_0 = t_{j-1}$, define t_j, S_j by (5.8), and restrict U^{j-1} to $[0, S_j]$. Apply Proposition 4.1 with $h_* = h_{j-1}$, $x_{\text{prev}} = x_{j-1}$, and $x_{\text{tar}} = x_j$. Its hypotheses follow from Section 5.4 and the estimates in §5.5. This supplies the ray m and transverse velocity v: for $j > J$ their activation data satisfy (4.8), while for $j = J$ we use $v(0) = q$. Choose the amplitude so that the leading gradient increment at t_j has size h_j:

$$\alpha_j = \frac{\delta_j h_j}{|m(t_j, 0)||v(t_j, 0)|}. \tag{5.18}$$

Use ℓ_j, k_j, δ_j and the boundary parameter $L = C_1 B_c + 1$ in the packet construction. Let X be the particle map of U^{j-1} and write $F(t, y) = D_a X(t, \ell_j y)$, as in (3.5). The bound (3.16), with $K_h = k_{j-1}^{C_*}$, controls the derivatives of X. Using $F^{-1} = \text{cof}(F)^T$ and (3.2) gives the coefficient and inverse multiplier bounds in §3.2 (i), with growth constants that are fixed powers of K_h. Coercivity follows from (5.17) and the choice of L. Claim 4 of Proposition 4.1 supplies a positive function g, with $g(t_0) = 1$, for which the solution map from $a_\perp(s)$ to $a_\perp(t)$ in $A = FR_\perp a_\perp$ has norm at most $P_j^c g(t)/g(s)$, uniformly for $|y| \leq 1/2$ and $t_0 \leq s \leq t \leq S_j$, and $\sup_{[t_0, S_j]} \alpha_j g \leq P_j^c$. It also gives

$$\alpha_j \leq P_j^c e^{-b x_{j-1}}.$$

Together with the packet smallness condition verified in §5.5, these bounds verify the hypotheses of Proposition 3.1. Applying that proposition constructs an exact smooth odd Euler solution U^j on $[0, S_j]$, with particle-map bound (3.16) at frequency k_j.

Preserving the global and initial bounds. We next verify (5.4) and (5.5). On the packet support for $\tau \geq 1$, Claim 3 of Proposition 4.1 and (5.18) give $\alpha_j |m||v| \leq C\delta_j h_j$. Insert this bound, $|f'_{\delta_j}| \leq C\delta_j^{-1}$, and $\|M_{j-1}\|_\infty \leq C_M h_{j-1}$ into (3.13) and (3.14) to obtain

$$\|M_j - M_{j-1}\|_\infty \leq Ch_j + k_j^{-1/4},$$
$$\|H_j - H_{j-1}\|_\infty \leq CC_M h_j h_{j-1} + k_j^{-1/4}.$$

In (3.14), the matrix $m \otimes m/|m|^2$ is positive semidefinite. Since $\chi_1 \geq 0$ and $m \cdot M_{j-1} v > 0$, the leading Hessian increment is negative semidefinite wherever $f'_{\delta_j} \geq 0$. Where $f'_{\delta_j} < 0$, the bound $f'_{\delta_j} \geq -C$ gives

$$\lambda_{\max}(H_j) \leq \lambda_{\max}(H_{j-1}) + CC_M\delta_j h_j h_{j-1} + k_j^{-1/4}.$$

For labels with $\chi_1(y) = 0$, both leading increments vanish, while their remainders retain the global bound $k_j^{-1/4}$. Throughout the history interval, and after activation while $0 \leq \tau < 1$, both absolute increments are bounded by $C(1 + C_M)b_j + k_j^{-1/4}$. By §5.5, the sum of $CC_M\delta_i h_i h_{i-1}$, $C(1 + C_M)b_i$, and $k_i^{-1/4}$ over $i \geq J$, together with the seed Hessian increment, is less than 1. Adding these upper bounds to $\lambda_{\max}(H_B) \leq K_B$ gives

$$\lambda_{\max}(H_j) \leq K_B + 1.$$

The triangle inequality, the preceding bounds for M_{j-1}, H_{j-1}, and these increment estimates give (5.4), by the choices of C_M, C_H and $h_j \gg h_{j-1}^2$. At time zero, the same estimates give $\|M_j(0) - M_{j-1}(0)\|_\infty \leq C(1 + C_M)b_j + k_j^{-1/4}$ for every $j \geq J$, including $j = J$, for which $t_0 = \tau = 0$. The sum is smaller than the margins left in the exterior and core initial gradient bounds (5.5), so those bounds persist with a strict margin. These statements use the exact pressure expansion (3.14); no spatial locality of the pressure is assumed.

Transferring the primary frame and activation conditions. For $t \in [t_j, S_j]$, write $m(t) = m(t, 0)$ and $v(t) = v(t, 0)$, and define $p_2(t) = m(t)/|m(t)|$ and $q_2(t) = v(t)/|v(t)|$. Oddness of U^{j-1} gives $X(t, 0) = 0$, so the physical origin has label $y = 0$ and phase $\theta = k_j m_0 \cdot y = 0$. Since $\chi_1(0) = 1$ and $f'_{\delta_j}(0) = \delta_j^{-1}$, (3.13) gives

$$M_j = M_{j-1} + h(t)q_2(t)p_2(t)^T + E_j(t), \qquad |E_j| \leq k_j^{-1/4},$$

$$h(t) = \frac{\alpha_j|m(t)||v(t)|}{\delta_j}, \quad h(t_j) = h_j.$$

At stage $j + 1$, U^j will be the parent solution and U^{j-1} will supply the older field. The ray and velocity equations for M_{j-1} imply that p_2, q_2 satisfy (4.2) with $B = M_{j-1}$, and $h'/h = -(M_{j-1})_{p_2p_2} - (M_{j-1})_{q_2q_2}$. The parameter-transfer formula (4.10) shows that the relative errors in successive values of a are absolutely summable: they are bounded by a fixed multiple of

$$x_j^{-4} + \frac{\beta}{x_j^2} + \frac{\sqrt{\beta}}{x_j^3} + e_*\Theta^{40}.$$

Indeed, $\beta \leq 2x_{j-1}^{-2}$, $x_j \geq x_{j-1}$, and $e_*\Theta^{60} \leq x_{j-1}^{-10}$ bound this expression by Cx_{j-1}^{-4}. Its sum can be made arbitrarily small by the final choice of x_0, as in §5.5. Starting from $a = 1$, the product of the successive ratios a_{new}/a therefore stays in $[1/2, 2]$. The other transfer estimate gives directly $\beta_{\text{new}} x_j^2 \in [1/2, 2]$; these errors need not be multiplied across stages.

To verify (4.7) for stage $j + 1$, evaluate the following quantities at $t = t_j$ and set $\overline{B} = M_{j-1}(t_j, 0) + E_j(t_j)$. The compression estimate (4.11) and (5.14) give

$$p_2 \cdot M_{j-1}p_2 \leq -\frac{c\sqrt{h_{j-1}}}{x_{j-1}x_j} + C(G_* + E_*) < -1.$$

Since $|E_j| \leq k_j^{-1/4} < 1$, this implies $\overline{B}_{p_2p_2} < 0$, while

$$\frac{\|M_{j-1} + E_j\|}{h_j} \leq \frac{C_M h_{j-1} + 1}{h_j} < \zeta.$$

Thus (4.7) holds at the next activation. The bounds (5.4) on $[0, t_j]$, together with $t_j^{-1} \le h_j$, verify (4.6) for the next stage with $t_0 = t_j$, $h_* = h_j$, and $h_{\text{old}} = h_{j-1}$. Every part of Section 5.4 now holds with j replaced by $j + 1$, closing the induction.

6. The limiting initial datum and finite-time breakdown

The preceding section constructed exact solutions U^j whose central gradients $|\nabla U^j(t_j, 0)|$ grow without bound. These solutions have different initial data. It remains to obtain an initial datum u_0 whose solution breaks down, as required by Theorem 1.1. In this section we complete that step. First, Section 6.1 uses the initial increment estimates to obtain convergence in every fixed Sobolev norm. Next, Section 6.2 shows that a smooth solution from the limiting datum persisting past the accumulation time would inherit the growing gradients by stability. Finally, Section 6.3 proves that $\|\nabla u(t)\|_\infty$ is unbounded as $t \uparrow T_*(u_0)$ and that $\int_0^{T_*(u_0)} \|\operatorname{curl} u(t)\|_\infty \, dt = \infty$.

6.1. **Smooth convergence at the initial time.** The first packet contributes a fixed smooth initial increment. To prove summability, it remains to estimate the normal stages. For each $j \ge J$, split $U^j(0) - U^{j-1}(0)$ into the oscillatory and mean increments defined in (3.60). For every fixed integer m, apply (3.17) to these two increments. Substituting the scales (5.3) and $\alpha_j \le P_j^c e^{-bx_{j-1}}$ bounds the logarithms of their H^m norms, respectively, by

$$-bx_{j-1} + \frac{mx_{j-1}}{j^2} + \frac{mx_{j-1}}{j^{7/2}} + C_m \log P_j, \tag{6.1}$$

$$-\frac{2x_{j-1}}{j^2} + \frac{mx_{j-1}}{j^{7/2}} + C_m \log P_j. \tag{6.2}$$

Since $\log P_j = o(\log k_j) = o(x_{j-1}/j^2)$, for sufficiently large j depending on m these bounds are at most $-bx_{j-1}/2$ and $-x_{j-1}/j^2$, respectively. Both exponentials are summable. The mean increment $u_{\text{mean},0}$ is independent of θ, so differentiating it does not produce the phase factor k_j^m. This explains why the derivative contribution mx_{j-1}/j^2 in (6.1) is absent from (6.2).

The sequence $U^j(0)$ is therefore Cauchy in every H^m. By Claim 3 of Proposition 3.1, the oscillatory initial increments are supported in $|a| \le \ell_j/2$, and the mean initial increments are supported in $|a| \le 2$. Together with the fixed support of the base datum, these supports lie in a fixed ball. The limit

$$u_0 = \lim_{j\to\infty} U^j(0) \quad \text{in every } H^m(\mathbb{R}^3) \tag{6.3}$$

is consequently smooth, compactly supported, odd, and divergence-free.

6.2. **Non-persistence of the smooth solution.** The times t_j in (5.8) increase and are bounded by S_{base}, as verified in §5.5. Consequently their accumulation time

$$T_\infty = \lim_{j\to\infty} t_j \quad \text{satisfies} \quad 0 < t_J \le T_\infty \le S_{\text{base}}. \tag{6.4}$$

Let u be the maximal smooth Euler solution with initial datum u_0. Local existence gives $T_*(u_0) > 0$. Suppose, for contradiction, that $T_*(u_0) > T_\infty$. Then all Sobolev norms of u are bounded on $[0, T_\infty]$. Compare U^j with u on $[0, t_j]$, where U^j exists, and write $w = U^j - u$. Its equation is

$$\partial_t w + u \cdot \nabla w + w \cdot \nabla u + w \cdot \nabla w + \nabla(p_j - p) = 0, \qquad \operatorname{div} w = 0.$$

Since $\operatorname{div}(u+w)=0$, each derivative word I satisfies $\langle \partial^I w, (u+w)\cdot\nabla\partial^I w\rangle_{L^2}=0$. The pressure term also has zero inner product with $\partial^I w$. Summing over $|I|\le 3$, Sobolev product estimates bound the terms containing u by $C\|u\|_{H^4}\|w\|_{H^3}^2$, and the nonlinear terms containing only w by $C\|w\|_{H^3}^3$. Thus

$$y'(t)\le C_u y(t)+Cy(t)^2, \qquad y(t)=\|w(t)\|_{H^3}, \tag{6.5}$$

where C_u depends on $\sup_{[0,T_\infty]}\|u\|_{H^4}$ and is independent of j. While $y\le 1$, $y(t)\le y(0)e^{(C_u+C)T_\infty}$. For sufficiently large j, the right side is less than $1/2$, since $y(0)\to 0$. This excludes a first time at which $y=1$, so the bound holds on all of $[0,t_j]$, proving

$$\sup_{0\le t\le t_j}\|U^j(t)-u(t)\|_{H^3}\longrightarrow 0.$$

At t_j, the leading increment $h_j q_2 p_2^T$ has operator norm h_j. The triangle inequality and (5.4) therefore give

$$\|M_j(t_j,0)\|\ge h_j - C_M h_{j-1} - k_j^{-1/4}\longrightarrow\infty.$$

But $H^3\hookrightarrow C_b^1$ and the preceding convergence imply $\|M_j(t_j,0)-\nabla u(t_j,0)\|\to 0$. Since ∇u is bounded on $[0,T_\infty]$, this contradicts the lower bound. Hence

$$0<T_*(u_0)\le T_\infty\le S_{\mathrm{base}}<\infty. \tag{6.6}$$

The maximal lifespan may end before T_∞; this upper bound is sufficient for finite-time breakdown. The comparison on the intervals $[0,t_j]$ was made under the contradiction assumption $T_*(u_0)>T_\infty$.

6.3. **Continuation criteria at the endpoint.** Set $T=T_*(u_0)<\infty$. We now prove the two divergence assertions of Theorem 1.1: the velocity gradient satisfies $\limsup_{t\uparrow T}\|\nabla u(t)\|_\infty=\infty$, and the vorticity satisfies $\int_0^T\|\operatorname{curl} u(t)\|_\infty\,dt=\infty$.

Suppose, for contradiction, that $B:=\sup_{0\le t<T}\|\nabla u(t)\|_\infty<\infty$. By (5.2),

$$\sup_{0\le t<T}\|u(t)\|_{H^3}\le\|u_0\|_{H^3}e^{C_3BT}\le M_3:=\max\{1,\|u_0\|_{H^3}e^{C_3BT}\}.$$

The local theory in Section 5.1 supplies a time $\delta_0=\delta_0(M_3)>0$, uniform for smooth divergence-free data of H^3 norm at most M_3. For each $s\in[0,T)$, let v_s be the solution starting from $v_s(s)=u(s)$ on $[s,s+\delta_0)$. Fix an embedding constant $C_{\mathrm{emb}}>0$ such that $\|\nabla f\|_\infty\le C_{\mathrm{emb}}\|f\|_{H^3}$, and set

$$\delta=\min\left\{\frac{\delta_0}{2},\frac{1}{2C_3C_{\mathrm{emb}}M_3}\right\}>0.$$

The energy inequality at order three gives

$$\|v_s(t)\|_{H^3}\le\frac{M_3}{1-C_3C_{\mathrm{emb}}M_3(t-s)}\le 2M_3 \qquad (s\le t\le s+\delta).$$

For every integer $m\ge 3$, apply (5.2) at order m to u on $[0,s]$, using $\|\nabla u\|_\infty\le B$, and to v_s on $[s,s+\delta]$, using $\|\nabla v_s\|_\infty\le 2C_{\mathrm{emb}}M_3$. For $s\le t\le s+\delta$, this gives

$$\|v_s(t)\|_{H^m}\le\|u(s)\|_{H^m}e^{2C_mC_{\mathrm{emb}}M_3\delta}\le\|u_0\|_{H^m}e^{C_mBT+2C_mC_{\mathrm{emb}}M_3\delta}.$$

Thus all higher Sobolev norms remain finite on the interval of length δ, which does not depend on m or s. Choose $s\in(\max\{0,T-\delta/2\},T)$. Uniqueness gives $v_s=u$ on $[s,T)$, while $s+\delta>T$, so v_s gives a smooth extension past T. This contradicts maximality. Since u

is smooth on every compact subinterval of $[0,T)$, a finite limsup of $\|\nabla u(t)\|_\infty$ at T would make B finite. It follows that

$$\limsup_{t\uparrow T_*(u_0)} \|\nabla u(t)\|_\infty = \infty.$$

To prove the vorticity assertion, we first derive a logarithmic bound for $\|\nabla u\|_\infty$ in terms of $\|\omega\|_\infty$ and $\|u\|_{H^3}$. Write $\omega = \operatorname{curl} u$. The whole-space Biot–Savart identity is $u = \operatorname{curl}(-\Delta)^{-1}\omega$. Thus $\nabla u(x)$ is the sum of a fixed linear function of $\omega(x)$ and a principal-value integral against $\omega(x-z)$, with kernel $K(z)$. This kernel is homogeneous of degree -3, with smooth angular dependence and zero spherical mean. For $0 < d < 1/2$, split this integral into $|z| < d$, $d < |z| < 2$, and a smoothly cut off far region. The zero spherical mean allows us to subtract $\omega(x)$ in the inner principal-value integral. The embedding $H^3 \hookrightarrow C^{1,\lambda}$, $0 < \lambda < 1/2$, gives $|\omega(x-z) - \omega(x)| \le C\|u\|_{H^3}|z|^\lambda$. Integrating against $|K(z)| \le C|z|^{-3}$ yields

$$\left| \int_{|z|<d} K(z)[\omega(x-z) - \omega(x)]\, dz \right| \le C d^\lambda \|u\|_{H^3}.$$

The middle region contributes $C\|\omega\|_\infty \log(2/d)$. On the far region, write $\omega = \operatorname{curl} u$ and integrate by parts, moving the derivative onto the cut-off kernel. The resulting kernel belongs to L^2, so Cauchy–Schwarz bounds this contribution by $C\|u\|_2$. Choose $d = c(e + \|u\|_{H^3})^{-1/\lambda}$, with a fixed $c > 0$ small enough that $d < 1/2$. Then $d^\lambda \|u\|_{H^3} \le c^\lambda$ and $\log(2/d) \le C\log(e + \|u\|_{H^3})$. Combining the three region estimates with the local term, and using conservation of kinetic energy, gives

$$\|\nabla u\|_\infty \le C\left[1 + \|u_0\|_2 + \|\omega\|_\infty \log(e + \|u\|_{H^3})\right]. \tag{6.7}$$

Set $Z(t) = \log(e + \|u(t)\|_{H^3})$. Dividing the $m = 3$ case of (5.2) by $e + \|u\|_{H^3}$ gives $Z' \le C\|\nabla u\|_\infty$, since $\|u\|_{H^3}/(e + \|u\|_{H^3}) \le 1$. Applying (6.7) therefore yields

$$Z' \le C(1 + \|u_0\|_2) + C\|\omega\|_\infty Z.$$

If $\int_0^{T_*} \|\omega(t)\|_\infty\, dt$ were finite, Gronwall's inequality would bound Z, hence $\sup_{t<T} \|u(t)\|_{H^3}$. The embedding $H^3 \hookrightarrow C^1$ and the preceding restart argument would then extend u past T, contradicting maximality. Therefore the datum (6.3) satisfies

$$\int_0^{T_*(u_0)} \|\omega(t)\|_{L^\infty(\mathbb{R}^3)}\, dt = \infty.$$

The datum in (6.3), the finite lifespan in (6.6), and the two continuation conclusions above prove Theorem 1.1.

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